11TH LASIGE WORKSHOP

13 May 2026

Poster Session

driven by excellence

Fundação para a Ciência e a Tecnologia

Ciências ULisboa

ULISBOA

This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025

Use of Human Activity Recognition for the Prevention of Traffic Accidents Involving Professional Drivers

Agda Costa[1,2], Wilson Melo Jr[2], Alan Oliveira[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] PPGMT, Instituto Nacional de Metrologia, Qualidade e Tecnologia, Brasil

cyber-physical systems

Abstract

The use of Human Activity Recognition (HAR) applied to vehicular telematics as a tool for preventing traffic accidents involving professional drivers. The proposed approach integrates onboard sensors, wearable devices, and IoT systems to collect data related to driving behavior and the driver's physical state. These data are analyzed using Artificial Intelligence techniques, enabling the identification of risk patterns. The proposal contributes to the development of intelligent monitoring systems, with potential applications in both individual safety and fleet management, while also incorporating mechanisms to preserve data privacy.

Motivation

Traffic accidents represent a critical issue, largely driven by human factors such as fatigue, distraction, and stress. These factors are estimated to be involved in a significant proportion of incidents, particularly among professional drivers exposed to long working hours. In this context, there is a growing demand for technological solutions capable of acting preventively by identifying risky behaviors in real time. The use of vehicular telematics combined with HAR emerges as a promising approach to monitor drivers and anticipate critical events, contributing to accident reduction and improved road safety.

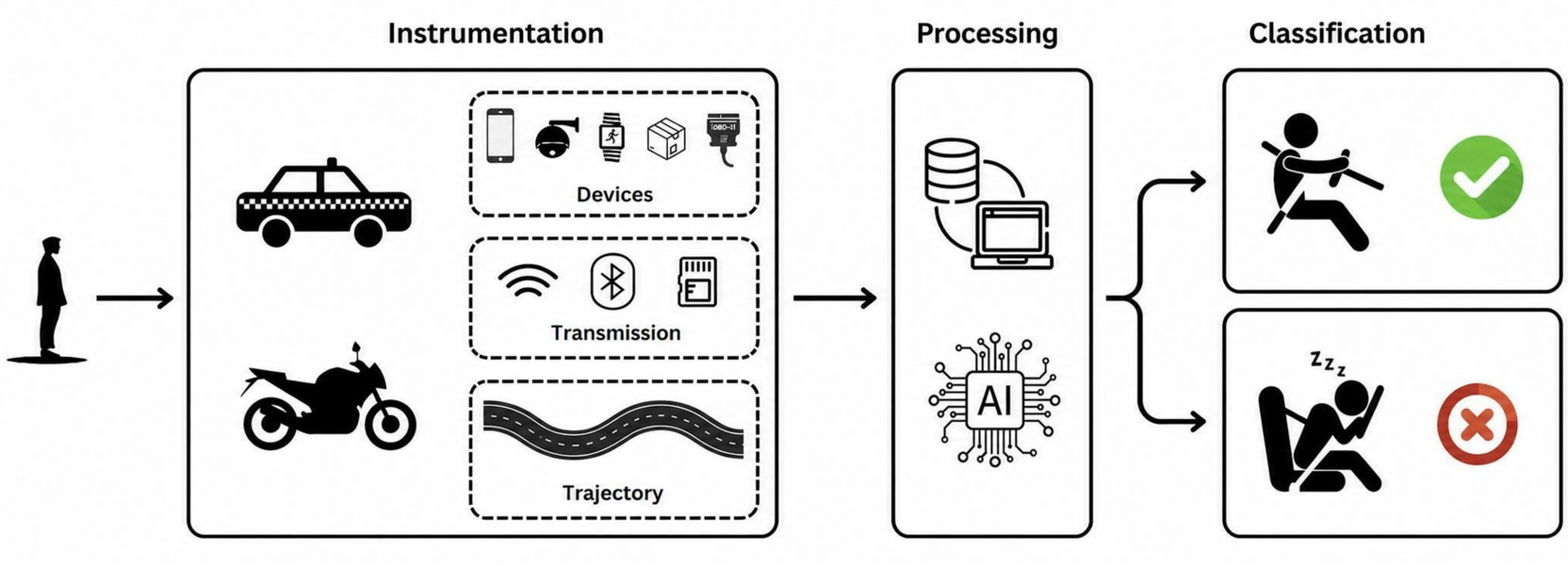
Objective

Develop methods and tools based on Human Activity Recognition (HAR) to identify risky driving behaviors and mitigate traffic accidents involving professional drivers, while preserving data privacy.

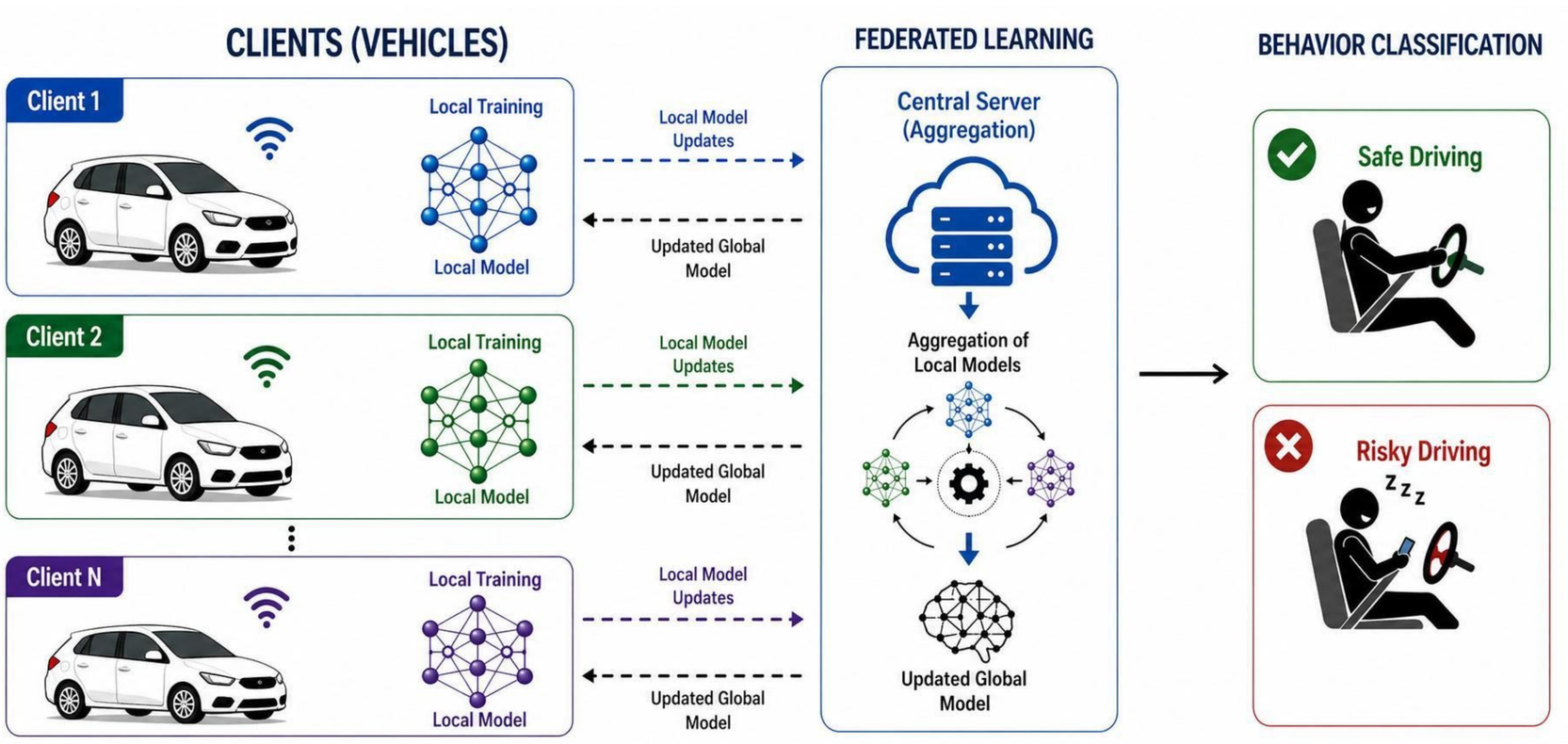
Criteria for Risky Driving

Criterion	Condition
Abnormal Acceleration	Δ speed ≥ 15 km/h within 5 seconds
Sharp Turn	Angular variation ≤ 0.7 radians
Weaving	Angle > 15° **&** lateral acceleration > 0.3 m/s² **&** ≥ 3 times
Classification	≥ 1 criterion → **Risky** / none → **Safe**

Centralized Learning

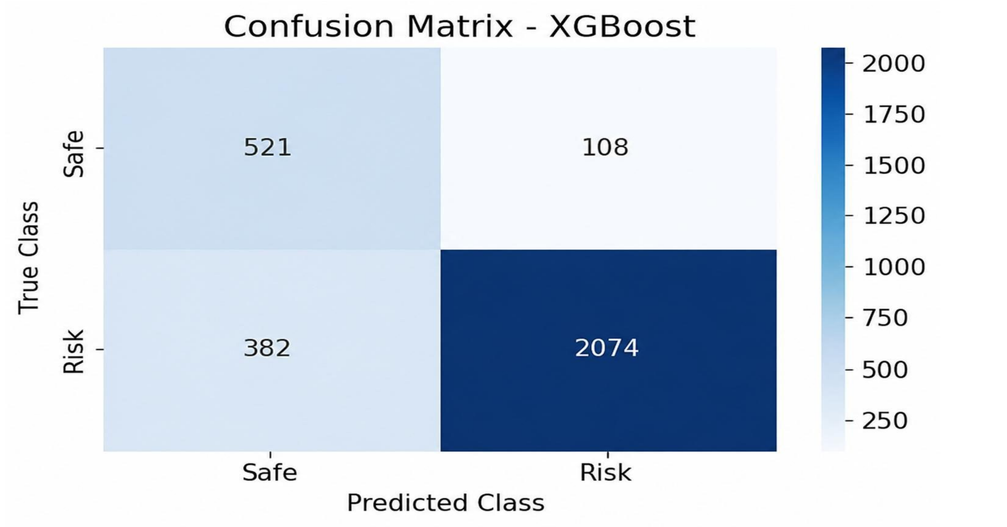


Federated Learning



📊 Model Performance (XGBoost)

🖥️ **Centralized Training ➡️ Accuracy: 84.12%**
🌐 **Federated Training ➡️ Accuracy: 82.90%**
📉 **Comparison ➡️** Reduction of only **1.22%** in the federated scenario.



Preliminary Results and Next Steps

Preliminary results indicate that the use of HAR for detecting risky driving behavior is promising. However, the identification of such patterns still requires improvement. To address these limitations, further experiments will be conducted, including the evaluation of additional parameters, aiming to increase accuracy and contribute more effectively to road safety.

References

- Chellapandi et al. 2023 CHELLAPANDI, V. P. et al. A survey of federated learning for connected and automated vehicles. In: IEEE. 2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC). [S.l.], 2023. p. 2485–2492.
- Gatteschi et al. 2021 GATTESCHI, V. et al. Comparing algorithms for aggressive driving event detection based on vehicle motion data. IEEE Transactions on Vehicular Technology, IEEE, v. 71, n. 1, p. 53–68, 2021.
- Richter et al. 2022 RICHTER, M. A. et al. Smart cities, urban mobility and autonomous vehicles: How different cities needs different sustainable investment strategies. Technological Forecasting and Social Change, v. 184, p. 121857, 11 2022. ISSN 00401625. Yurdem et al. 2024 YURDEM, B. et al.
- Federated learning: Overview, strategies, applications, tools and future directions. Heliyon, Elsevier, v. 10, n. 19, 2024.

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025. Additional funding was provided by the – Coordination for the Improvement of Higher Education Personnel (CAPES) through a Sandwich Doctorate grant (Process No. 88881.219262/2025-01), and by the Research Support Foundation (FUNDEP) for the Mover project (Process No. 29271.01.01/2023.03-00).

Protecting the security of interoperable digital twins in smart environments.

André Belo, Vinícius Cogo, Pedro Ferreira

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physic systems

Motivation: Scale vs. Security

Ecosystem Growth

DTs are expanding across energy management, smart facilities, and industrial systems, evolving from isolated to vast ecosystems

The Interoperability Mandate

Heterogeneous DTs, built on different platforms and data models, must seamlessly exchange information and services to function

Expanded Attack Surface

Increased connectivity exposes the network to unauthorized access, threatening data integrity, privacy, and service availability

The I2DT Vision

	Siloed Platforms	The I2DT Ecosystem
Architecture	Isolated Digital Twins plataforms tied to specific data models	Heterogeneous use cases supported through standardized interaction.
Security Integration	Security treated as an optional, domain-specific add-on	Security embedded as a fundamental, cross-platform enabler for interoperability
Risk Profile in Smart Environments	Vulnerable to operational disruptions and economic loss during cross-platform data exchange	Protected by robust authentication, authorization, and privacy mechanisms enforced consistently

Three Main Objectives

Analyze

Evaluate security mechanisms within existing DT platforms, placing primary focus on the Eclipse Ditto microservice architecture

Design

Map the exact security requirements needed for secure cross-domain interoperability, including authentication, authorization and secure communication

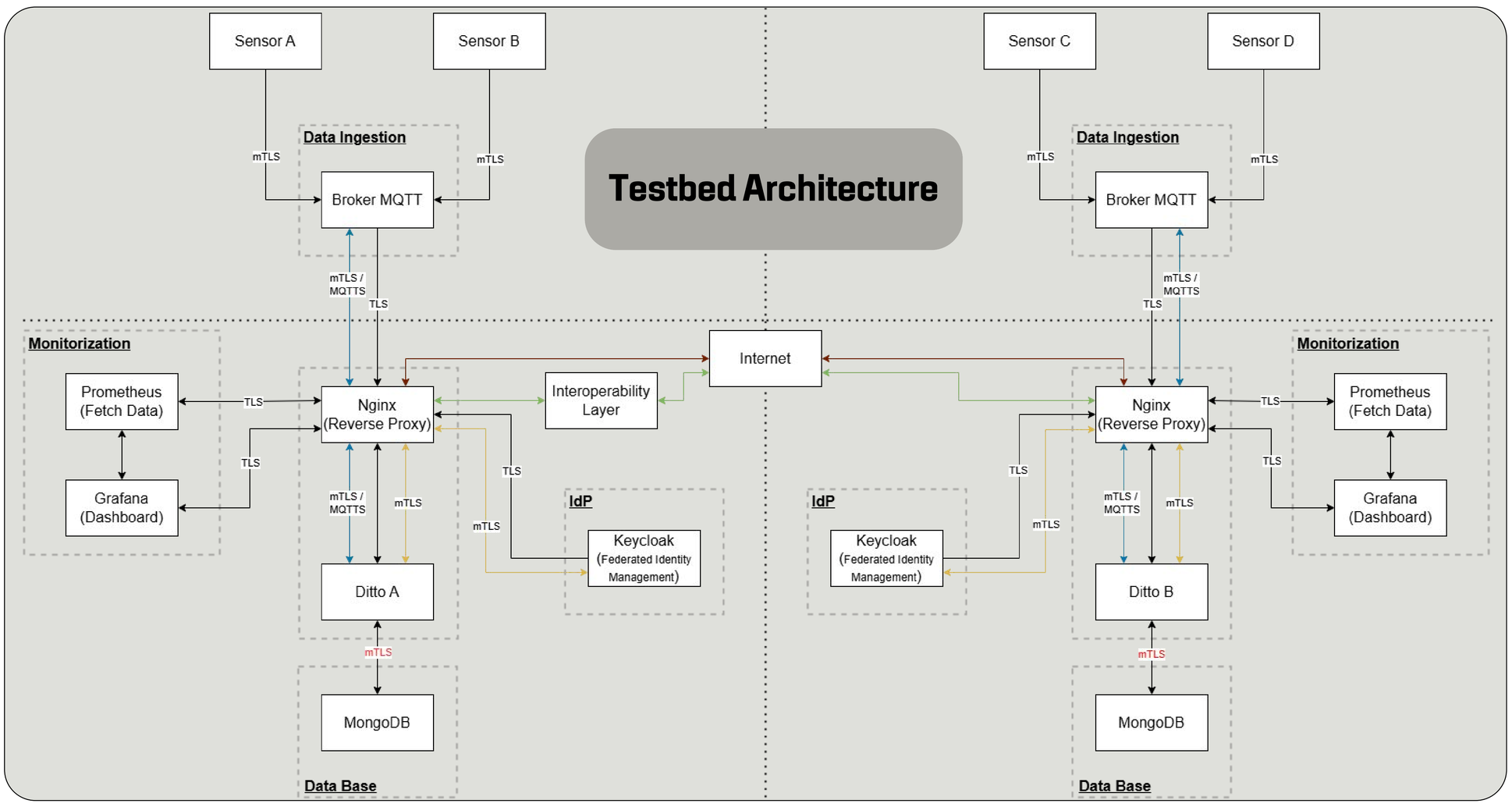
Evaluate

Implement a local DT test environment, integrated with physical sensors, to configure and practically evaluate these security mechanisms

Mechanisms to Implement

	Mechanisms	Base Technology
Authetication	Tokens JWT, OAUTH 2.0 / OIDC	Keycloak + Eclipse Ditto
Authorization	Access Control (ABAC/RBAC) mapped through JWT claims	Ditto's built-in Policies + Keycloak Roles
Communication	Encrypted channel (mTLS, TLS, HTTPS, MQTTS)	Network Infrastructure
Privacy and Data	Field-level access and Queryable Encryption	Ditto Policies + MongoDB

Testbed Architecture



Core Components

Data Ingestion
Broker MQTT for receiving telemetries from the sensors

Identity Management
Keycloak for the standard-based authentication flows

Digital Twin Platform
Eclipse Ditto for the state management, policies application, APIs and forwarding

Persistence Layer
MongoDB instance for maintaining the state, behavior, and meta data of the physical assets

Reverse Proxy
Nginx to enable secure management of incoming API traffic and route requests

Container Modularity
Enables service-level debugging and controlled interaction between components

Acknowledgements: This work was partially supported by P2030 through project I2DT, ref. COMPETE2030-FEDER-00389100, an ITEA4 European project (ref. 22025), and by FCT through the LASIGE Research Unit, ref. UIDB/00408/2025. (DOI: https://doi.org/10.54499/UIDB/00408/2025).

U LISBOA · Ciências ULisboa · fct Fundação para a Ciência e a Tecnologia · LASIGE · ITEA 4 · I2DT

Scheduling Energy for Electrical Vehicles in Condominiums

Daniel Costantino[1], Alan O. de Sá[1], Pedro M. Ferreira[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

cyber-physical systems

1. PROBLEM INTRODUCTION

As **electric vehicle adoption grows**, managing charging patterns becomes essential to ensure grid stability and reliable energy distribution.

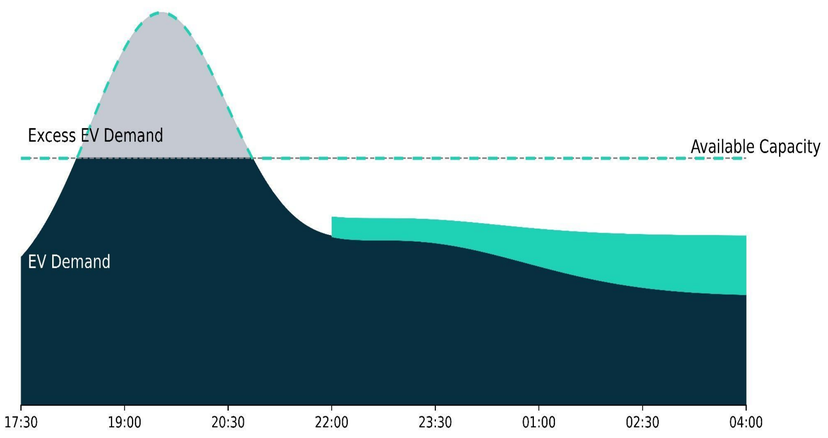
The electrical grid faces increasing capacity constraints as multiple electric vehicles charge simultaneously.

This **concurrent demand** creates peak load conditions, which can strain infrastructure, reduce system efficiency, and increase the risk of localized overloads.

2. PROPOSED SOLUTION

Historical user-level consumption data can be used to estimate a **stochastic demand profile**.

This enables the formulation of a demand-side management problem in which loads are **temporally shifted** to minimize peak load and improving overall load distribution.



3. SOLUTION FORMULATION

An **optimization problem** is formulated to satisfy all user demands, subject to the condominium's contracted power constraint, with the objective of minimizing peak load.

Sets

$$\mathcal{N} = \{1, \ldots, N\} \qquad \text{set of EVs}$$
$$\mathcal{T} = \{1, \ldots, 24\} \qquad \text{set of time slots}$$

Decision Variables

$$x_{i,t} \in \{0, 1\} \qquad 1 \text{ if EV } i \text{ charges at time } t$$
$$P^{\text{peak}} \geq 0 \qquad \text{maximum aggregate power}$$

Optimization Problem

$$\min_{x_{i,t},\, P^{\text{peak}}} \quad P^{\text{peak}}$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{N}} P_i^{\max} x_{i,t} \leq P^{\text{peak}}, \qquad \forall t \in \mathcal{T}$$

$$\sum_{i \in \mathcal{N}} P_i^{\max} x_{i,t} \leq P^{\text{contr}}, \qquad \forall t \in \mathcal{T}$$

$$\sum_{t \in \mathcal{T}_i} P_i^{\max} x_{i,t}\, \Delta t = E_i^{\text{req}}, \qquad \forall i \in \mathcal{N}$$

$$x_{i,t} \in \{0, 1\}, \qquad \forall i \in \mathcal{N},\ t \in \mathcal{T}$$

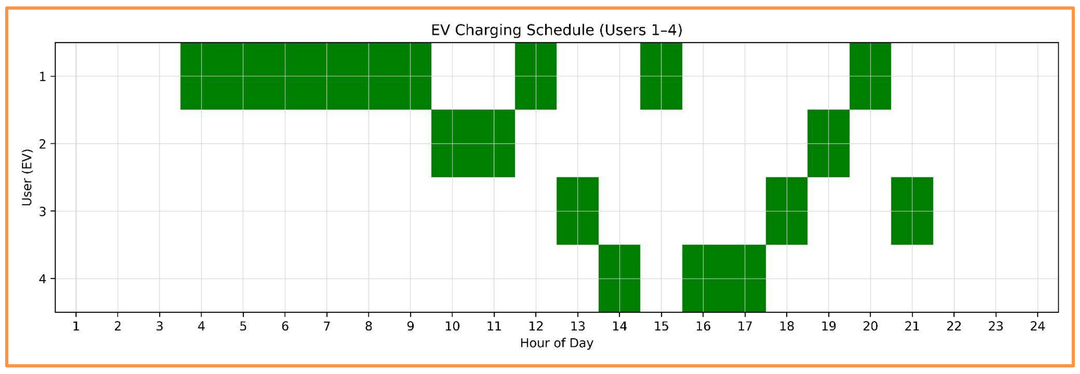
Parameters

$P_i^{\max}$: maximum charging power of EV i

P^{contr} : contracted power limit

E_i^{req} : required energy demand

$\Delta t = 1$ hour



4. FUTURE WORK

- The current scheduling solution assumes a **static allocation** over the daily horizon, neglecting user-specific availability constraints.
- A more robust approach would introduce a **dynamic model** that incorporates stochastic user availability, defined as a probability distribution over time. The objective function can then be reformulated to maximize expected availability-weighted utility across time slots, conditioned on user arrival and departure processes.

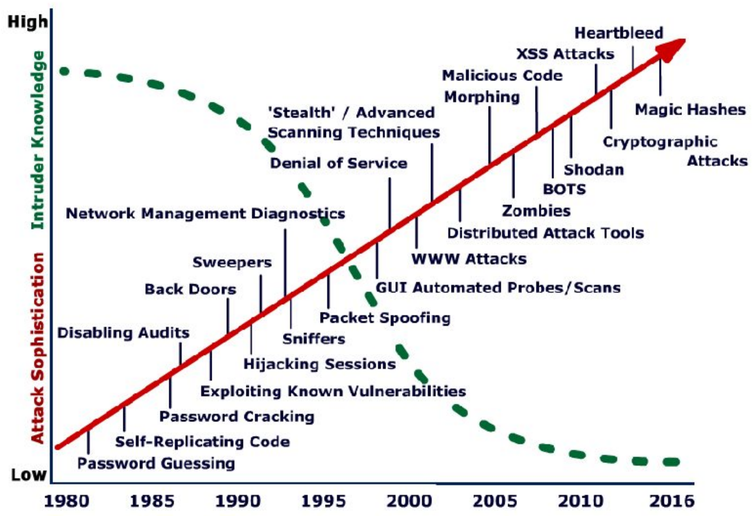
Acknowledgements: This work was supported by FCT through project 56 - ATE: Aliança para a Transição Energética (Ref. C644914747-00000023), and the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025

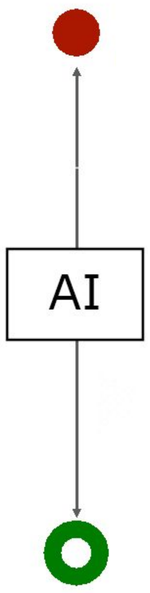
Mitigating Intelligent Cyberattacks Through Adversarial Machine Learning Techniques

Gonçalo Fernandes, Alan Oliveira, José Cecílio

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physical systems





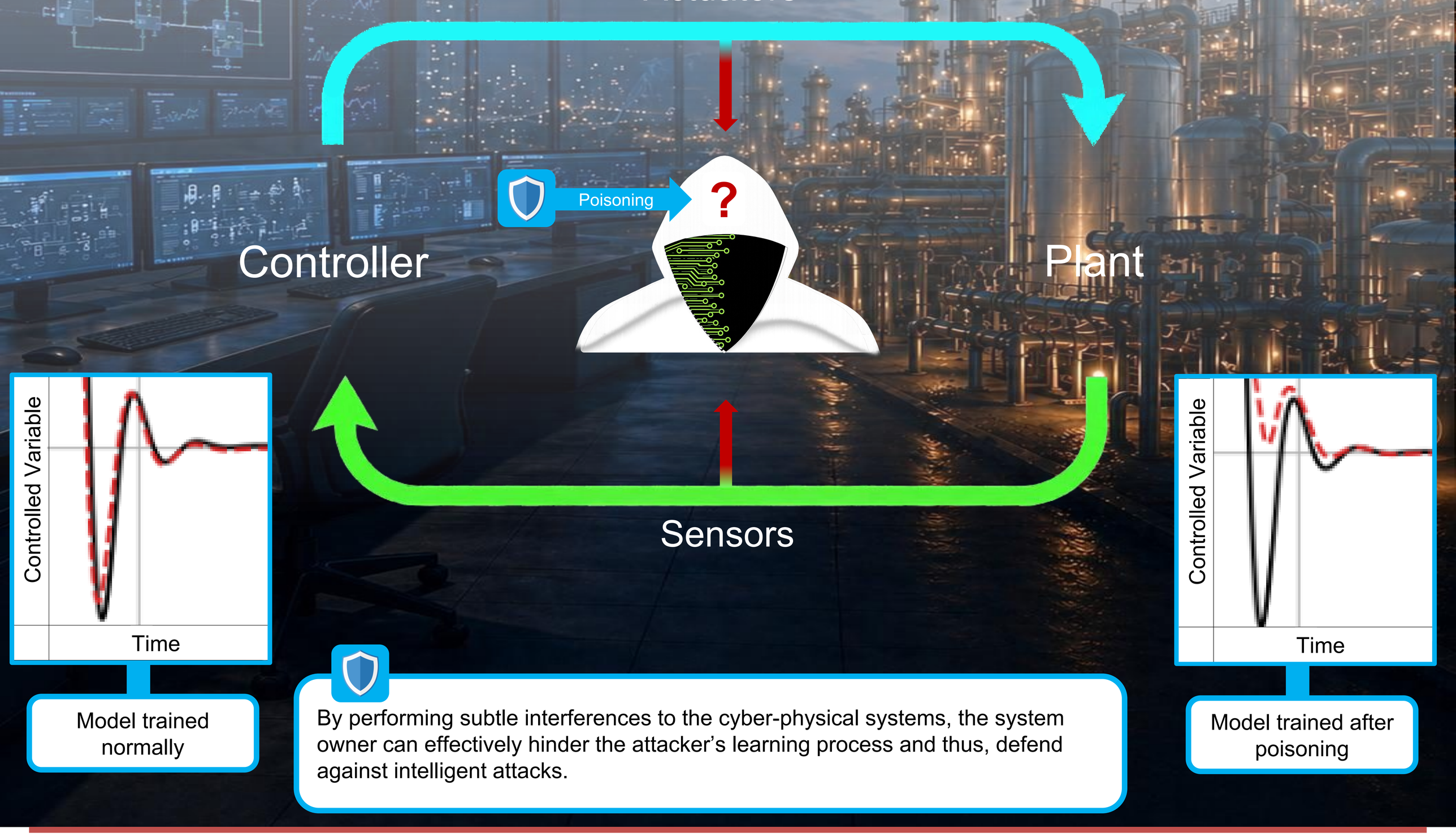
- Intelligent attacks on cyber-physical systems using Machine Learning models are more and more prevalent.
- Can adversarial ML (AML) techniques mitigate intelligent attacks by impairing the attack agent learning process?

O1 - Produce a simulation environment of a CPS to collect datasets, simulate attacks, and the correspondingcountermeasures developed.

O2 - Create datasets to train the ML-based intelligent at-tacks, using the simulation environment produced in O1.

O3 - Execute ML-based System Identification attacks, train-ing their models using the dataset obtained in objective O2.

O4 - Study the use of poisoning techniques as counter-measures against ML-based system identification at-tacks, assessing their performance in terms of accuracydegradation caused to the attacker models.



Model trained normally

By performing subtle interferences to the cyber-physical systems, the system owner can effectively hinder the attacker's learning process and thus, defend against intelligent attacks.

Model trained after poisoning

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025 which includes the LASIGE Seed project ref. LASIGE-SP-2025.01).

“SIMBAY-GRASP”, Simulation-Based Bayesian Inference for Object Perception in Robot Grasping

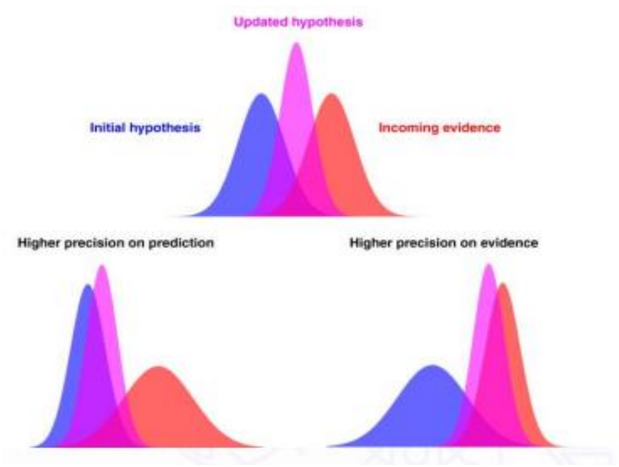
Guilherme Duarte[1], João Bimbo[2]

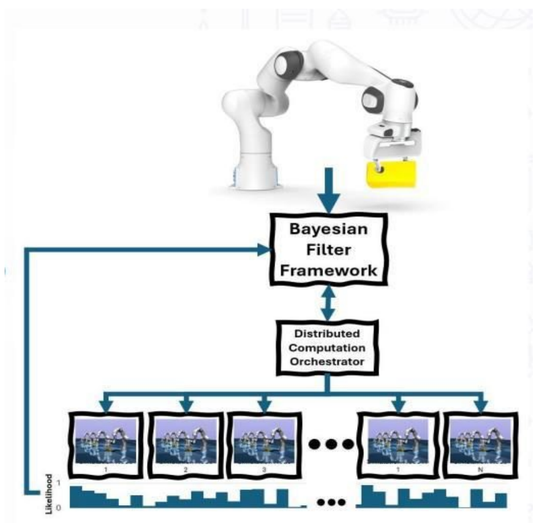
[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physical systems

Introduction

- **The Challenge:** Operating in unstructured environments requires robots to manage severe physical and sensory uncertainty.
- **Our Solution:** A cognitive architecture that estimates unknown object properties by bridging Bayesian Inference with massive parallel simulations.





Method (Bayesian Filter)

Initialize

Create simulated worlds (particles),each representing a possible state of the world

Predict

Propagate the real robot's control inputs across all simulation instances

Update

Compare simulated sensor data against real-world observations to assign particle weights (likelihoods).

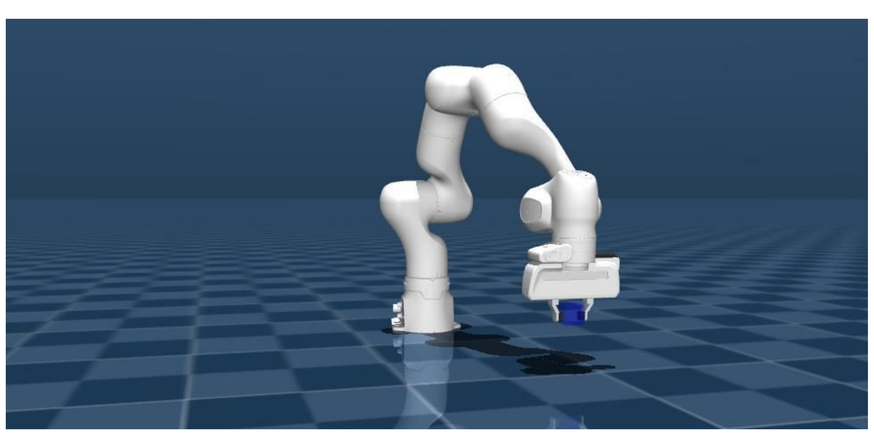
Resample

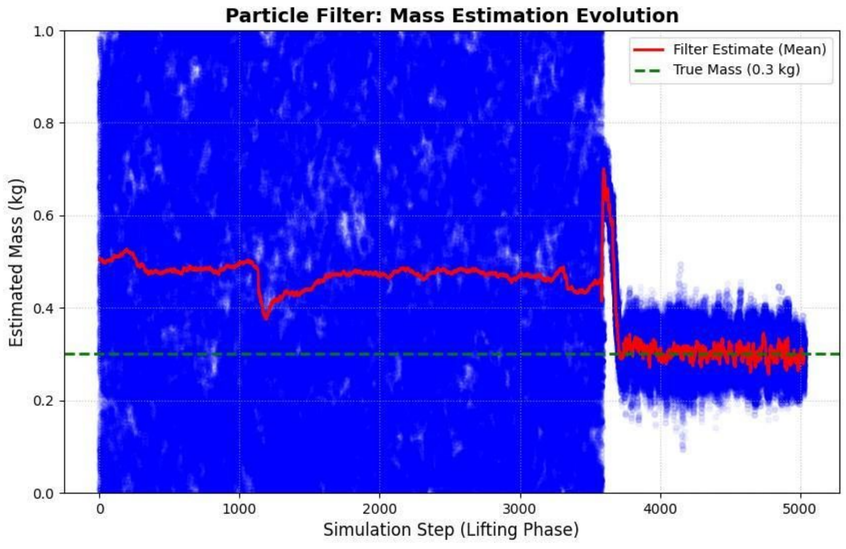
Periodically refresh the particle set to focus on high-probability regions

A versatile architecture: infer any physical property simply by swapping the likelihood function

Mass Estimation

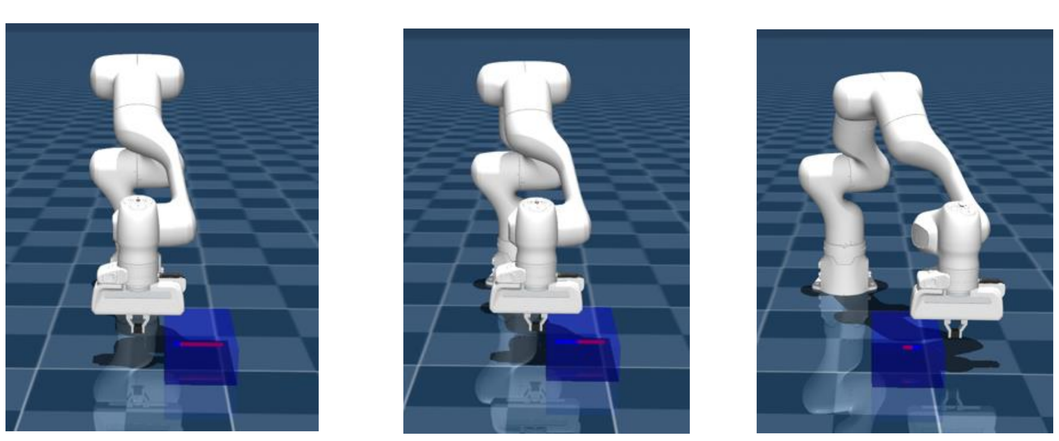
To infer mass, the measurement model evaluates the continuous external joint torques. Particles are assigned higher likelihoods when their simulated torque readings closely match the physical robot's sensors. Upon lifting the object, the filter rapidly converges to a highly accurate mass estimate, demonstrating strong robustness against underlying sensor noise.

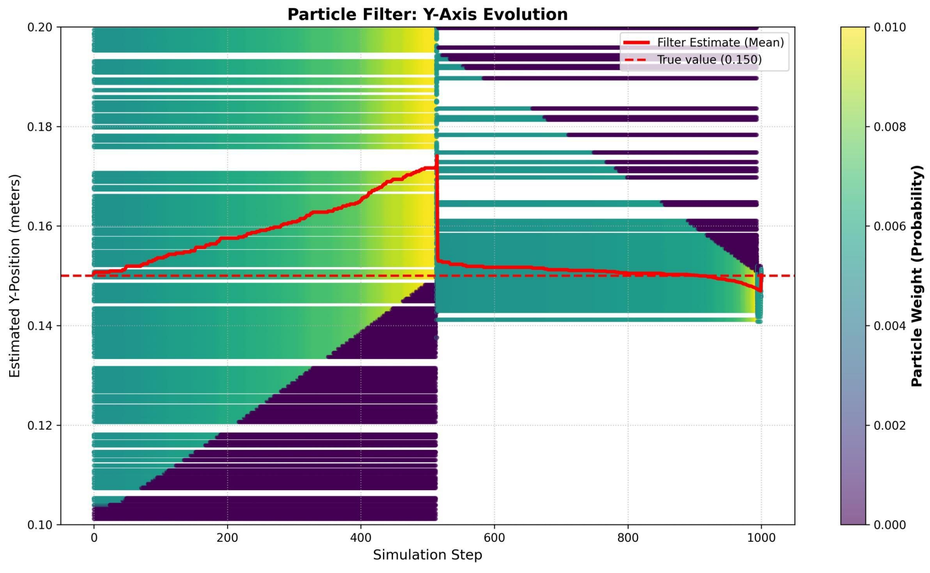




Position Estimation

This framework utilizes a binary contact measurement model (contact vs. no-contact). As the robot performs exploratory sweeps through free space, the simulations mimic its exact trajectory. Particles that predict a collision when the real robot feels nothing, or vice versa, are heavily penalized. This process iteratively eliminates impossible object positions, allowing the swarm to converge on the true spatial coordinates.





Acknowledgements: This work was supported by FCT through project “SIMBAY-GRASP”, “Simulation-Based Bayesian Inference for Object Perception in Robot Grasping”, Candidatura Ref.: 1573.1 – 2 BI EM SIMBAY, and the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Platform for Analysing Home Appliance Efficiency Based on Electric Consumption

Guilherme Sousa, Ana Paula Afonso, Pedro M. Ferreira

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

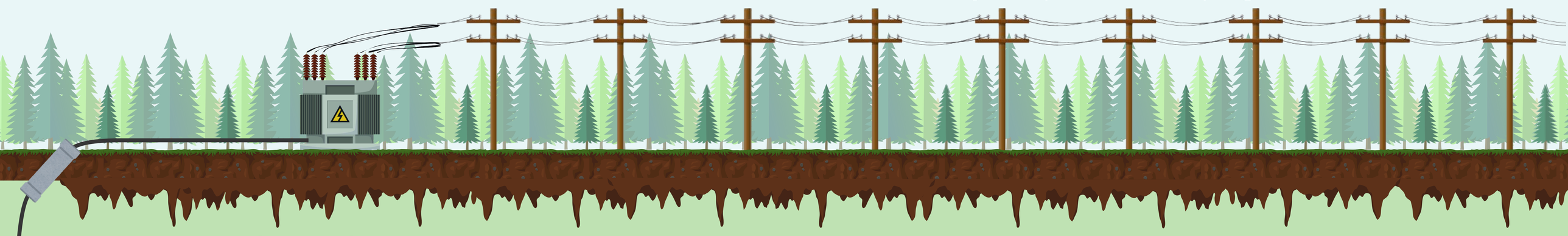
LASIGE cyber-physical systems

Motivation & Problem

- Energy consumption in households is high
- Existing apps: Locked to brands
- Limited AI transparency
- Only premium appliances supported

Previous Work

- Machine learning estimates laundry load using only energy data
- No additional sensors required
- Enables efficiency calculation (energy per load)
- Low cost and easy setup with smart plugs



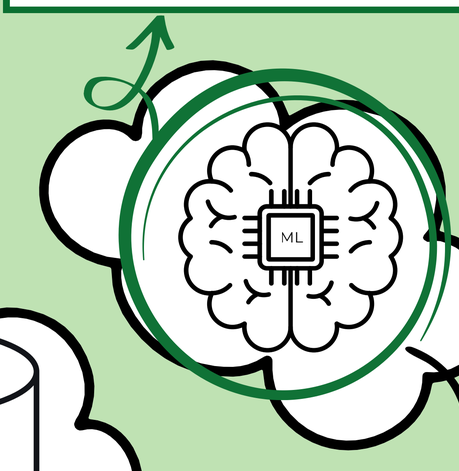
Objectives

Create a digital platform that provides energy efficiency recommendations to the public.
Platform components:

- Mobile app: View consumption information* and receive efficiency recommendations
- Website & Cloud-backend: Manage users, data storage, and run ML models
- Smart plug integration: Collect real-time appliance data
- Online service: Accessible to users and supports data collection for improvement

*: Energy use (E), cycle duration, and efficiency (E/M - energy per unit of laundry).

Machine learning model** that estimates the laundry load mass (M) inside the washing machine, using only energy consumption data collected from the smart plug.

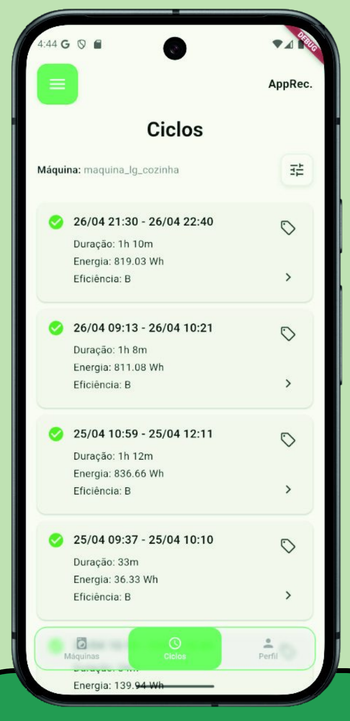


Cleans data by detecting cycles and deleting data outside of the cycles, also calculates the duration of the cycle and total energy spent

App Features

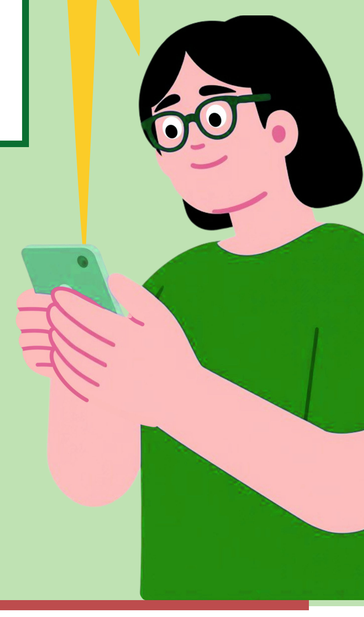
- View efficiency, energy use, and duration for each wash cycle
- Get recommended settings to improve efficiency over time

Add, and view your favorite machines!



Track usage cycles for your machines!

Details for every cycle!



**Jasiūnas, Ž., Julião, T., Cecílio, J., da Graca, G. C., & Ferreira, P. M. (2025). A soft sensor to assess the energy performance of laundry washing machines. Applied Energy, 383, 125349.

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

Integration of simultaneous localization and mapping (SLAM) system with adaptability and terrain perception in a quadruped robot (Pupper)

Author: João Borrega[1], Coordinator: João Bimbo[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

cyber-physical systems

Objective

Goal: Build a navigation architecture for the Stanford Pupper.

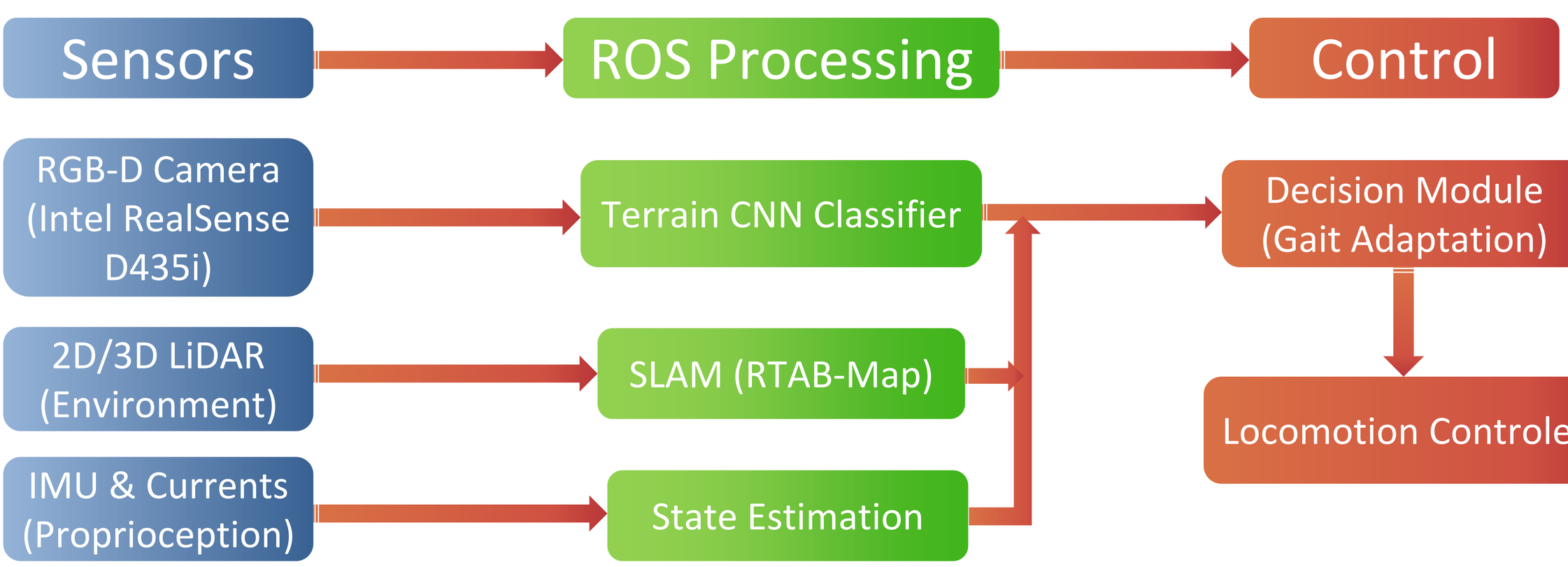
Integration: SLAM + Terrain Perception.

Dynamic Adaptability: Real-time adjustments of gait/speed based on terrain information.

- **Intel RealSense D435i (RGB-D):** High-resolution depth sensing and terrain classification (CNN).
- **LiDAR Scanner:** Crucial for robust 360 degrees spatial awareness and accurate SLAM.

Proposed Hardware Configuration

System Architecture



Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

Construction Robots: Cooperative Transport of Payloads Using Drones

Manuel Lourenço, Luís Pinto, João P. Matos-Carvalho

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physical systems

MOTIVATION

- **Cooperation**: Distributing weight across multiple agents enables heavy lifting, not possible with only one drone
- **Adaptability**: Flexible cable connections allow transport of irregular shapes
- **Precision**: Active control of both payload position and orientation (6-DOF)

CHALLENGES

- Coupled multi-body system
- Outdoor environment
  - Wind
  - Sun
- Suspended payload pendulum effect
- Lightweight system

CONTROL STRATEGIES

- **Formation Control**: PID-based regulation of the distance and angle between UAVs
- **Orientation Alignment**: Synchronizing headings to prevent cable torsion
- **Load Sway Damping**: Actively counteracting the "Pendulum Effect" caused by acceleration

PROPOSED SOLUTION

Using cameras and AprilTags:

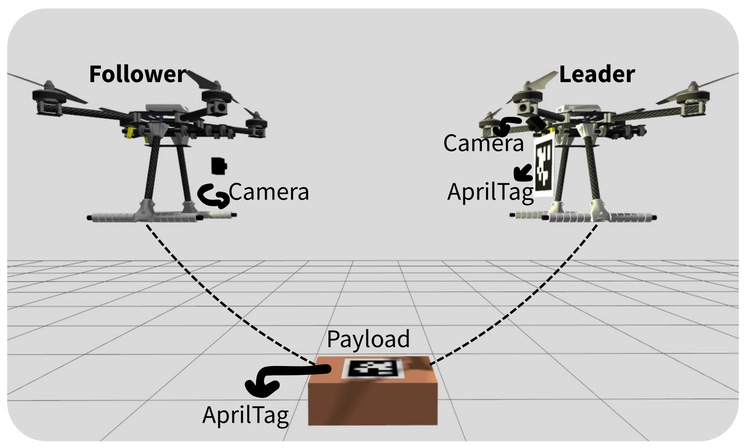
- Ligthweigth
- Cheap solution
- Works outdoors
- Adaptable to different systems

Leader-Follower approach:

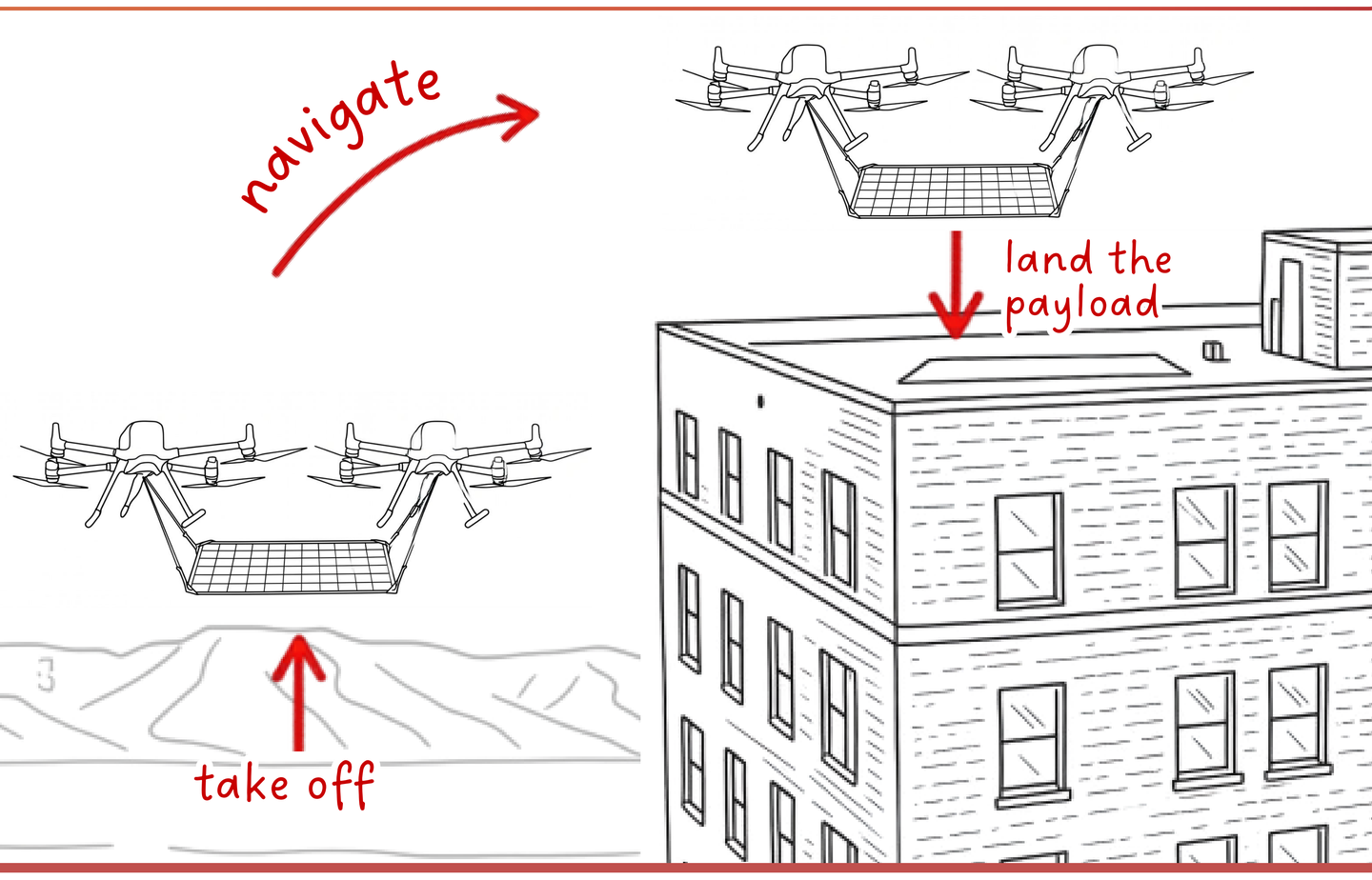
- Follower with a camera
- Leader with GNSS and an AprilTag
- Leader follows a path while Follower mantains formation

Anti-swing approach:

- Leader with a camera
- Payload with an AprilTag



TESTS IN SIMULATED & REAL ENVIRONMENTS



Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025.

Securing the Continuum Remote Attestation Middleware

Maria Gomes da Silva[1], António Casimiro[1], André Souto[1], Pedro M. Ferreira[1], Alan Oliveira[1], José Cecílio[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physical systems

WITHOUT SCRAM

Static Security Patch Attacks Succeed

DISCOVERY
Scans devices in the environment

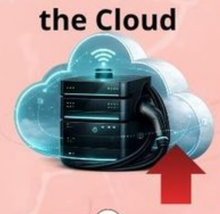
INITIAL ACCESS
Gains unauthorized access to a device

DATA MANIPULATION
Manipulates device measurements and data

CORRUPTED INPUT
Processes and trusts manipulated data

WRONG ACTIONS
Executes incorrect actions

ESCALATE TO CLOUD
Propagates corrupted state to the Cloud



CONSEQUENCES

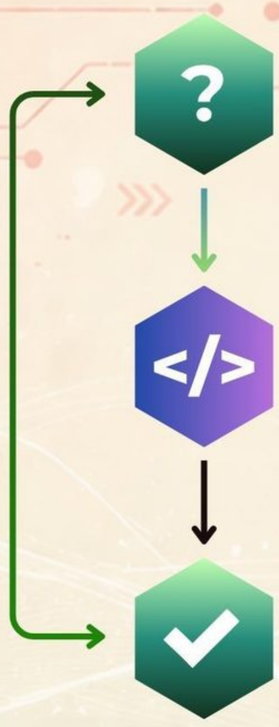
- Compromised devices
- Manipulated data and incorrect decisions
- Cascading failures and network disruption
- Finantial loss

CHAOS UNTRUSTED UNSAFE

SCRAM

SCRAM ACHIEVEMENTS

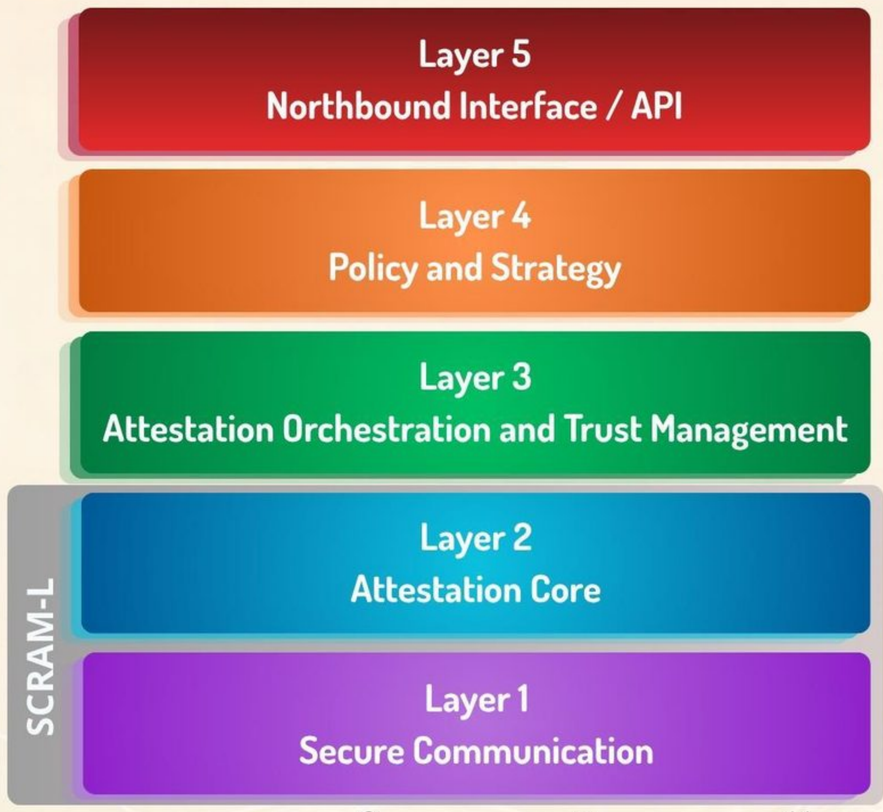
HOW IT WORKS



1. CHALLENGE
Request integrity proof

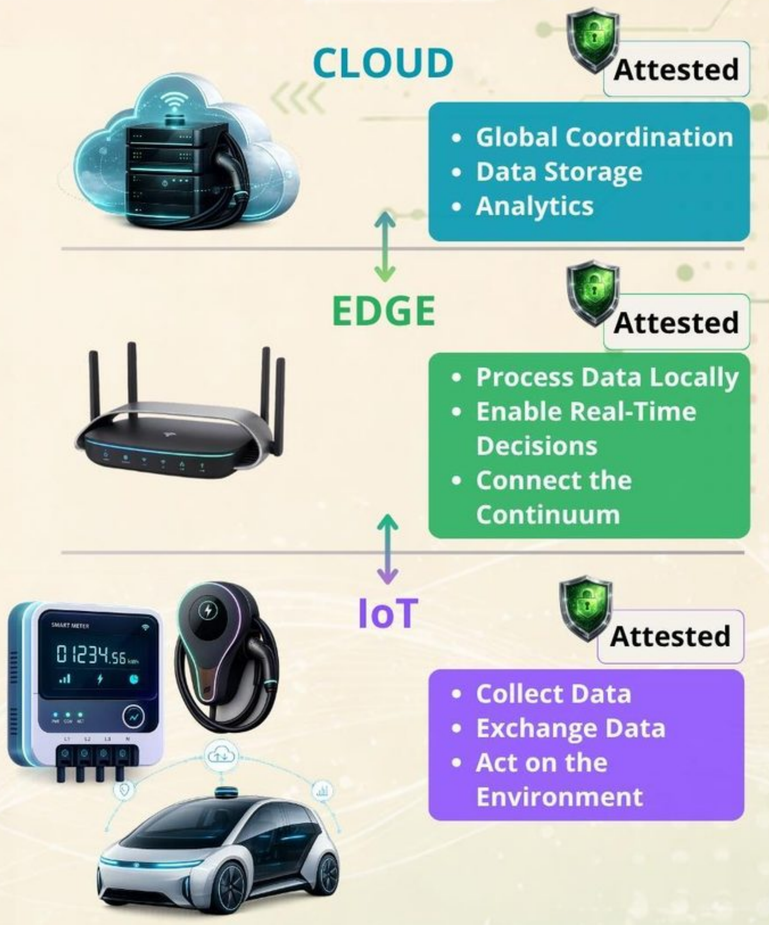
2. RESPONSE
Provide integrity evidence

3. DECENTRALIZED TRUST DECISION
Accept or reject



- Policy-driven attestation
- Context aware adaption
- Heterogeneous support
- Bimodal scalable deployment

THE CONTINUUM



ATTACK ANYWHERE - TRUST EVERYWHERE

WITH SCRAM

Continuous Integrity Verification

Attacks Detected and Blocked

DISCOVERY
Devices and context identified

ATTESTATION CHALLENGE
Device integrity challenged

INTEGRITY VERIFIED
Integrity verified, access granted

EDGE GUARDIAN ACTIVE
Untrusted entities isolated

COMMAND PROTECTION
Malicious commands blocked

CLOUD TRUST PRESERVED
Only trusted data propagated

OUTCOMES

- Integrity verified and enforced
- Malicious behavior detected and blocked
- Decisions based on trusted inputs
- Secure and resilient network operation

CONTROL TRUSTED SECURE

Acknowledgements: This work was funded by IAPMEI - Agência para a Competitividade e Inovação, I. P., and European Union - Next Generation EU within the scope of the project ATE: Aliança para a Transição Energética (Ref. C644914747-00000023), and by FCT through the LASIGE Research Unit, {https://doi.org/10.54499/UID/00408/2025}

Impedance control for Safe human robot interaction

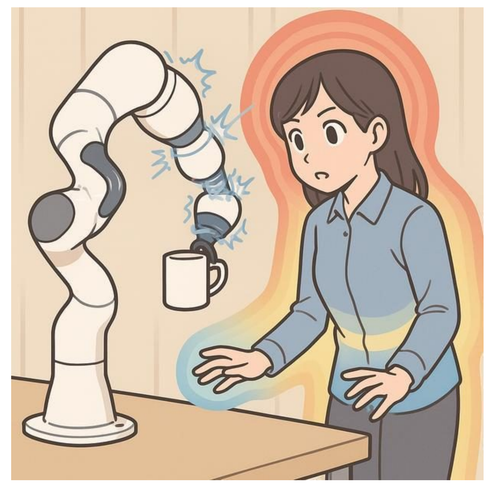
Marta Sousa, João Bimbo

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physical systems

Introduction

Robotic devices can assist individuals who have partially or completely lost motor functions, enabling them to perform tasks that would otherwise be difficult or impossible. The objective of this work is to develop and implement an impedance control algorithm for the FR3 robotic arm. By reducing joint stiffness and appropriately tuning damping parameters, the robot is expected to exhibit spring–damper-like behavior. In the event of unintended contact with a human, particularly with sensitive areas such as the head, the robot should safely yield, limit interaction forces, and gradually return to its nominal configuration.



Impedance Control

The Cartesian impedance controller regulates the dynamic interaction between the robot end-effector and the environment by enforcing a desired mass–spring–damper behavior in task space. The control objective is to track a reference pose while allowing compliant deviations when external forces are applied.
This is achieved by defining a virtual mechanical system where position errors generate restoring forces proportional to stiffness, while velocity errors are damped to ensure stability. The resulting interaction enables safe and adaptive behavior during human–robot contact.

Cartesian Impedance Control

$$F = M\left(\ddot{x}_d - \ddot{x}\right) + D\left(\dot{x}_d - \dot{x}\right) + K\left(x_d - x\right)$$

K → Stiffness
D → Damping
M → Virtual Mass

Torque

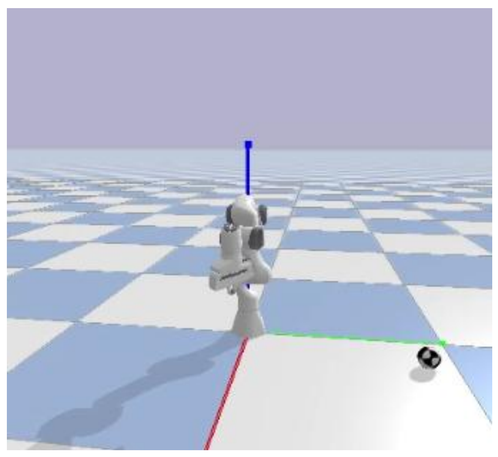
$$\tau = J^T F$$

J → Jacobian Matrix
F → Cartesian Force

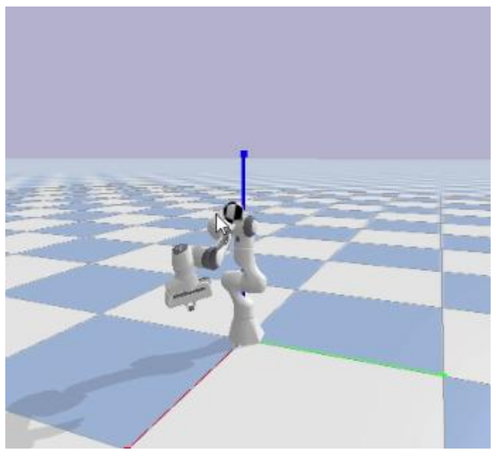
Velocity Mapping

$$\dot{x} = J\left(q\right)\dot{q}$$

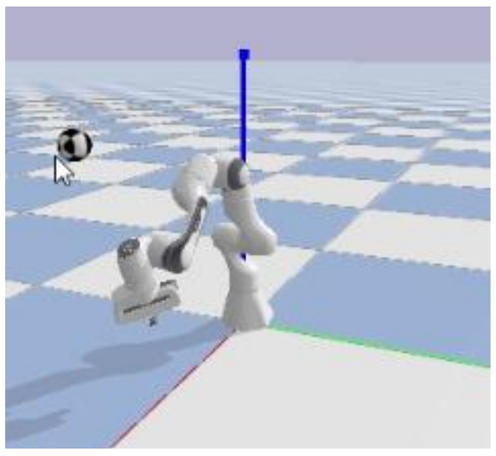
Simulation



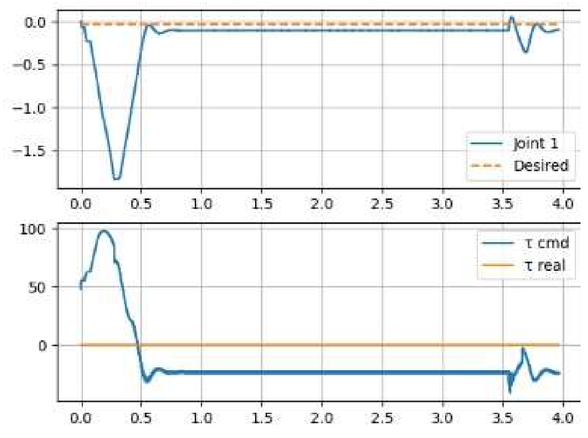
Robotic arm in the simulator alongside a sphere that represents a human body part



The sphere collides with the robotic arm, simulating a collision during a collaborative task



With the formulas above the robotic arm will reduce the stiffness during the collision, and once the sphere is away from task space return to original position



Graphics showing the alteration in the torque when the impact occurs.

Future Works

Future work will focus on extending the proposed impedance control framework to incorporate real-time dynamic information regarding human position and motion. By integrating data acquired from external vision systems and wearable sensors, the robotic manipulator will be able to autonomously adjust joint-level impedance parameters according to human proximity and the nature of the interacting object or body part.

This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025

U LISBOA Ciências ULisboa fct Fundação para a Ciência e a Tecnologia LASIGE

Dynamic Household Energy Use Profiling and Short-Term Forecasting

Nuno Costa[1], Pedro M. Ferreira[1], Alan Oliveira[1], José Cecílio[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE cyber-physical systems

Motivation

How to reduce energy-related expenditures?
This work focuses on improving energy demand forecasting through the development of more accurate user profiles.

Features

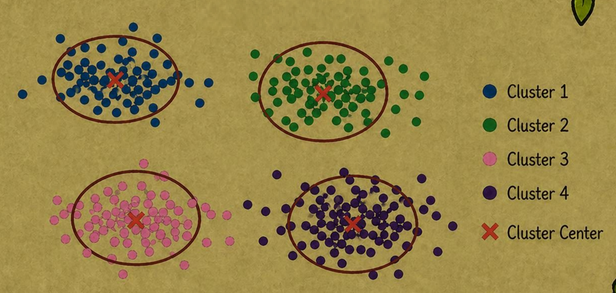
Energy Profiles

A profile represents how a user (individual profile) or a group (typical profile) consumes energy over a given period. The focus is daily profiles for residential households.

Metrics

- Silhouette Index
- Davie Bow Index
- Calinski-Harabasz index
- Inertia
- Raw, Norm AUC
- MAE RMSE Compactness
- Nearest Centroid Distance

Clustering



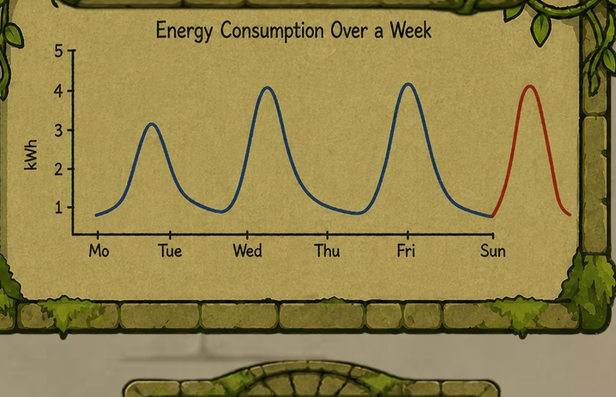
Models

- K means
- K means DTW
- K medoids
- K medoids mahalanobis
- EWKM
- DBSCAN
- AHC and GMM

Metrics

For each model and feature-hyperparameter combination, forecasting errors were stored in a day-by hour grid, from which MAE, RMSE, AUC and R^2 were computed.

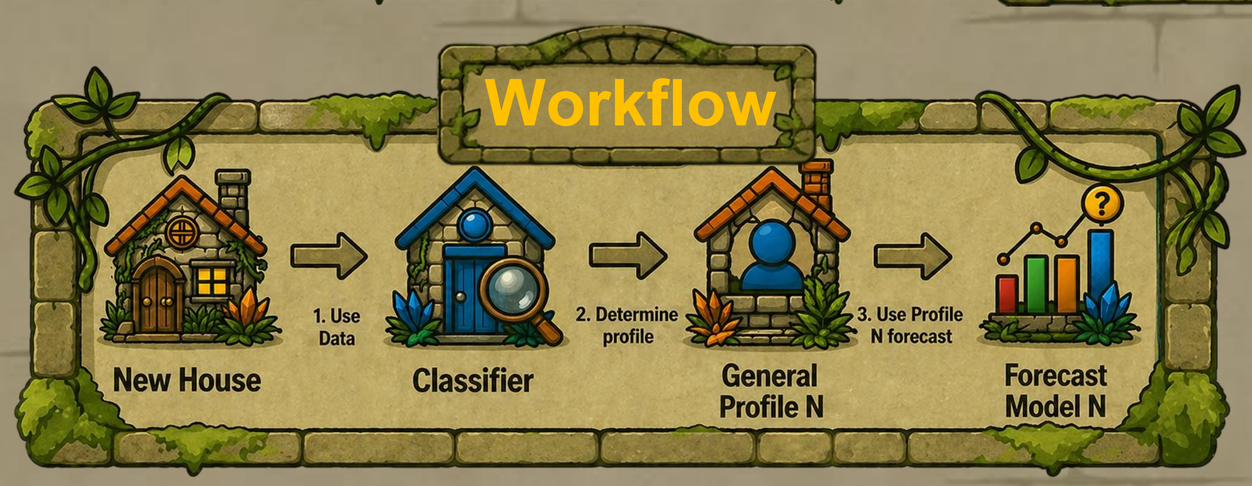
Forecasting



Models

The models selected for this stage were a subset identified in the literature that showed promising results. Specifically, we evaluated LSTMs (2-3 layers), xGBoost and RF.

Workflow



Cluster Centre's (Profiles) and Forecasting Models

More Accurate Profile Specific Daily Energy Forecasting

Acknowledgements: This work was funded by IAPMEI - Agência para a Competitividade e Inovação, I. P., and European Union - Next Generation EU within the scope of the project ATE: Aliança para a Transição Energética (Ref. C644914747-00000023), and by FCT through the LASIGE Research Unit, {https://doi.org/10.54499/UID/00408/2025}

Design and Development of an Underactuated Robotic Hand for Object Grasping

Simão Silva, João Bimbo

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

cyber-physical systems

Introduction

Goal: Developing a cost-effective underactuated robotic hand for object grasping.

Most robotic hands in the industry are very expensive, heavy and complex to repair, unlike the low-cost underactuated hand, which is built through accessible 3D printing technologies.

Roadmap

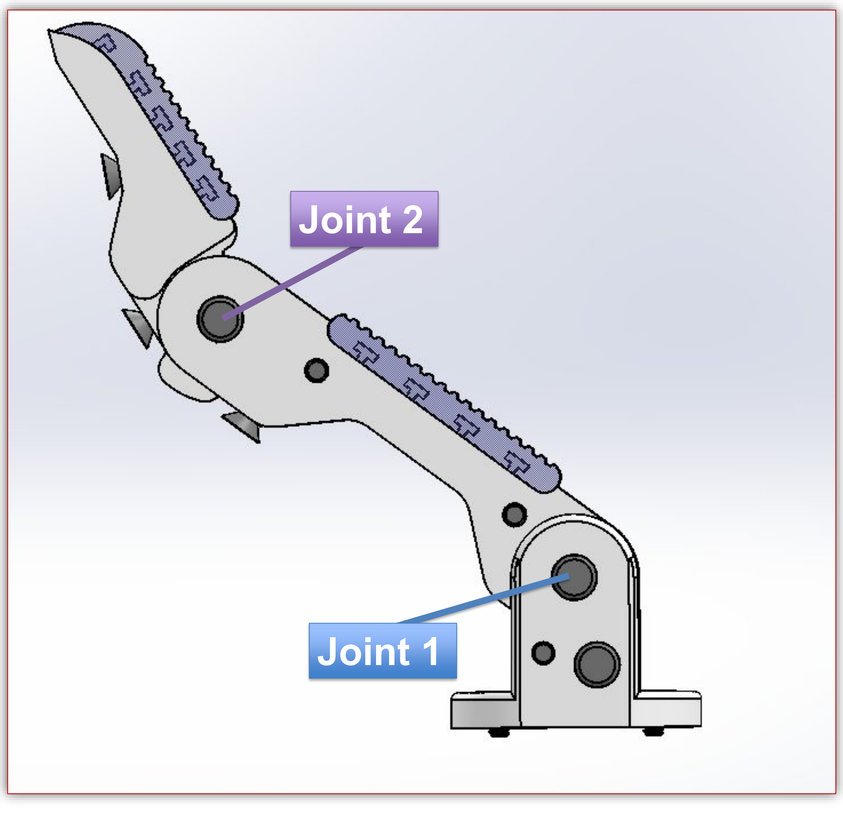
Baseline Replication
- Assemble and validate CAD models in SolidWorks.
- Fabricate structural components using 3D printing.
- Source Dynamixel motors and Open RB-150 controller.

System Integration
- Assemble physical underactuated tendon/pulley system.
- Establish PC-to-hardware communication via Dynamixel Wizard.
- Calibrate motor limits and starting positions.

Kinematic Validation
- Execute baseline grasping tests on standard geometries (spheres, cylinders).
- Evaluate payload capacity and conformal grip strength.

Proposed Modifications
- Execute baseline grasping tests on standard geometries (spheres, cylinders).
- Evaluate payload capacity and conformal grip strength.

Finger Kinematics

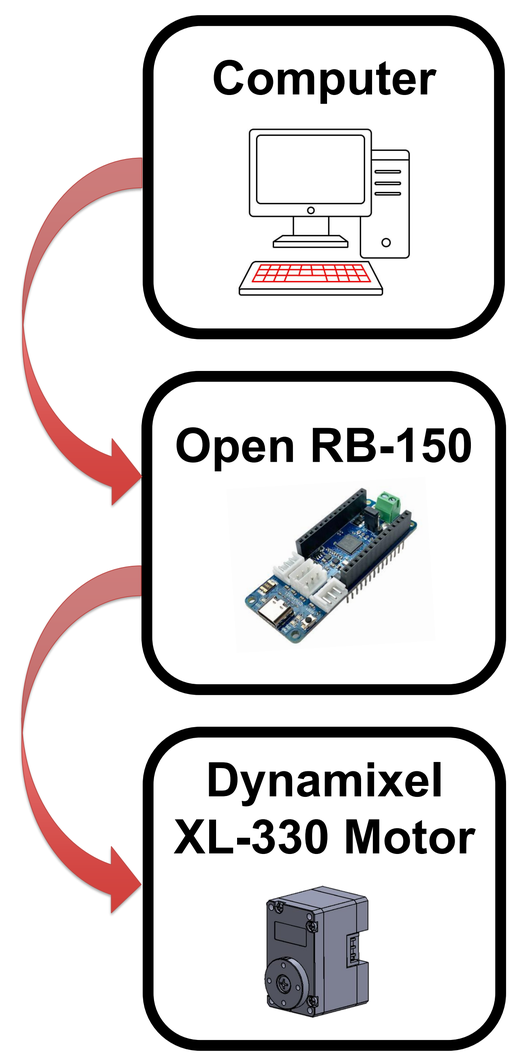


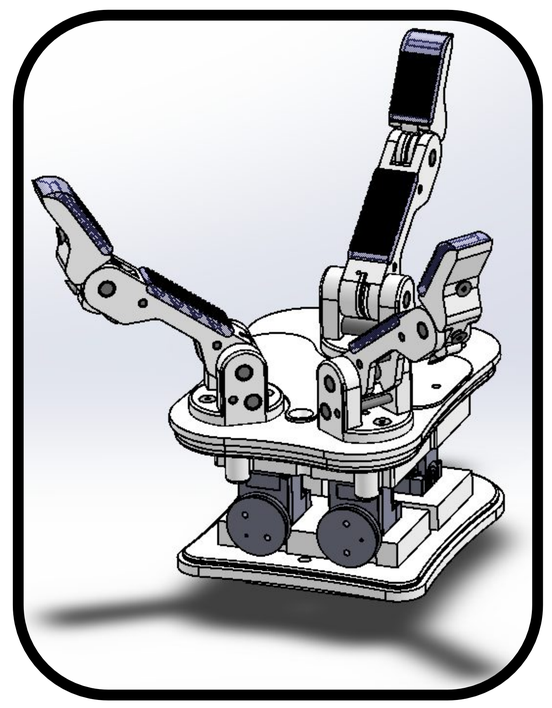
Methodology

This work is based on Yale's OpenHand Project Model O.

Structural Composition:
- 4 Dynamixel XL-330 Motors
- **Fingers:** 2 index fingers (vertical bending and horizontal rotation) and 1 thumb (vertical bending)

Actuation:
Underactuated design using traction system composed by 3 pulleys, wire and several gears.





Proposed Modifications:

- **Material Choice:** The gears need to be specially resistant to mechanical stress; the finger tips need to stick better to the objects, possibly meaning replacement for rubber or resin.
- **Sensor Integration:** Tactile sensors on the fingertips.
- **Kinematic Configuration:** Alter the DoF for a specific finger to make it more mobile. Several motors can be linked to the same finger.

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025.
DOI https://doi.org/10.54499/UID/00408/2025

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

IOT-NODE*n*

Network Optimization and Design Engine —
a benchmarking and topology framework for IoT in challenging environments

Tommaso Tragno[1], Leonardo Monteiro[2], José Cecílio[2], Alan Oliveira[2], Pedro Ferreira[2]

[1] Università degli Studi dell'Aquila, Italy · [2] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

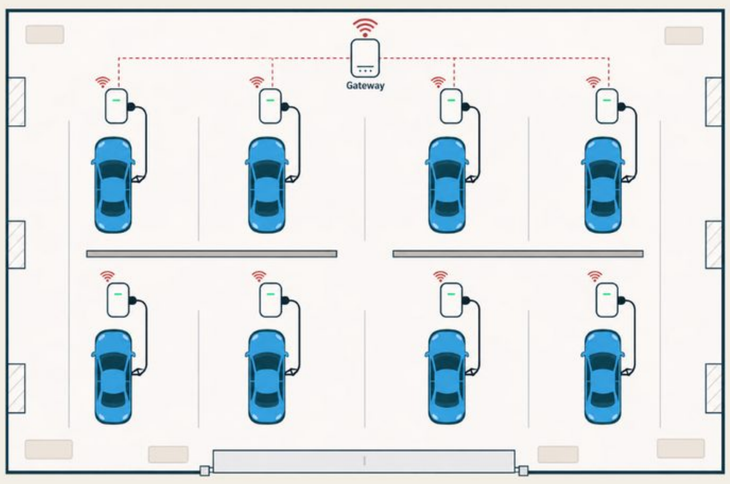
LASIGE cyber-physical systems

CONTEXT & GOALS

Motivation

IoT deployments in **challenging environments** (garages, tunnels, noisy areas) face variable signal attenuation and multipath interference.

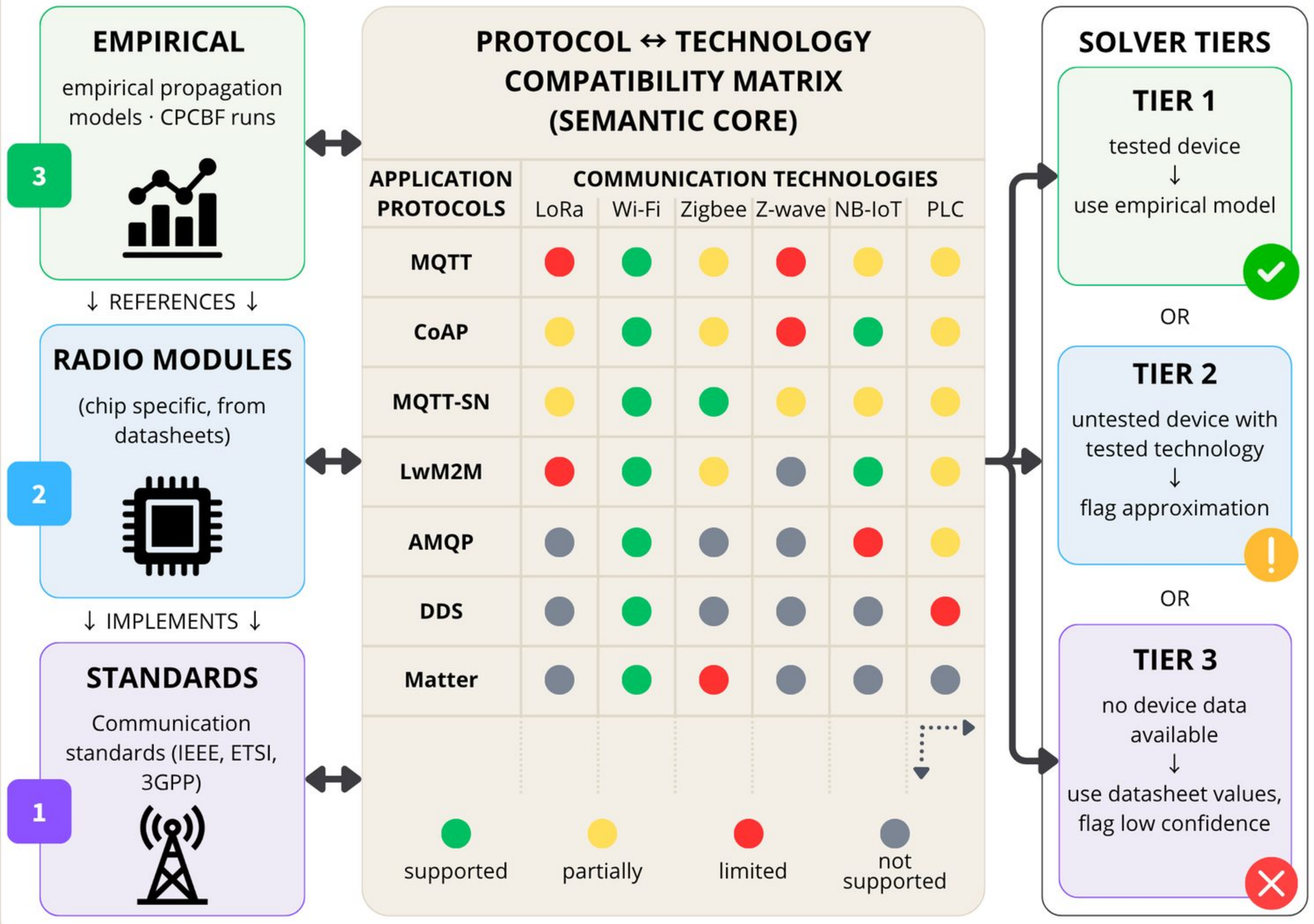
No framework today unifies **formal semantic modelling**, **automated topology, technology and protocol selection**, combined with **empirical validation** in one tool.



GOAL: provide reliable connectivity to IoT devices in challenging environments (e.g.: E.V. chargers in underground garages)

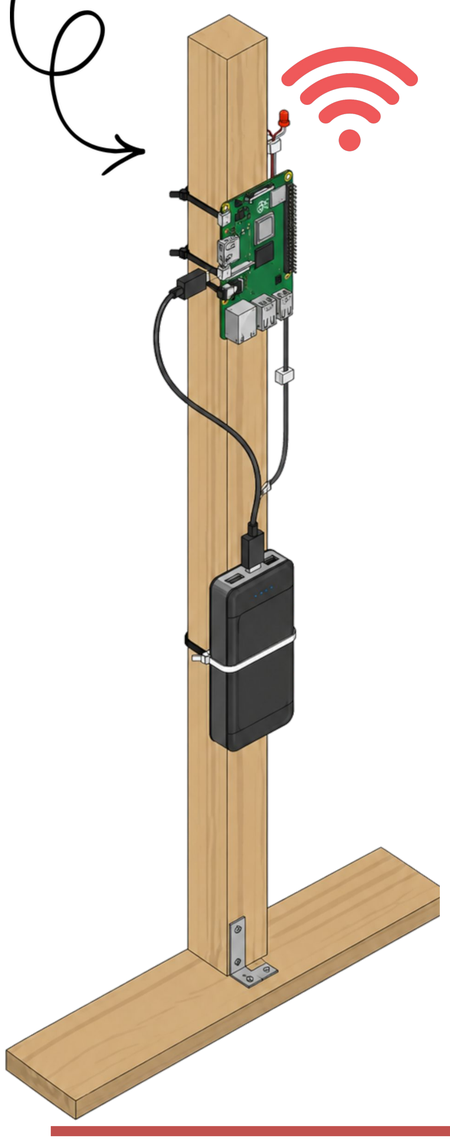
SYSTEM ARCHITECTURE

Topology Design Assistant

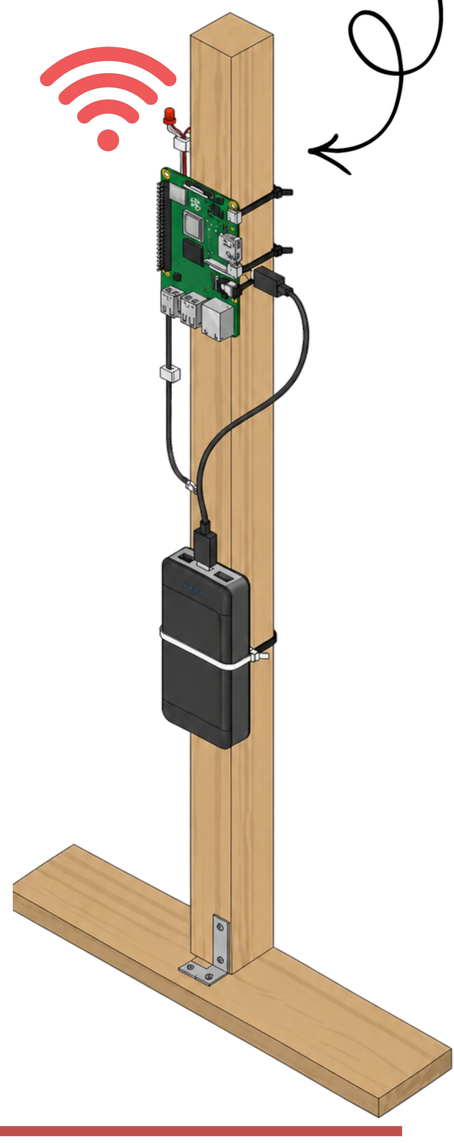


EVALUATION & RESULTS

Our Experimental Setup: a Wall Box Emulator



Raspberry Pi 4B + battery on a wood pole



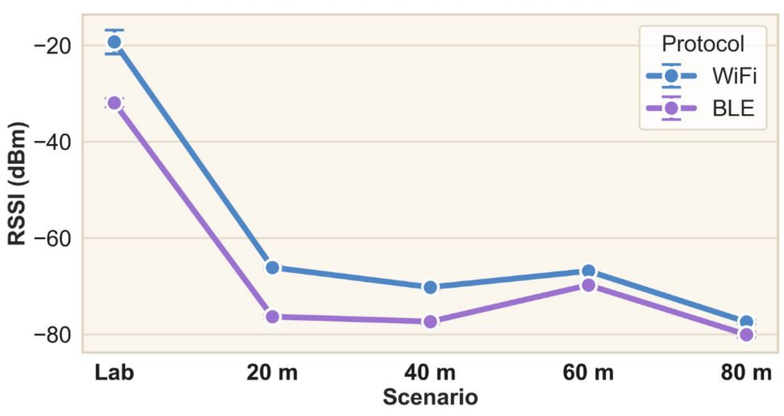
Evaluation Environment

Protocol-agnostic HAL C++ framework to collect:

RSSI dBm · **RTT** µs · **Throughput** bps

Collected at 20 / 40 / 60 / 80 m intervals

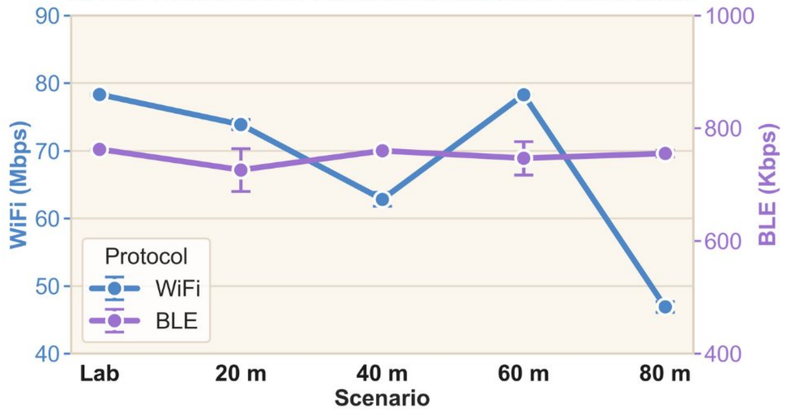
RPI 4B RSSI ACROSS VALIDATION SCENARIOS



Hardware Platforms

PLATFORM	STACK
Raspberry Pi 4B Linux SBC	Wi-Fi · BLE
Arduino MKR WAN 1300 SAMD21 + SX1276	LoRa
Arduino MKR WiFi 1010 SAMD21 + ESP32	Wi-Fi · BLE
Seeed Wio-WM1110 nRF52840 + LR1110	LoRa · BLE · Zigbee
TP-Link TL-PA7010 Wired Adapter	PLC · Powerline

RPI 4B THROUGHPUT ACROSS VALIDATION SCENARIOS



This work was funded by IAPMEI - Agência para a Competitividade e Inovação, I. P., and European Union - Next Generation EU within the scope of the project ATE: Aliança para a Transição Energética (Ref. C644914747-00000023), and by FCT through the LASIGE Research Unit, {(https://doi.org/10.54499/UID/00408/2025%7D)https://doi.org/10.54499/UID/00408/2025}

U LISBOA · Ciências ULisboa · fct Fundação para a Ciência e a Tecnologia · LASIGE · SMART ENERGY LAB · PRR Plano de Recuperação e Resiliência · REPÚBLICA PORTUGUESA · Financiado pela União Europeia NextGenerationEU

Probabilistic DAG-based Consensus for High-Throughput Blockchain

Henrique Vale, Hasan Heydari, Alysson Bessani

LASIGE, Faculdadede de Ciências, Universidade de Lisboa, Portugal

LASIGE dependable and secure decentralized systems

What Is Consensus?

Consensus refers to the problem in which several participants in a distributed system attempt to agree on a common value.

Consensus

Agreement
All correct nodes agree on the same value

Termination
Every correct node eventually decides

Validity
The agreed value was proposed and is valid

Probabilistic Consensus

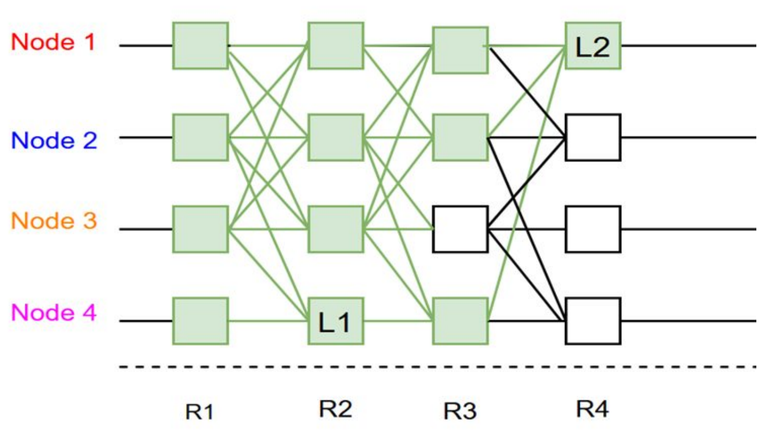
Probabilistic Agreement
All correct nodes agree on the same value with high probability

Probabilistic Termination
Every correct node eventually decides with probability 1

What is a DAG-based Consensus Protocol?

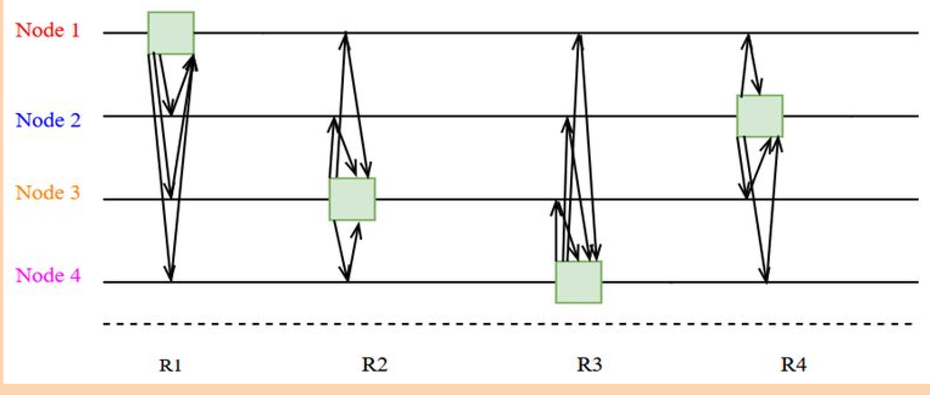
A DAG-based consensus protocol organizes blocks of transactions in a DAG (directed acyclic graph) rather than a single, linear chain. In each round, instead of waiting for a "leader" to pack transactions into a block, every node can add a new block to the DAG.

DAG-based Protocols



+Much better throughput
+No single point of failure

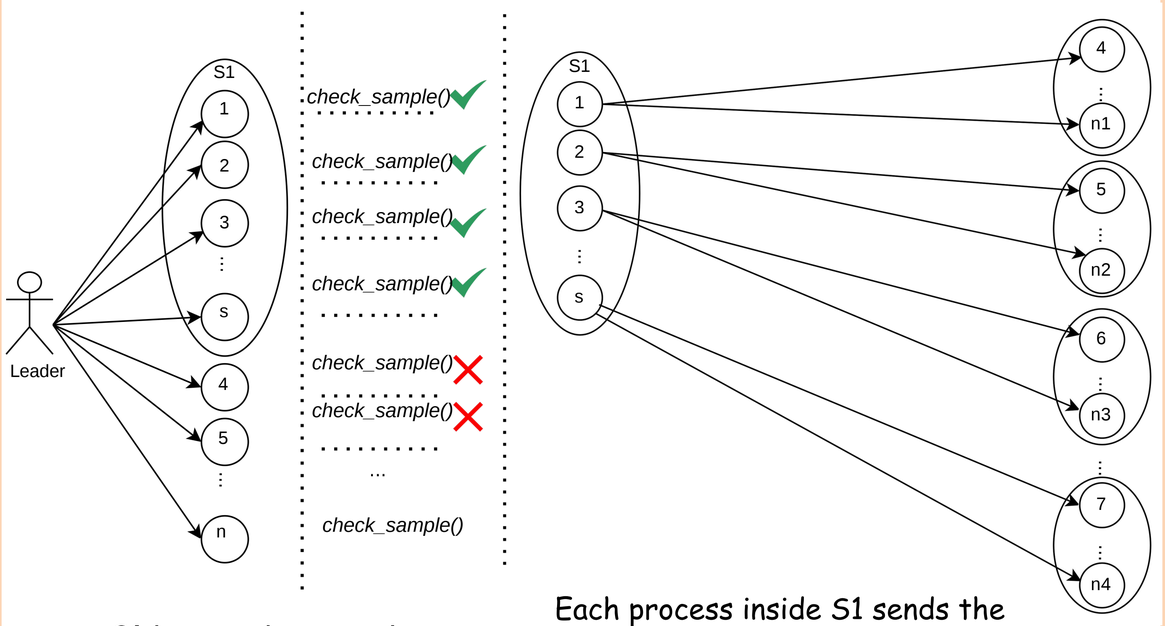
Leader based Protocols



Problem!

Limited to the network's bandwidth due to **High Communication Complexity: $O(n^4)$**. These protocols exchange significantly more messages and blocks per round.

Solution: Probabilistic Reliable Broadcast (PRBC)



S1 is a random sample with this size: $c1\sqrt{n}$

Each process inside S1 sends the message to its specific sample with an excepted size of: $c2\sqrt{n}$

Approach: Reduce Communication Complexity

Develop a novel Probabilistic reliable broadcast (PRBC) to reduce the number of times we send large messages containing the block.

Integrate this new PRBC into an implementation of Sailfish, a Dag-based Protocol.

Standard Reliable Broadcast

- **Standard RBC:** Exhausts bandwidth by forcing all-to-all transmission of massive block payloads.
- **Novel PRBC:** Restricts heavy payloads to two targeted hops, exchanging only lightweight hashes elsewhere.

Workplan

Development

Integrate this Probabilistic RBC into a chosen DAG-based Protocol called Sailfish

Evaluation

Evaluate and compare the newly implemented protocol with others

Document

Write protocol proofs

Patching and Improvement

Fix latency issues and bugs found with the current implementation of the protocol

Acknowledgements: This work was supported by FCT through project SMaRtChain, ref. 2022.08431.PTDC, DOI 10.54499/2022.08431.PTDC (https://dx.doi.org/10.54499/2022.08431.PTDC) and the LASIGE Research Unit, ref.UID/000408/2025 DOI https://doi.org/10.54499/UID/00408/2025.

Byzantine Fault-Tolerant Searchable Encryption

Rafael Correia, Alexander Davidson, Bernardo Ferreira

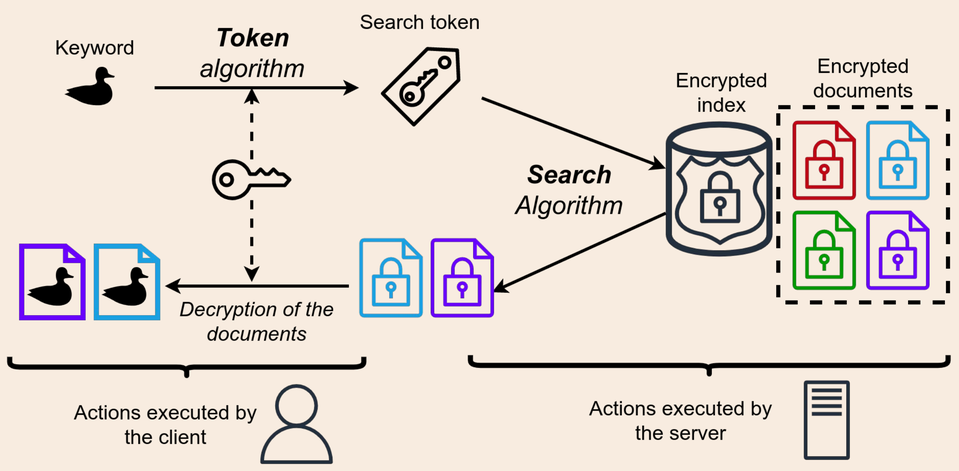
LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

dependable and secure decentralized systems

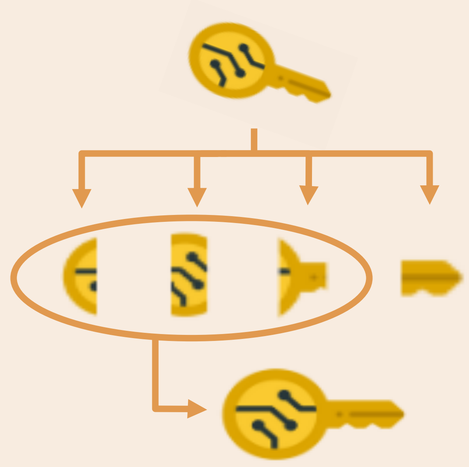
Background

What is Searchable Encryption?

It's a cryptographic protocol that allows searching encrypted text documents!



Threshold Cryptography



Threshold cryptography allows the distribution of a secret key across many nodes, and it takes only a **subset** of **shares** to recover the key

BFT-SMR

Byzantine Fault-Tolerant State Machine Replication allows the system to continue operating **correctly** even in the presence of invalid or malicious replicas

Motivation

Searchable Encryption designed for single client & server

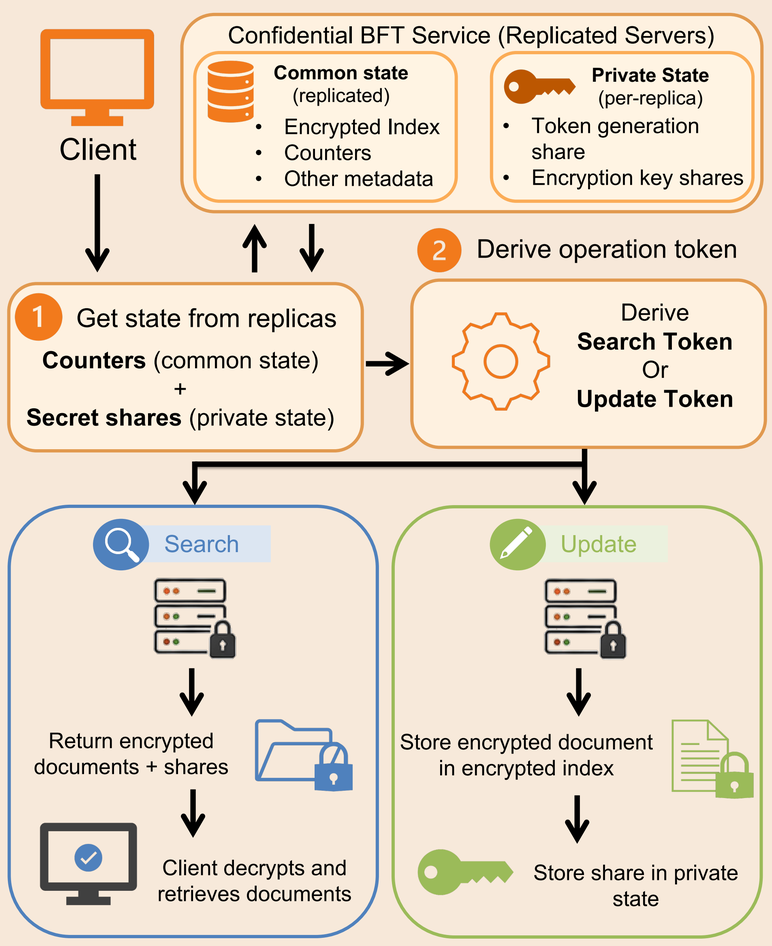
Real-world systems have **multiple concurrent clients!**

Real-world systems may **crash or be corrupted** by hackers!

Idea: Extend Searchable Encryption to support multiple concurrent clients and Byzantine Fault Tolerant replication!

Methodology

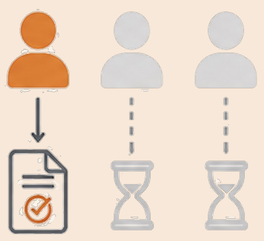
- Use a black-box **Confidential BFT service** to replicate server state
  - **Confidentiality:** data remains protected unless more than f secret shares are exposed
  - **Correctness and availability:** the replicated service tolerates up to f Byzantine replicas with n ≥ 3f + 1
- Store sensitive client state as **secret shares** across replicas
- Clients obtain **threshold-protected state** from replicas to efficiently derive **search and update material** during each operation

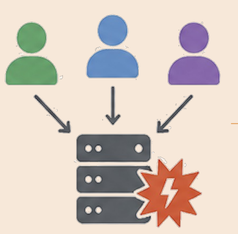


Open Problems

Client state synchronization solved, but there is still issue of **conflicting updates**

Current Prototype	Open challenge
One active client at a time	Multiple active clients efficiently?
(obstruction-freedom)	*(wait-freedom)*





Objective: Maximise concurrency and throughput while minimizing latency

Conclusions

Multi-client BFT Searchable Encryption can be very interesting for practical applications

Achieved

Multi-client SE over BFT replication provides **consistency** and **availability**, while secret sharing **protects client state** across replicas

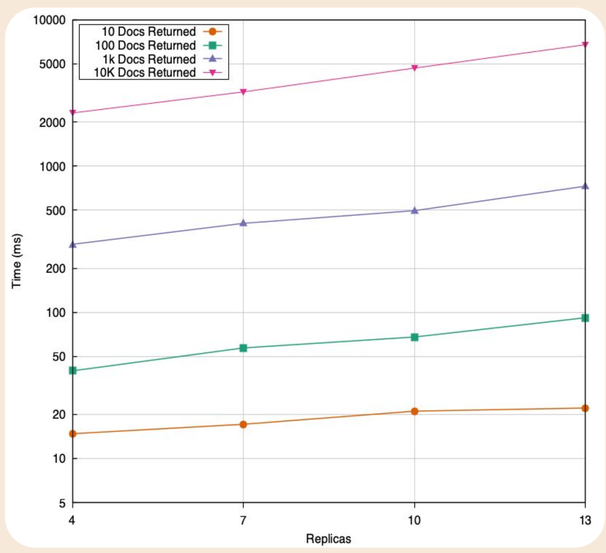
Main Challenge

The hard part is not BFT integration but handling **conflicting updates** from **multiple active clients**

Initial Results

Initial experiments show **promising performance** despite the overhead of BFT replication.

Search Latency



Acknowledgements: This work was supported by FCT through the project APOSTLE, ref. 10.54499/2023.12254.PEX (https://doi.org/10.54499/2023.12254.PEX) and the LASIGE Research Unit, ref. UID/00408/2025 DOI https://doi.org/10.54499/UID/00408/2025.

Language-Grounded Shared Autonomy for Dexterous Co-Manipulation

Sajjad Hekmat[1], Georgia Chalvatzaki[2], João Bimbo[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] Faculty of Computer Sciences, Technical University of Darmstadt, Germany

LASIGE data and systems intelligence

1. Motivation

- Dexterous co-manipulation remains difficult for general robots
- Pure teleoperation is cognitively demanding
- Pure autonomy is brittle in contact-rich collaboration
- Human partners use short contextual commands: "Hold this nut", "Keep it steady", "Move it left"
- Goal: combine telepresence, language grounding, shared autonomy, and dexterous contact control

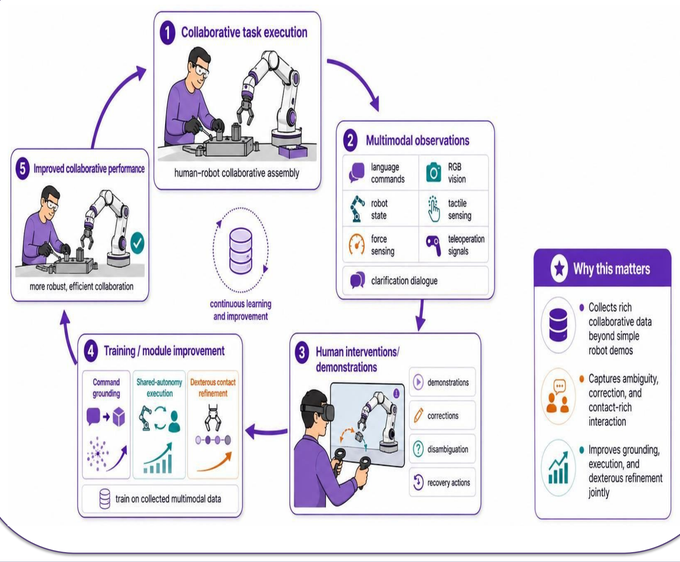
2. Research Questions

- How should short, context-dependent collaborative commands be grounded into explicit robot roles, referents, and subgoals during dexterous co-manipulation?
- What is the best architecture for combining shared-autonomy telepresence with VLA-based macro control and a separate contact-aware dexterous expert?
- Which multimodal data streams and data-collection strategies most improve contact-rich collaborative performance: language, vision, proprioception, tactile/force feedback, and teleoperation corrections?

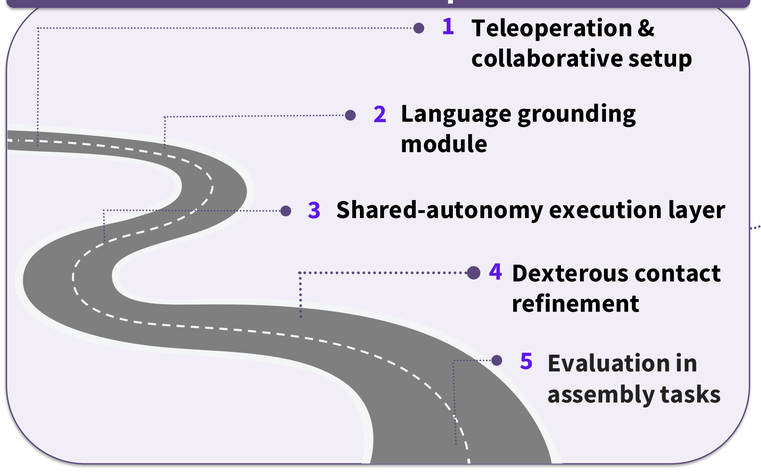
3. Core Idea

- Ground collaborative language using scene, task, and dialogue context.
- Execute macro actions through shared autonomy and VLA-based visuomotor control.
- Refine contact-rich manipulation with a specialized dexterous expert using tactile/force feedback and motor primitives.

4. Expected Contribution



5. Road Map



6. System Architecture for Dexterous Co-Manipulation

INPUTS

- Speech / NL command
- Task state
- Dialogue history
- Workspace camera
- Wrist camera
- Robot state
- VR-AR / teleop cues
- Tactile / force

LAYER 1 - COLLABORATIVE GROUNDING

Perception & Context Builder | Collaborative Command Grounding | Dialogue & Clarification Manager

Intent | Referent | Role | Subgoal | Confidence

- Grounds short natural commands into structured collaborative intent, referents, roles, and subgoals.

LAYER 2 - SHARED-AUTONOMY EXECUTION

Semantic Verifier | Physical Feasibility & Subgoal Generator | Shared-Autonomy Policy / VLA Executor

Executable subgoal | Action chunk velocities / positions / gripper

- Verify, plan, and execute grounded collaborative subgoals through shared autonomy and VLA-based visuomotor control.

LAYER 3 - DEXTEROUS CONTACT REFINEMENT

Dexterous Expert

Grip-force adjustment | Local alignment

- Refine contact-rich phases using tactile, force, proprioceptive, and local visual feedback.

OUTPUTS

- Cooperative motion
- Shared-autonomy assistance
- Dexterous contact execution

Ongoing Works

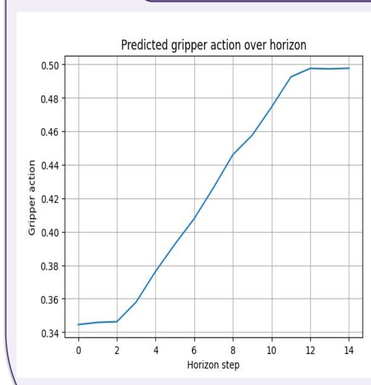
You Are Here!

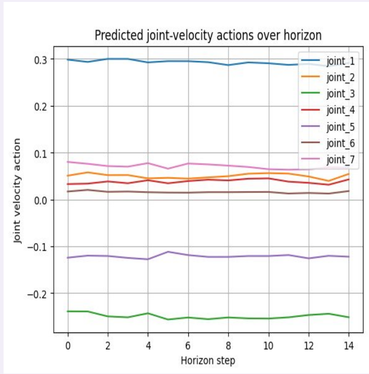
A. Visual-Language-Action-Models Testing:

- Vision: RGB-D video stream of the shared workspace, capturing the relative positions and velocities of the robot, the collaborator, and industrial objects.
- Language: Natural language instructions from the local collaborator (e.g., "Hold the plate while I tighten the screw").
- Haptics & State: Proprioceptive data from the Franka arm, including joint torques and the forces exerted by the collaborator.

$$a_t = \pi(V_t, L, H_t)$$

Prompt: "Pick up the red bottle"





B. Human-Like Dexterity via DMPs:

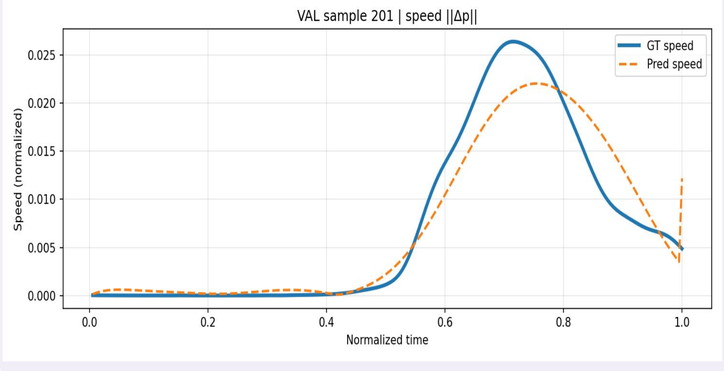
- Dynamic Movement Primitive is a Spring-Damper derived differential equation:

$$\tau\dot{z} = \alpha_z\big(\beta_z(g-y) - z\big) + f(x)\ , \quad \tau\dot{y} = z$$

- To encode the "style" of human reaching from teleoperation data, we extract basis function weights using:

$$f(x) = \frac{\sum_{i=1}^{N} \Psi_i(x)\, w_i}{\sum_{i=1}^{N} \Psi_i(x)}\, x$$

- During the Robot Learning phase, a generative model is trained on these weights to reconstruct human-like trajectories.



Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025,
DOI https://doi.org/10.54499/UID/00408/2025

Words Matter: Explainable AI for Hate Speech and Fake News Detection

Ana Luísa Brito[1], Soraia M. Alarcão[1], Vânia Mendonça[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE data and systems intelligence

COVID-19 PANDEMIC

Conspiracy Theorists Burn 5G Towers Claiming Link to Virus

Fake AI videos of Iran war are spreading online

Fake news, online hate swell Indonesia anti-Rohingya sentiment

Fact Check. Partido dos Trabalhadores distribuiu kit gay nas creches no Brasil?

Armed Man Threatens D.C. Pizzeria Targeted By Fake News Stories

RACISMO

Extrema-direita usou crime em Espanha para espalhar mentiras e ódio online

How India's Ruling Party is Using AI to Boost Hate Speech in States Near Bangladesh

"Negros em Portugal avisam: 'Vamos destruir o vosso país'". Jornal digital "Notícias ao Minuto" publicou este artigo?

The Niger uranium claim: How false documents fuelled America's $1.7tn war in Iraq

Why Words Matter?

- Hate Speech incites violence, polarizes communities, targets vulnerable groups.
- Fake News undermine trust in institutions, distort discourse, manipulate public opinion.
- Misinformation and Hate spread fast. By the time we fact-check the damage is already done.

Motivation

Hate Speech and Fake News are Increasingly Intertwined
- Fake News can contain and amplify Hate Speech;
- Hate can lead to the creation of Misinformation;
- Most detection approaches treat them in isolation.

AI-Generated Content Brings New Challenges
- Generative AI amplifies misinformation and hate propagation;
- Detection models trained on human-authored documents perform worse.

European Portuguese is Underrepresented
- Scarce, non-uniform datasets;
- Lack of multimodal resources;
- Limited explainability support.

Other Limitations
- Implicit language is difficult to detect;
- Evolving narratives outpace annotation.

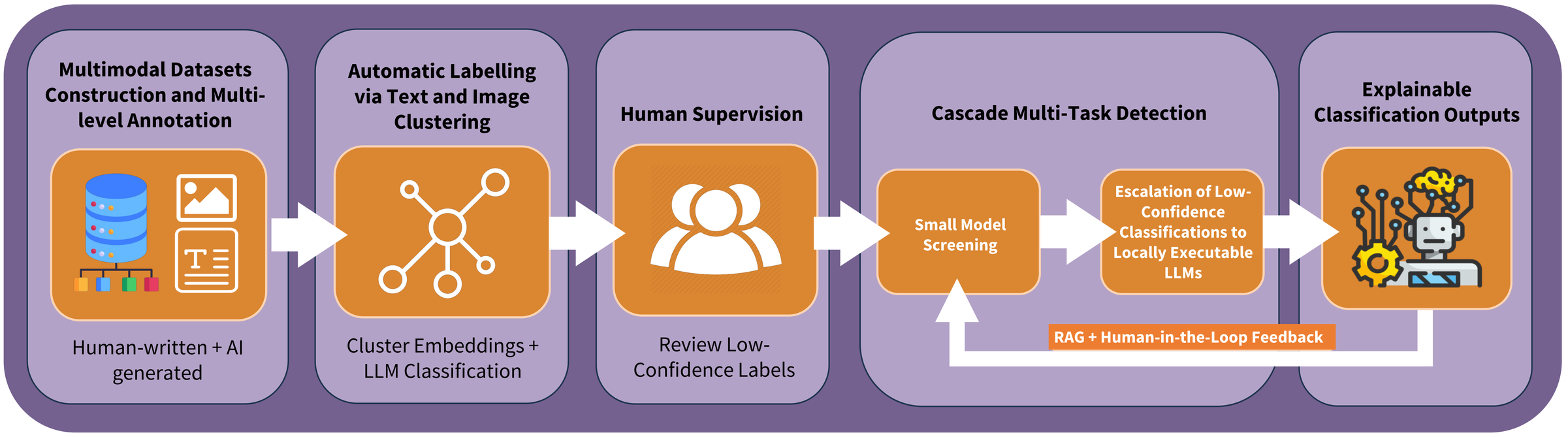
Our Proposal

Combat Data Scarcity: Two European Portuguese Datasets
- Multimodal and multi-level automatic labelling;
- Covering human-written and AI generated content.

Identify Implicit Language: Cascade + Multi-Task Detection
- Capture shared and task-specific features;
- All instances will be processed by smaller models;
- Instances with low-confidence predictions (implicit language) will be escalated to LLM-based models.

Stay Up-to-Date: Retrieval Augmented Generation (RAG) and Human-in-the-Loop Feedback
- Ground predictions in up-to-date external evidence and in human judgement.

Explain Everything: Built-in Explainability from the Start
- Explainable dataset annotations and model construction;
- Validation through user studies.



Acknowledgements:This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Automated Feature Engineering using Genetic Programming

Gabriel Meirinho, Filipe Gouveia, Alcides Fonseca

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

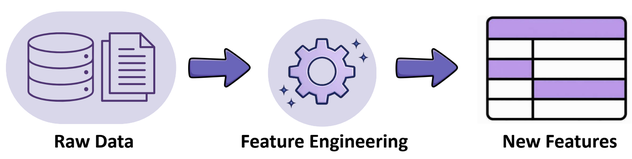
LASIGE data and systems intelligence

Motivation

Automatically discovering high-quality features can improve **model performance** and give **faster insights**.

Why is it important?

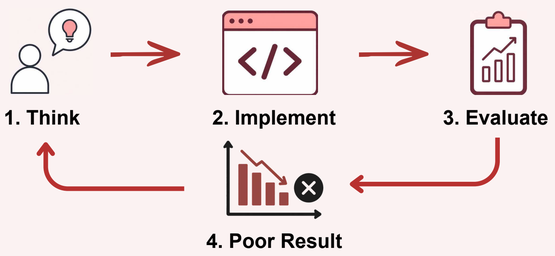
- Improve Performance
- Capture Patterns
- Improve Interpretability



Problem

Manual feature engineering is **time-consuming**, **error-prone**, and **hard to scale**.

The manual process



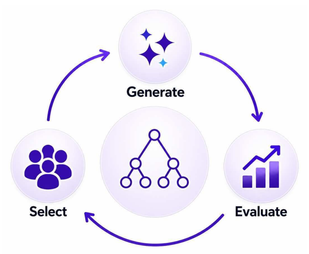
- Trial and error
- Human bias and inconsistent quality

Our Proposal

We propose an **automated** approach that **searches for high-quality features** while explicitly **managing trade-offs**.

1. Explore the search space

Grammar-based search



- Exploration of large **feature spaces**
- Less **human bias**
- More **diverse solutions**

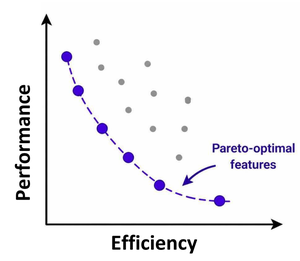
2. Evaluate features

Multi-objective evaluation

- **Performance** (Cross-validated AUC)
- **Efficiency** (Computational Cost)

3. Balance trade-offs

Pareto-optimal solutions



- Find features with **high performance** and **low computational cost**

Grammar

Domain-specific grammar defines the search space for **valid expressions**. This enables the incorporation of **domain knowledge**.

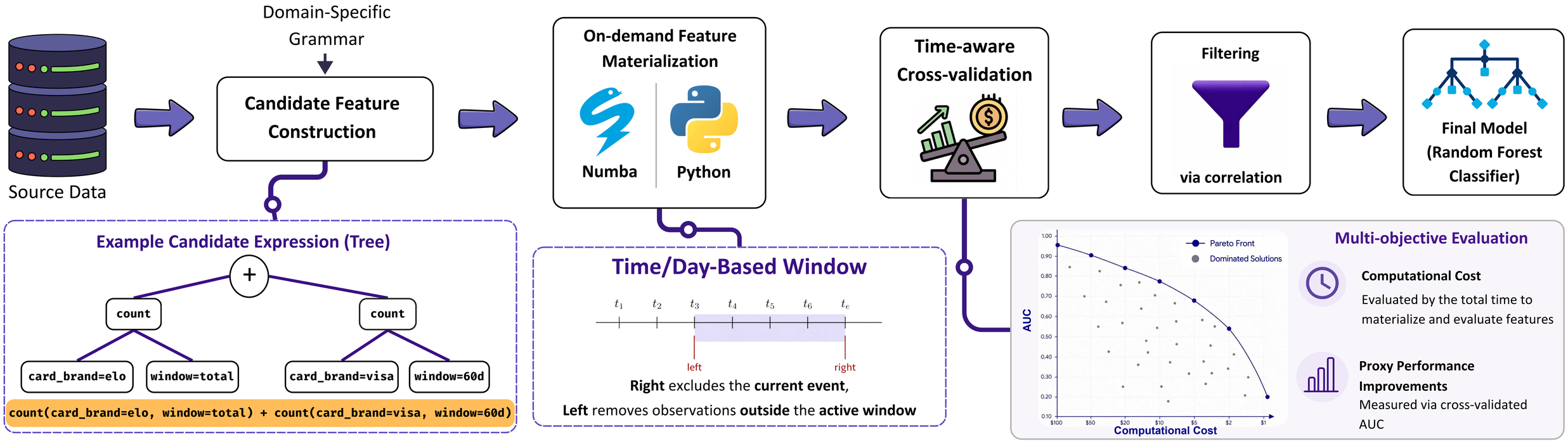
Grammar (BNF-style)

```
expr ::= mean(field, conditions, window)
       | sum(field, conditions, window)
       | count(conditions, window)
```

Examples

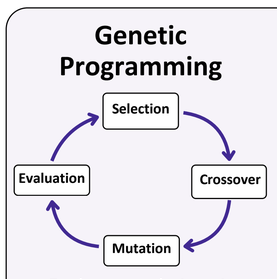
```
mean(amount, card_brand=visa, window=7d)
sum(amount, status=accepted, window=5h)
count(status=approved, credit_risk=high, window=2w)
```

Pipeline



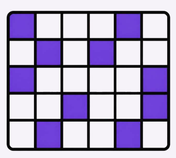
Search Strategies

Genetic Programming



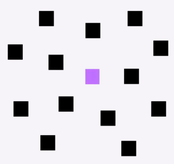
Evolves population via selection, crossover and mutation

Enumerative



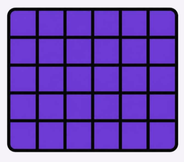
Systematic enumeration within depth and size limits

Random Search



Randomly samples expressions from the search space

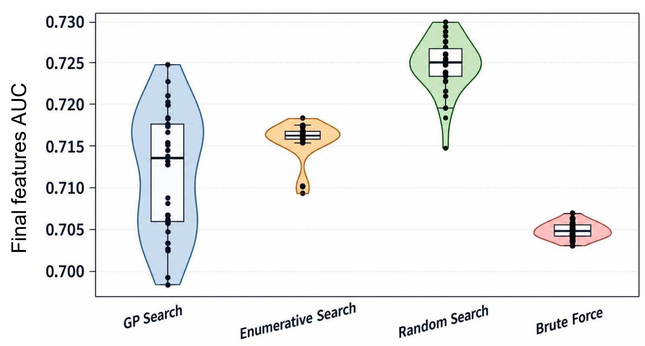
Brute Force



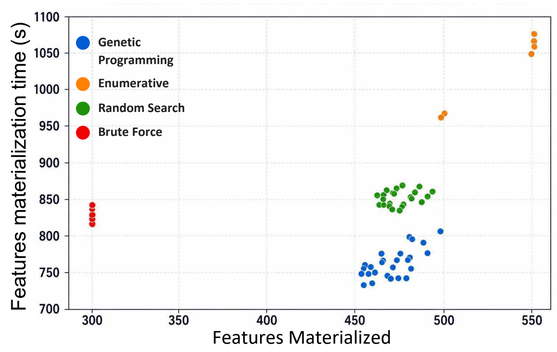
Model uses all features discovered during the search

Results & Evaluation

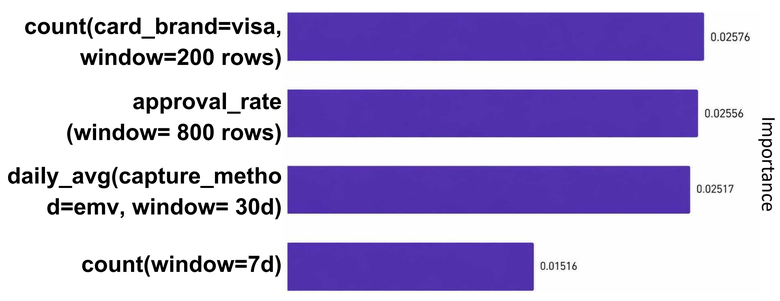
Final Feature Set AUC



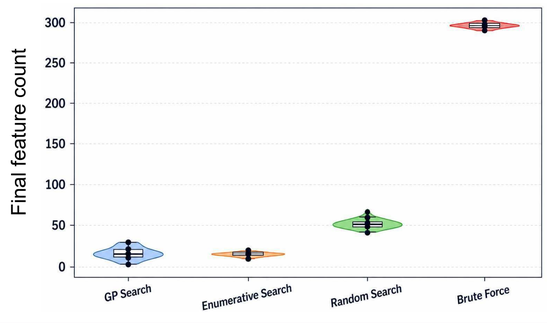
Features Materialized vs Materialization Time



Feature Examples of Random Search



Final Feature Count



Key Findings

- Random search achieved the highest AUC among the evaluated search strategies
- Genetic Programming produces competitive features with lower materialization cost
- No strategy dominated both objectives

Future Work

- Extend the grammar with richer domain constraints and cost-aware operators
- Compare generated features against manually designed domain features

This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025. This work was also funded by the European Union - NextGenerationEU through the PRR - Recovery and Resilience Plan under the LASIGE Research Unit, ref. UID/PRR/00408/2025, DOI https://doi.org/10.54499/UID/PRR/00408/2025 and ref. UID/PRR2/00408/2025.

Hybrid Recommender system for Biological Control Agents against plant pathogens

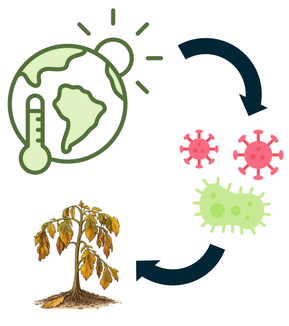
Gonçalo Loureiro[1], Ricardo Ramiro[2] , Manisha Sirsat[2] , Márcia Barros[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] InnovPlantProtect, Elvas, Portugal

LASIGE data and systems intelligence

The Beginning

- ✓ Food production has become one of the major global concerns, driven by rapid population growth and the in creasingly visible effects of climate change. Around 40% of global annual losses is caused by pathogens.



Biological Control Agents (BCA's)

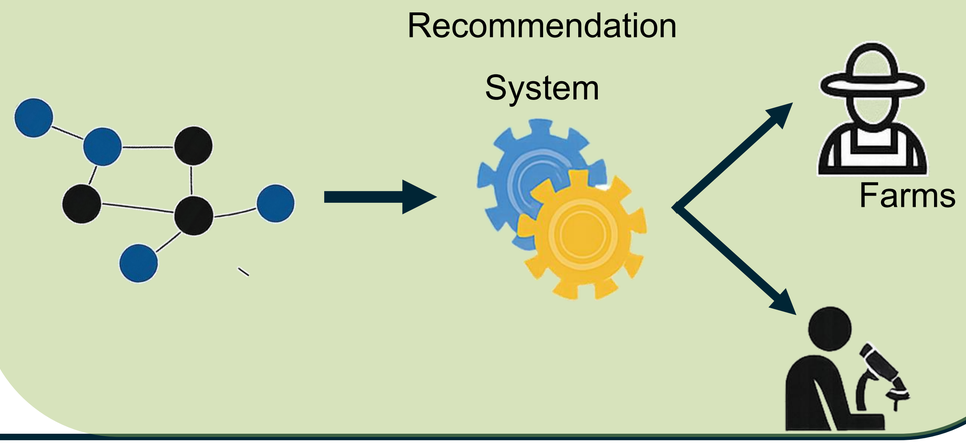
Solution?

PROBLEM!

- Farmers rely on unsustainable solutions
- Fragmented and disorganized data
- High time and cost in finding new BCAs

Objective

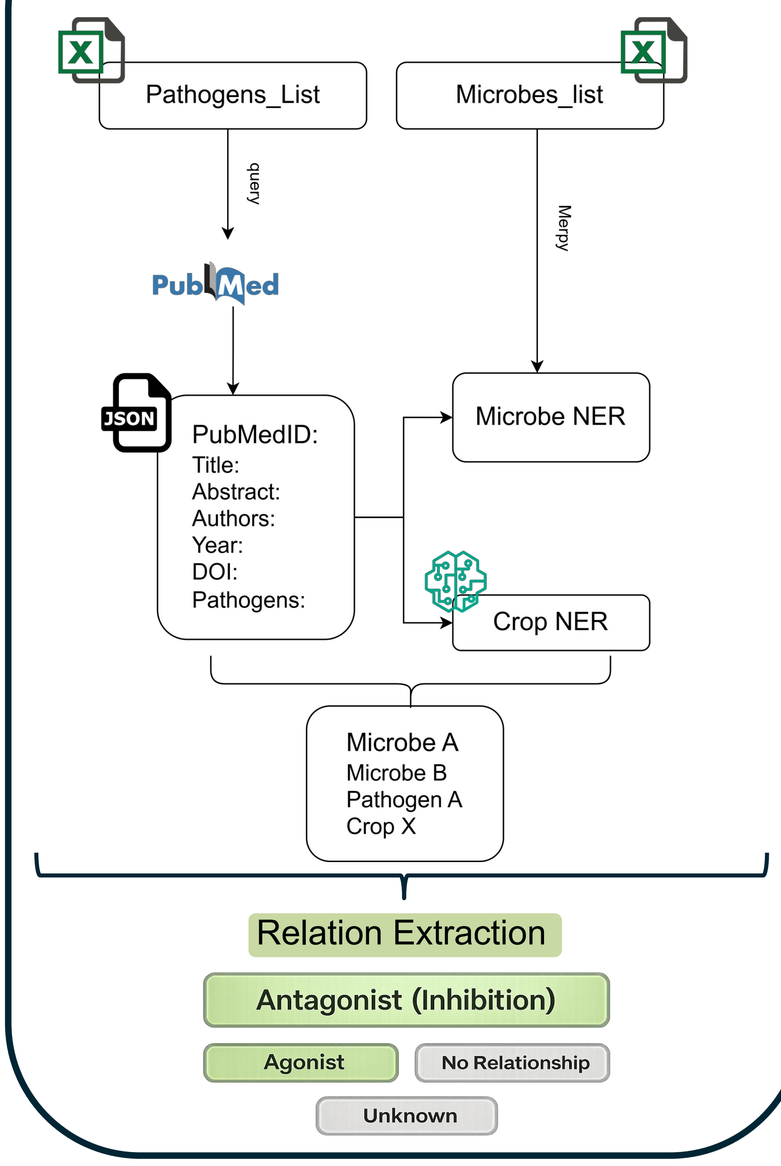
- ✓ Extract relationships between pathogens and microbes and develop a recommender system that suggests which microbes are likely capable of inhibiting a given pathogen.



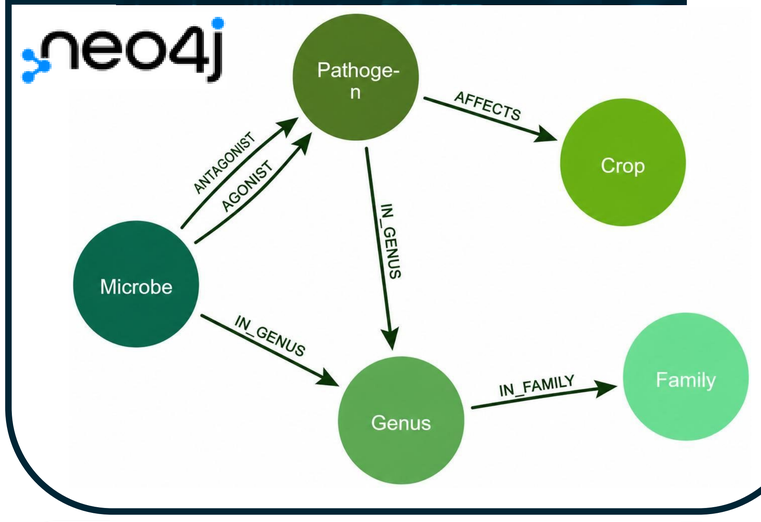
Methodology

- ✓ Scientific articles will be collected from the PubMed database, focusing on each plant pathogen.
- ✓ Named Entity Recognition (NER) and relationship extraction techniques will be applied to the retrieved articles.
- ✓ The extracted relationships between pathogens and microbes will be compiled into a document containing pathogen–microbe classifications.
- ✓ This information will be used to construct a knowledge graph in Neo4j.
- ✓ The resulting knowledge graph will serve as the foundation for the recommendation system.

1 Data Pipeline

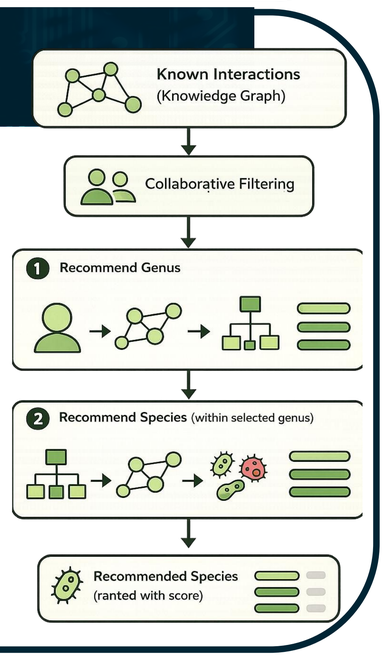


2 Knowledge Graph



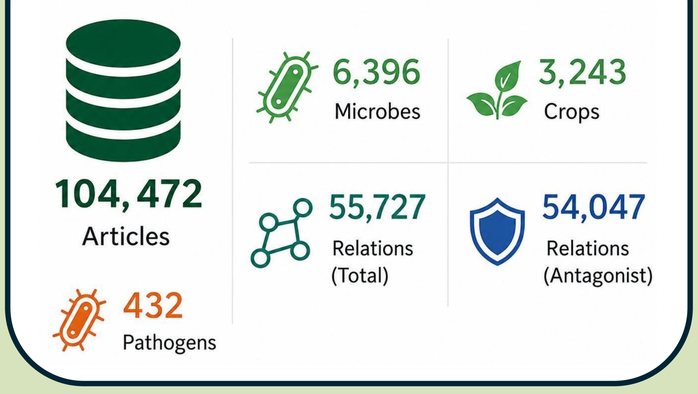
3 Recommender System

Designed to identify microorganisms with antagonistic potential against plant pathogens by combining interaction data.

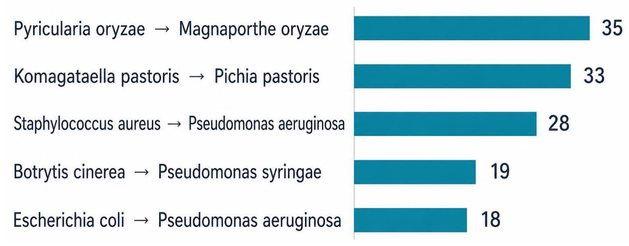


Results

DATABASE OVERVIEW



TOP 5 AGONIST MICROBE–PATHOGEN PAIRS



Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the project Recommender system for biological control agents against plant pathogens, ref. 2023.14580.PEX, DOI https://doi.org/10.54499/2023.14580.PEX, and under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

Multimodal Anomaly Detection for Human-Robot Interaction

Guilherme Ribeiro[1], Iordanis Antypas[1], Leonardo Bizzaro[1,2], João Bimbo[1], Nuno Cruz Garcia[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] University of Padova, Italy

Motivation

Assistive robots operate in dynamic, unpredictable settings where most anomaly detection systems—designed for stable, single-modality data—fail to generalize. To ensure safety and trust, robots must interpret anomalies contextually, we integrate **visual, physical, and relational cues** that mirror how humans perceive irregularities. This work proposes a **multimodal perception framework** that fuses complementary sensory data to detect anomalies in a **fault-tolerant and human-aware** manner. Unlike prior approaches that reconstruct a single feature vector, we reconstruct a **shared multimodal representation**, capturing the interplay across modalities. This enables the robot to reason about context, reduce false alarms, and adapt reliably to changing real-world conditions.

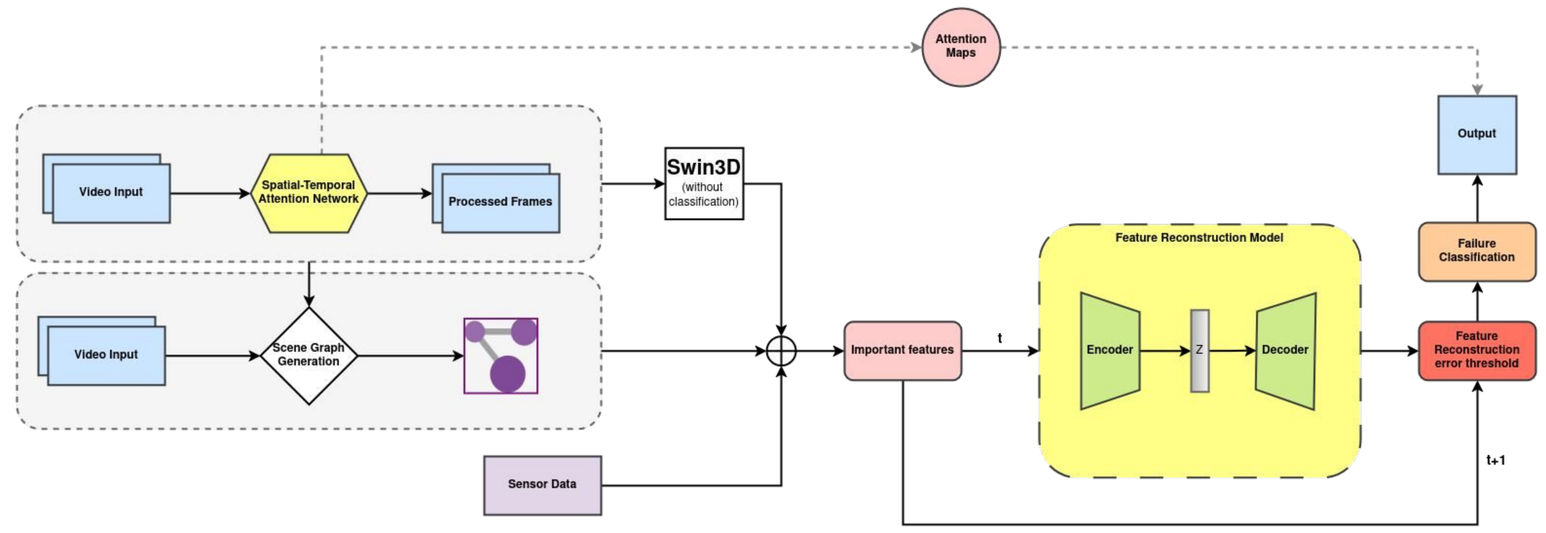


Figure 1 – Framework Overview *Proposed multimodal anomaly detection pipeline. Visual, relational, and sensor data are fused into a unified feature vector, reconstructed through an autoencoder. Anomalies are inferred from reconstruction error, enabling unsupervised and context-aware failure detection.*

Figure 2 – Example of a Natural Failure in the Dataset
Example trial from the Human–Robot Interaction dataset showing a ***handover failure****.*
(Top) the participant prepares to grasp the cup from the robot; (Bottom) the cup slips during the handover.

Dataset

We recorded a **Human–Robot Interaction Action Recognition dataset** to train and evaluate our anomaly detection models. Participants handed a cup to the robot, which placed it on another object. Natural failures such as slips, misgrasps, and misplacements occurred during interaction, eliminating the need for simulated errors.
Videos (1080×720 @ 15 fps) and synchronized robot sensor data were collected and segmented into **32-frame clips** organized by action class. Normal clips were used for training, while anomalous clips served for evaluation. Each sample includes aligned **RGB-D**, **sensor**, and **metadata**, enabling multimodal feature reconstruction and anomaly analysis.

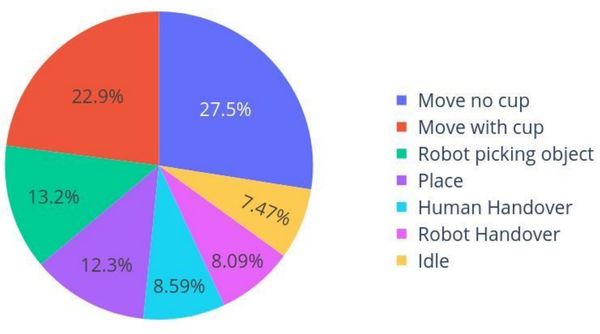


Figure 3 – Dataset Class Distribution *Distribution of the recorded Human–Robot Interaction dataset across seven action classes.*

Method

The proposed framework performs multimodal anomaly detection through a four-stage pipeline.

(1) Environment Perception: visual, relational, and sensor modalities capture external motion, spatial context, and internal robot dynamics.

(2) Multimodal Fusion: features are concatenated into a unified vector representing cross-modal dependencies.

(3) Feature Reconstruction: an autoencoder learns to reconstruct normal behavior, encoding typical multimodal patterns.

(4) Anomaly Inference: reconstruction error between original and reconstructed features is used as an unsupervised anomaly score.

Samples exceeding a statistical threshold are flagged as anomalies.

This process enables **robust, interpretable, and context-aware detection**, allowing assistive robots to identify diverse failures across sensing modalities.

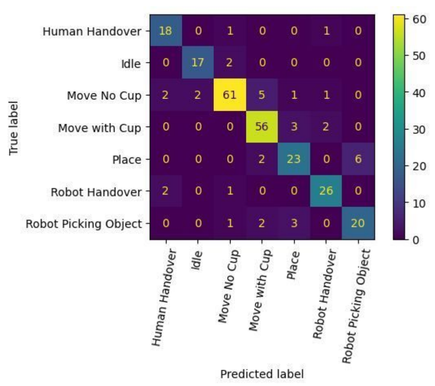


Figure 4 – Action Recognition Confusion Matrix
Confusion matrix for the action recognition model across seven action classes. The diagonal values indicate correct classifications, while off-diagonal values represent misclassification. The model demonstrates high accuracy for classes such as 'Move No Cup' and 'Move with Cup', with some confusion between visually similar actions like 'Place' and 'Robot Picking Object'.

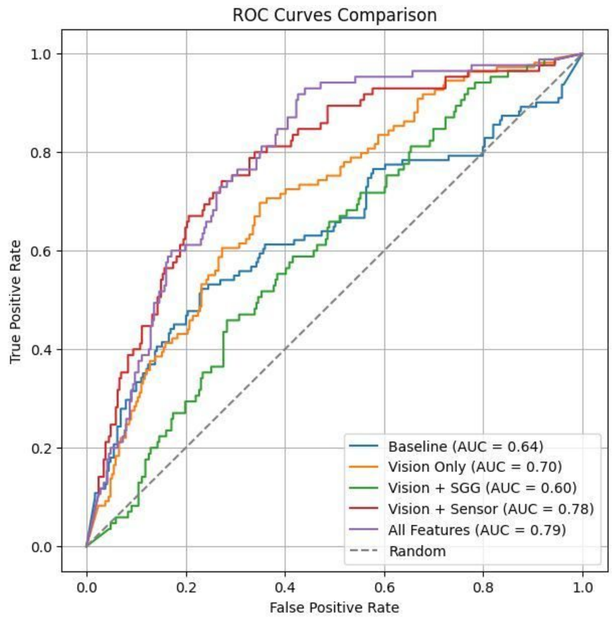


Figure 5 – ROC Curves Across Modalities
Comparison of model configurations showing progressive improvement in anomaly detection with multimodal fusion. Combining visual and sensor data achieves the highest AUC, confirming the benefit of integrating complementary modalities.

Experiments & Results

We compared four configurations—**Baseline**, **Video-only**, **Video + Sensor**, and **Full Multimodal**—to evaluate the contribution of each modality.

The baseline detected only clear visual anomalies, while the video-only model captured motion cues but missed internal faults.

Adding **sensor data** improved robustness, enabling detection of torque anomalies and reducing false alarms. Scene graph information offered contextual depth but limited quantitative gain.

The **full multimodal fusion** achieved the best overall discrimination (AUC↑ +0.09), confirming that combining external vision with internal sensing yields the most reliable, unsupervised anomaly detection.

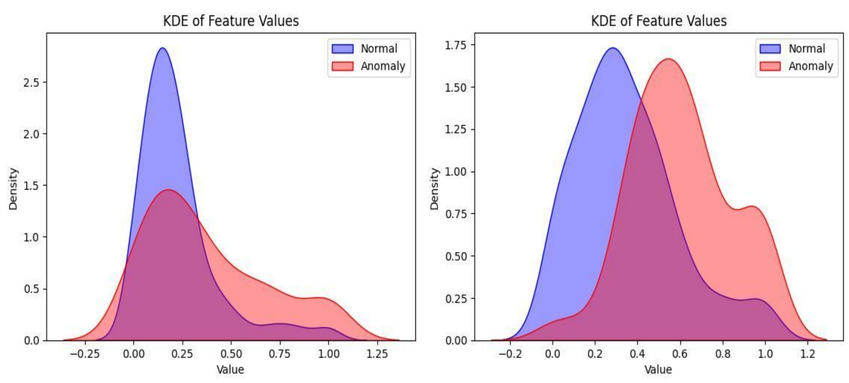


Figure 5 – Reconstruction Error Distributions
Kernel Density Estimation (KDE) of reconstruction errors for normal and anomalous samples.

Future Work

This study established a foundation for multimodal anomaly detection in Human–Robot Interaction. Next steps aim to extend the framework toward **anomaly localization and classification**, enabling robots to identify not only when but also **where and why** failures occur. Future datasets will include **richer interaction tasks**, **more participants**, and **diverse failure scenarios** to improve generalization.
Finally, developing **stronger baselines** and **real-time deployment** strategies will help transition this approach from research to **adaptive, safety-critical assistive systems**.

Code and paper available at:
https://github.com/Gui921/Multimodal-Anomaly-Detection-for-Human-Robot-Interaction

This work was supported by the Horizon Europe project 'HARIA- Human-Robot Sensorimotor Augmentation' (GA No. 101070292) (https://doi.org/10.3030/101070292), and the LASIGE Research Unit, ref. UID/00408/2025 - LASIGE

ATIVA-ER: Exploration of Language Models for Emotion Recognition in Neurodivergent Adults

Marcos Torres[1], Sofia Costa[1], Vânia Mendonça[1], Soraia M. Alarcão[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE data and systems intelligence

Motivation

- **Neurodivergence** corresponds to **variations** in the development of the human brain
- Includes conditions such as:
  - **Attention-Deficit/Hyperactivity Disorder** (ADHD)
  - **Autism Spectrum Disorder** (ASD)
  - **Intellectual Disability** (ID)
- Neurodivergent individuals are using **AI-powered Language Models** and **Assistive Technologies** to improve their quality of life
- However, these models often fail to adapt to their **way of communicating and expressing emotion**, hindering their effectiveness

Goals

Implementation of multiple **Transformer-Based Language** Models for the task of **Emotion Recognition** in **Neurodivergent-written Text**

Objective 1

- Custom **Emotion-annotated Dataset**
- Obtained from **Neurodivergent individuals**

Objective 2

- Implementation of **Transformer Models**
- Model fine-tuning with Custom Dataset

Methodology

Literature Review (Complete!)

- What approaches have been considered for Emotion Recognition?
- What types of Emotion Datasets are there?

Data Acquisition and Analysis (Complete!)

- Data obtained from several Subreddits
- What is the emotional content on each post?
- What are the most common keywords in posts of neurodivergent individuals?

Model Tuning (Ongoing!)

- Hybrid RoBERTa-CNN Model
- Multiple Transformer Baseline models, including BERT and DistilBERT
- Fine-tuned with custom dataset

Model Evaluation (Ongoing!)

- Traditional Metrics, such as Accuracy and F1-Score
- Uncertainty Metrics, such as Entropy

Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Image Recommendation Based on Affective Neural Networks with Graphs

Nuno Pereira[1], Ana Domingos[1], Soraia M. Alarcão[1], Márcia Barros[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

Motivation

- Most Recommender Systems ignore the affective dimension of content
- Emotions are treated as static attributes
- No modelling of relational structures between users, images and emotions
- Visual stimuli trigger highly subjective emotional responses
- Individual variability makes generic recommendations insufficient

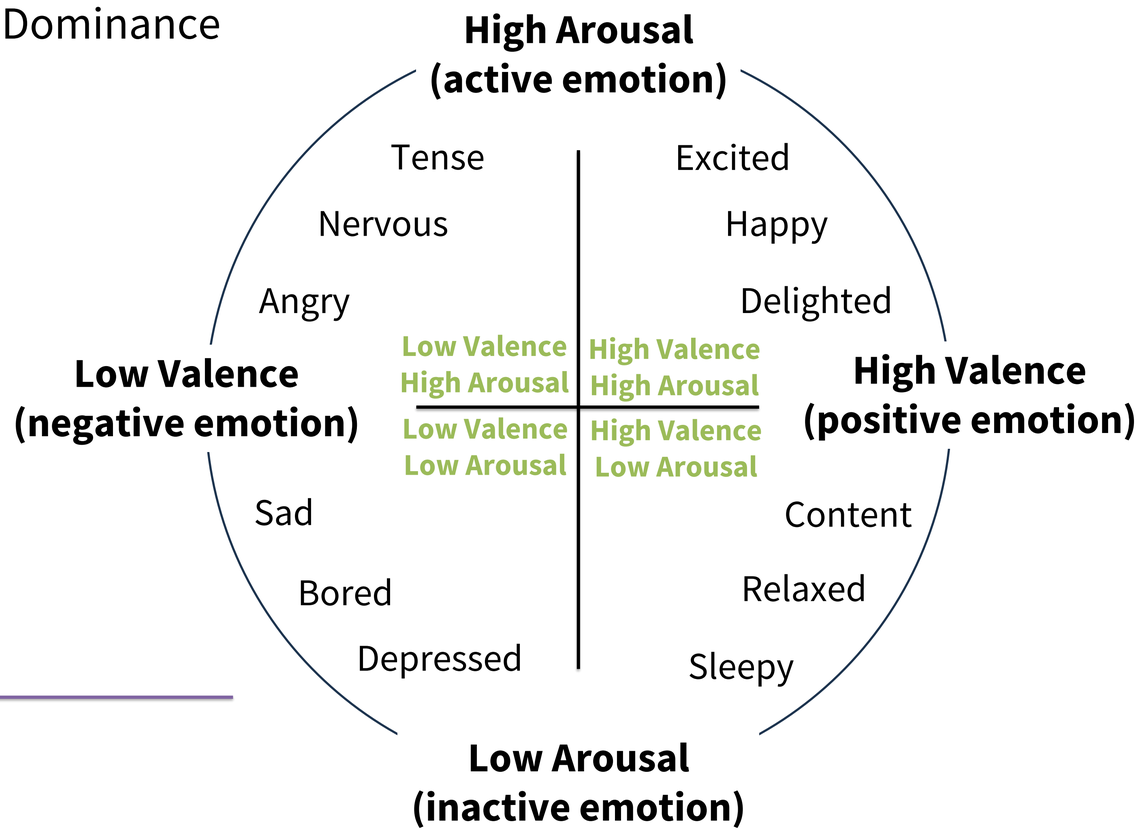
OBJECTIVE

Hybrid Recommendation System combining Graph Neural Networks, collaborative filtering, and affective embeddings.

Emotional Theory

Discrete Model (Ekman): Universal Emotions

Dimensional Model (Russell): Valence, Arousal and Dominance



Methodology

1) Data Analysis
Exploratory analysis of EmoRecSys

2) Graph Database - Neo4j
Cypher query language for graph traversal, similarity and embedding creation

3) Hybrid GNN RS
Feature-level fusion in a Knowledge Graph with Collaborative Filtering and affective VAD embeddings

4) Evaluation
Precision, Recall, MRR, Hit Rate

Please scan the QR Code and help us collecting more data!

EmoRecSys Dataset

422 users | **1158 images** | **6330 ratings**

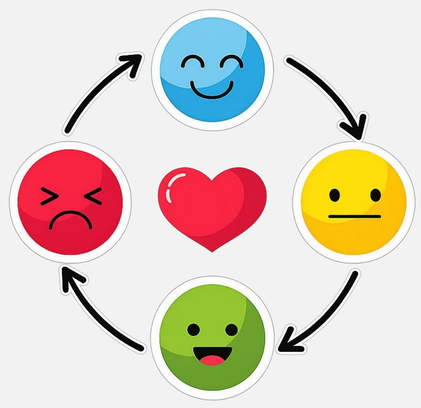
56.2% F · 43.1% M
Gender balance

18 – 24 years
More frequently answered

Bachelor's Degree
More frequently answered

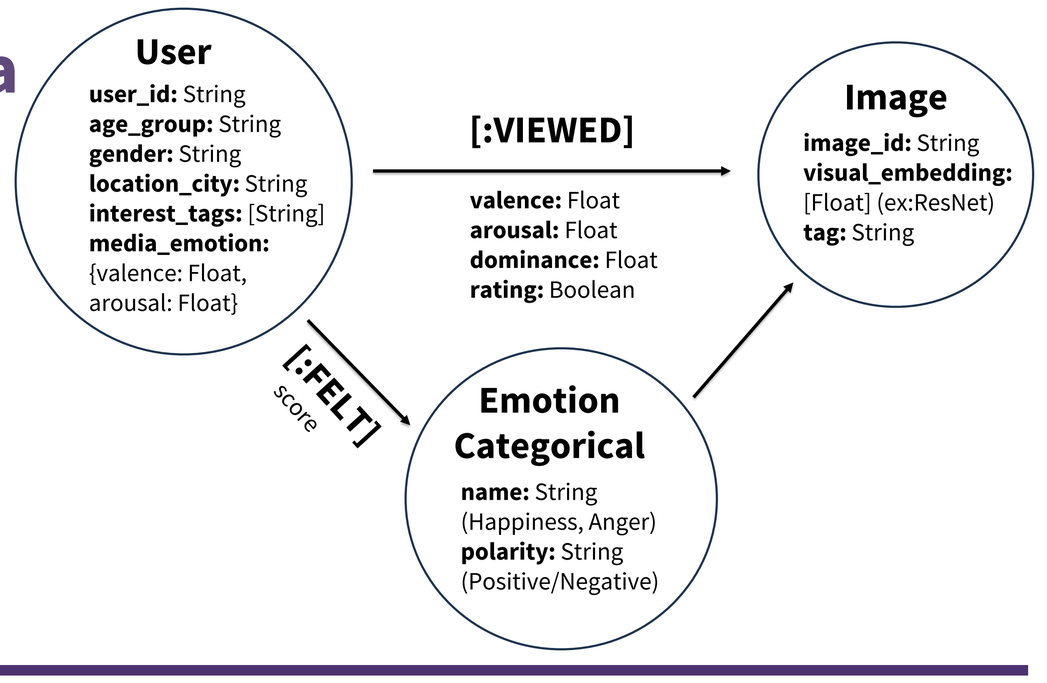
Comparison of Emotion Intensity by Gender (0-5)

EMOTION	FEMALE	MALE
Happiness	1.46	1.46
Neutral	0.98	**1.20**
Sadness	**0.71**	0.63
Surprise	0.48	**0.69**
Fear	**0.52**	0.45
Disgust	**0.29**	0.26
Anger	**0.20**	0.19



Knowledge Graph Schema

- Graph structure captures relational complexity beyond tabular models
- Typed edges preserve emotional context per user–image pair
- Heterogeneous nodes allow demographic and affective fusion
- Native support for GNN message-passing algorithms



Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Pattern-centric Learning Models for Heterogeneous Longitudinal Data

Rui Correia, Diogo F. Soares

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE data and systems intelligence

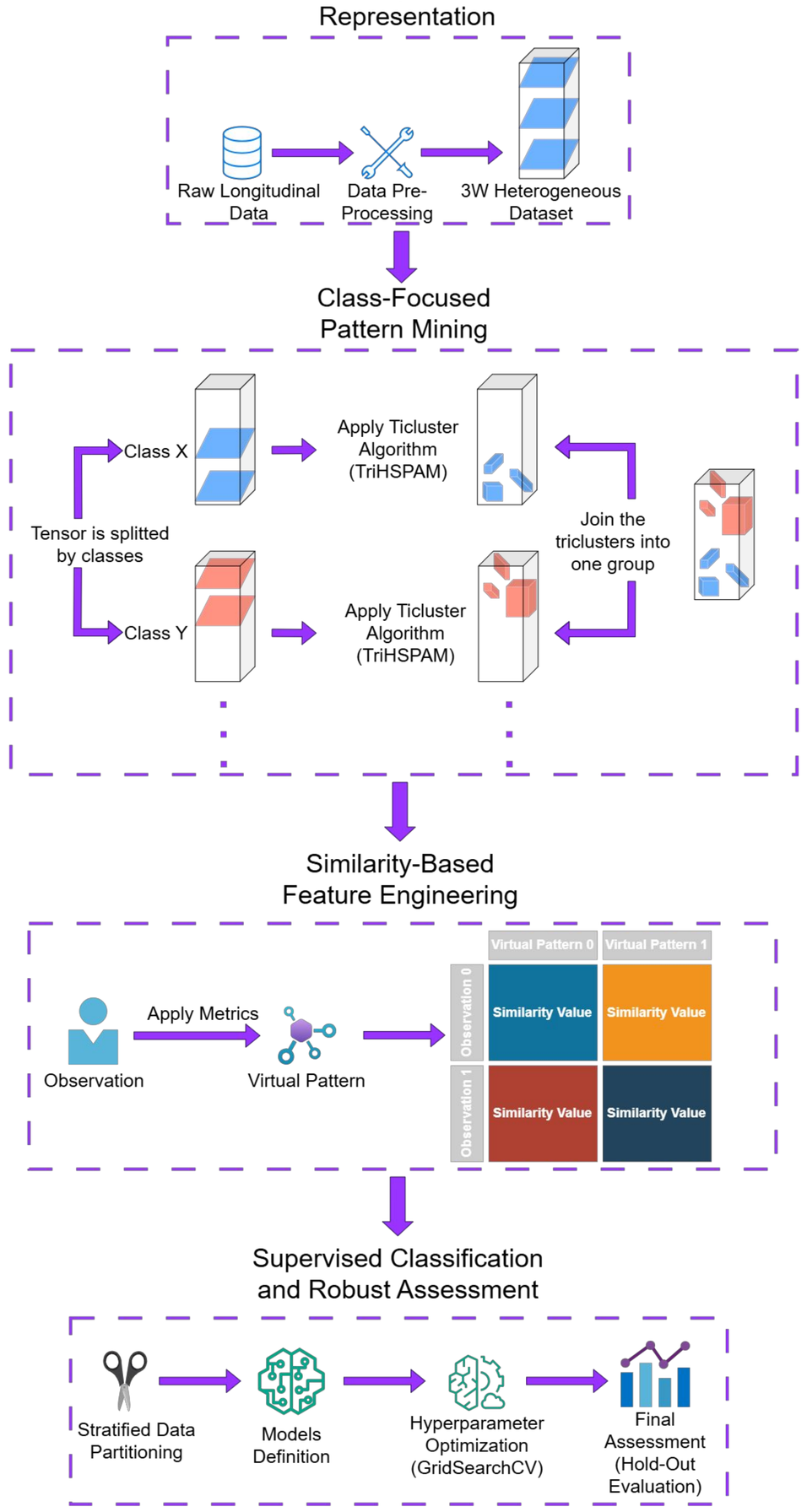
Introduction & Motivation

- ✓ **The Challenge:** Longitudinal and Multidimensional data (Observations × Features × Time) is too complex for direct use in traditional classifiers.
- ✓ **The Approach:** Use *triclustering* to extract coherent local patterns (subgroups with similar behaviors over time).
- ✓ **The Problem:** Translating variable-dimensional triclusters into fixed tabular features for Machine Learning.

Objectives

- ✓ Develop a pattern-based predictive classification pipeline for three-way data.
- ✓ Propose robust **Similarity Metrics** to convert triclusters into numerical features.
- ✓ Ensure high predictive performance and inherent model explainability.

Methodology & Architecture



Similarity-Based Transformation

The pipeline evaluates the proximity of a new observation (x) to the reference triclusters (T):

- **Baseline/Boundary:** Measures bounding box overlap via an Inclusion Ratio.

$$Sim_{Base}(x, \mathrm{T}) = \frac{|x \cap \mathrm{T}|}{|\mathrm{T}|}$$

- **Horizontal:** Evaluates the strict agreement of the underlying features ($\mathcal{J}$).

$$Sim_{Horiz}(x, \mathrm{T}) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \delta(x_j, \mathrm{T}_j)$$

- **Vertical:** Assesses the temporal alignment (K) of the state progression.

$$Sim_{Vert}(x, \mathrm{T}) = \frac{1}{|\mathrm{K}|} \sum_{k \in \mathrm{K}} \gamma(x_k, \mathrm{T}_k)$$

Dynamic Data Handling (Functions δ and γ adapt to variable types):

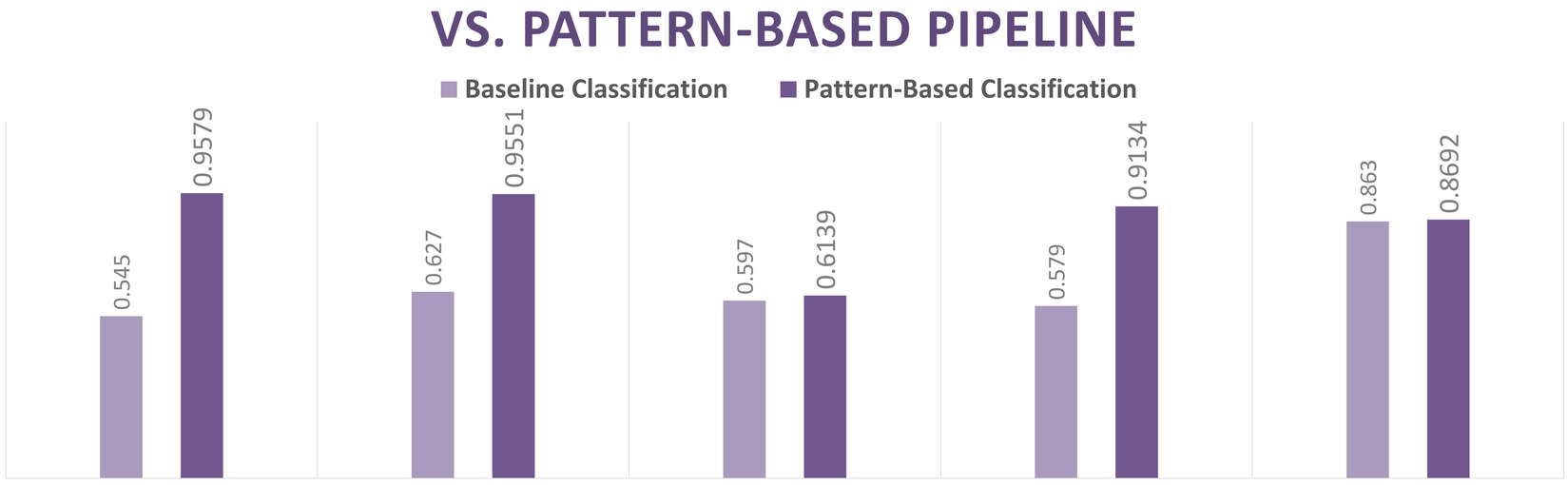
- **Numeric:** Continuous distance evaluation.
- **Ordinal:** Rank-Based distance evaluation.
- **Nominal:** Exact match.

Evaluation & Preliminary Results

Evaluation Framework: Dual-track validation on synthetic benchmarks and high-dimensional clinical data. Main KPIs: **AUC-ROC**, **F1-Score**, **Sensitivity** and **Specificity**.

Preliminary Outcomes:

- **Overall Superiority:** The pattern-based similarity pipeline outperforms traditional baseline classification.
- **Ongoing Optimization:** Fine-tuning extraction parameters to further enhance performance on highly complex datasets.



Acknowledgements: This work was supported by Portuguese national funds through FCT - Fundação para a Ciência e a Tecnologia, I.P., in the scope of LASIGE Research Unit, ref. UID/00408/2025 (https://doi.org/10.54499/UID/00408/2025), FCT-Tenure (1st edition), ref. 2023.15441.TENURE.047/CP00003/CT00026. This work was also partially funded by European Union, through NextGenerationEU and PRR - Portugal Recovery and Resilience Plan initiatives.

Data visualization of patient-centric disease dynamics: Insights into Parkinson's Disease

Diana Santos, Diogo Branco, Diogo F. Soares

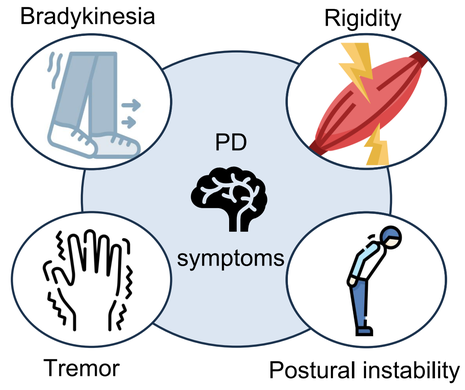
LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

health and biomedical informatics

Background

Parkinson's Disease (PD) is a progressive neurological disorder with no known cure, characterized by the deterioration of brain cells and the manifestation of motor and non-motor symptoms.
The disease was first described by the physician James Parkinson in 1817 and is now known for its four most common motor symptoms:

- Tremor;
- Rigidity;
- Bradykinesia;
- Postural instability.



Challenges and goals

- The heterogeneity of data presents a significant challenge to extract key features;
- Integrate multiple data modalities (motor, non-motor, sensor data);
- Characterizing the disease progression at the patient level is also complex;
- Develop patient-level visualizations in collaboration with clinicians, addressing the challenge of how best to present this information to them.

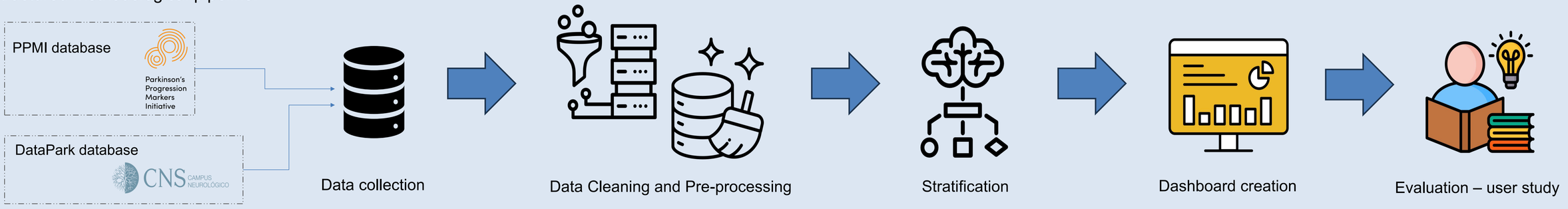
<u>Goals</u>: Apply data-driven models, create dashboards and apply interactive visualization tools to extract meaningful insights.

Research Questions

1. How do patient-centric visualizations that place individual patient data in the context of population and data-driven subgroups support clinicians' decision-making?
2. Which visualization techniques best support dashboards for extracting patient-level insights into progression, treatment and clinical manifestations in Parkinson's disease?
3. Which clinical dimensions should be visualized in interactive dashboards to best support clinicians in monitoring Parkinson's disease?

Methods

Structured methodological pipeline:



Preliminary Results

After the integration step the data is prepared to be visualized with different types of visuals. Figures 1 and 2 present examples of dashboards.

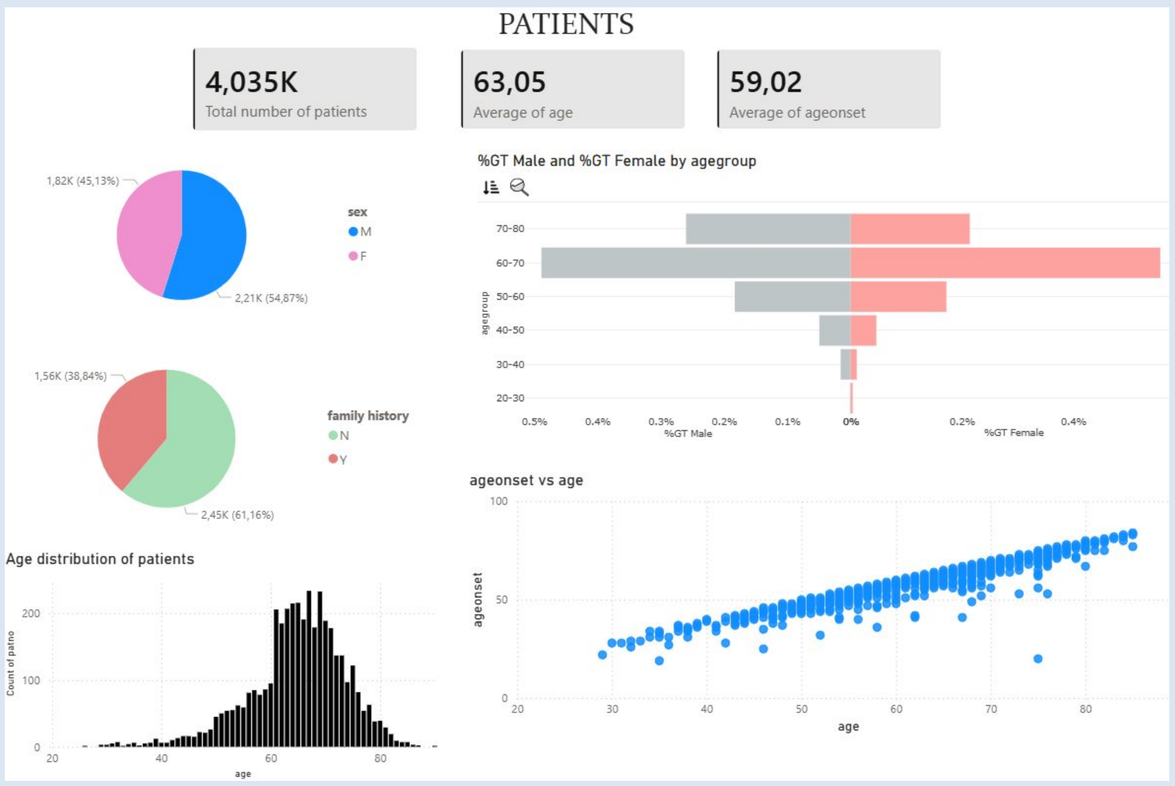


Figure 1 – Dashboard of patients from the PPMI[1] dataset.

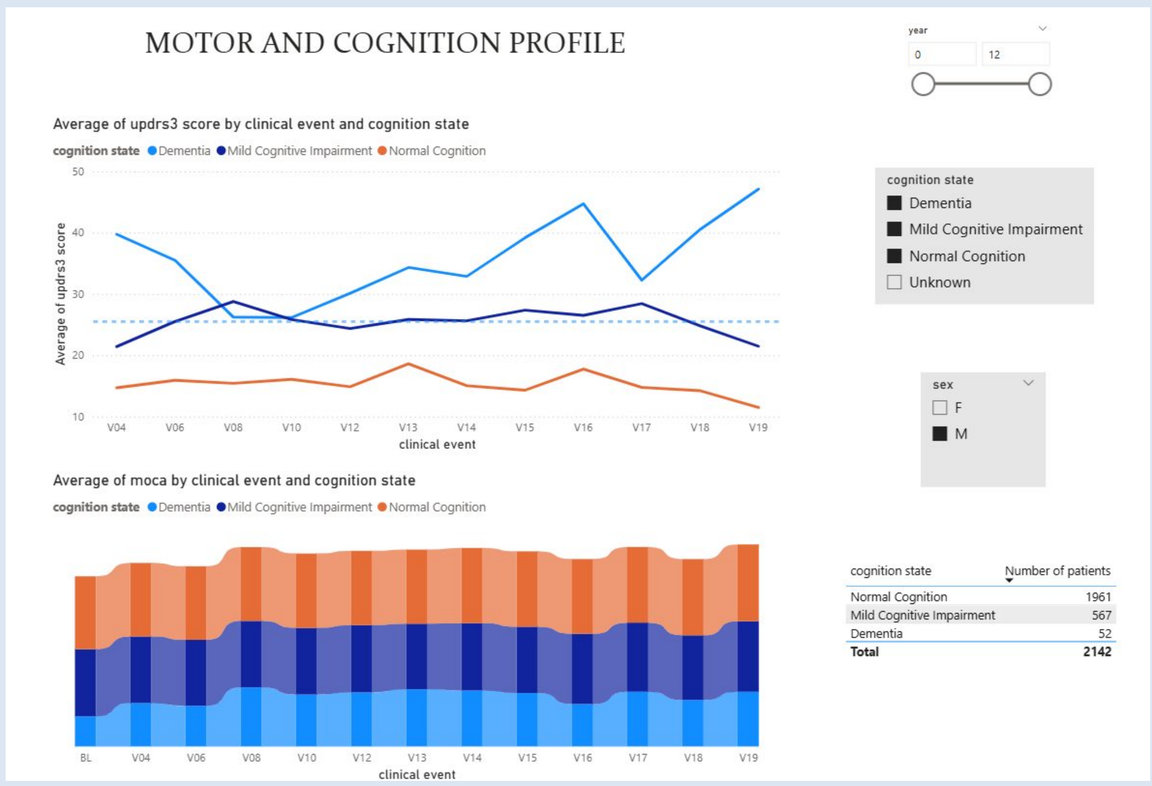


Figure 2 – Dashboard of motor and cognition profile of patients from the PPMI[1] dataset.

The icons presented in the poster are designed by Freepik; [1]clinical data from the international multicenter database Parkinson's Progression Markers Initiative.

Acknowledgements: This work was supported by Portuguese national funds through FCT - Fundação para a Ciência e a Tecnologia, I.P., in the scope of LASIGE Research Unit, ref. UID/00408/2025 (https://doi.org/10.54499/UID/00408/2025), FCT-Tenure (1st edition), ref. 2023.15441.TENURE.047/CP00003/CT00026. This work was also partially funded by European Union, through NextGenerationEU and PRR - Portugal Recovery and Resilience Plan initiatives.

Leveraging Language Models towards Generalisable Drug Repurposing Explanation Generation

Diogo Venes, Susana Nunes, Cátia Pesquita

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

health and biomedical informatics

Motivation

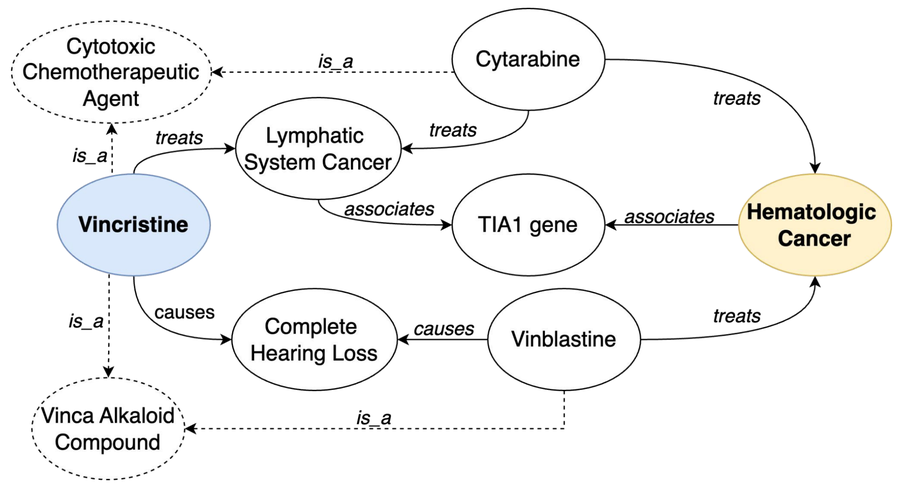
Knowledge Graphs represent entities and their relationships in a domain, and they can be explored by **explainability methods** to find logical **paths** between biomedical entities, **forming explanations** that can be used to justify **drug repurposing** opportunities, for example.

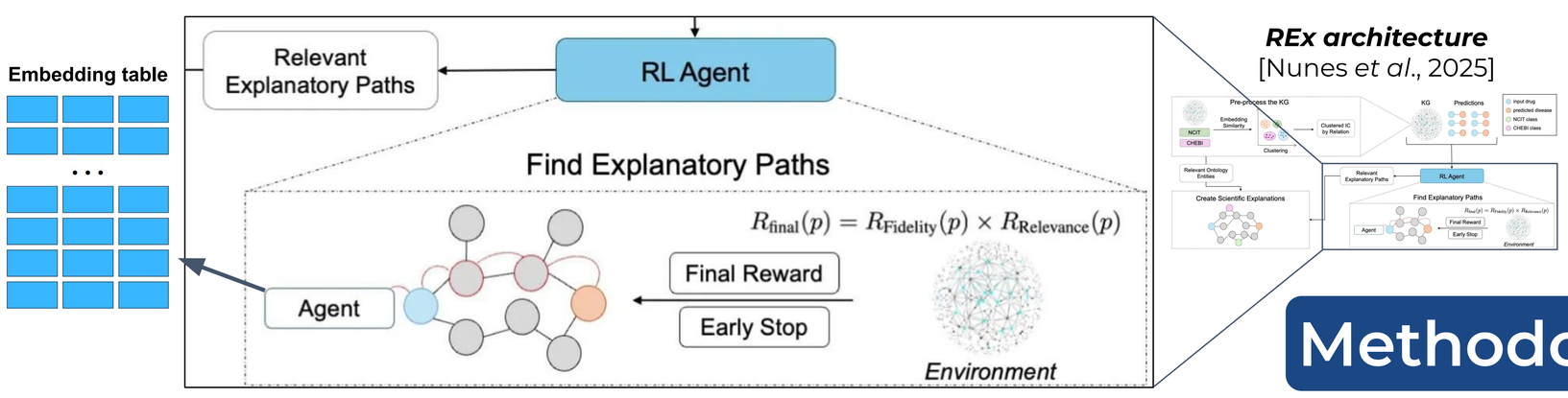
For these models to evolve into real-world tools, they must also be able to **generalise their knowledge** to ensure faster use in real-world applications.

In the state-of-the-art **REx** architecture, which we will extend, a reinforcement learning agent uses entity and relation **embeddings** to **make decisions** on where to move, which are tied to each individual element of a graph and stored in embedding tables.

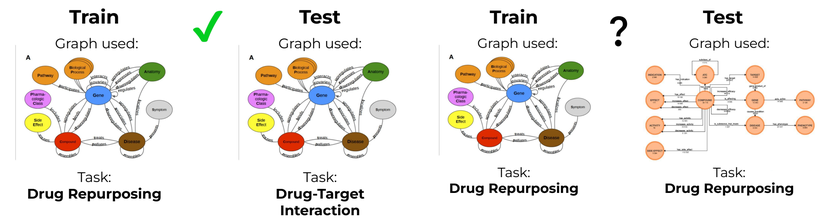
- **Embeddings for different graphs are in different embedding spaces**
- **Need for a common embedding space to have cross-dataset explanations**

Example of a drug repurposing explanation





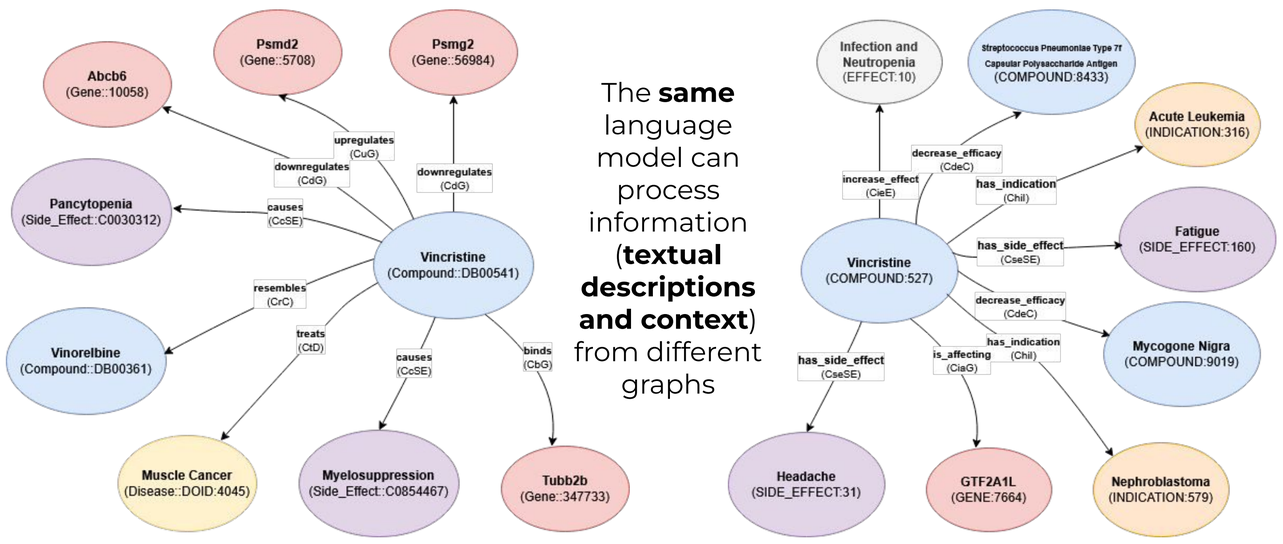
REx architecture
[Nunes *et al.*, 2025]



Proposal

Language as a common embedding space, allowing for cross-dataset generation of explanations.

Leverage textual information found across any KG.

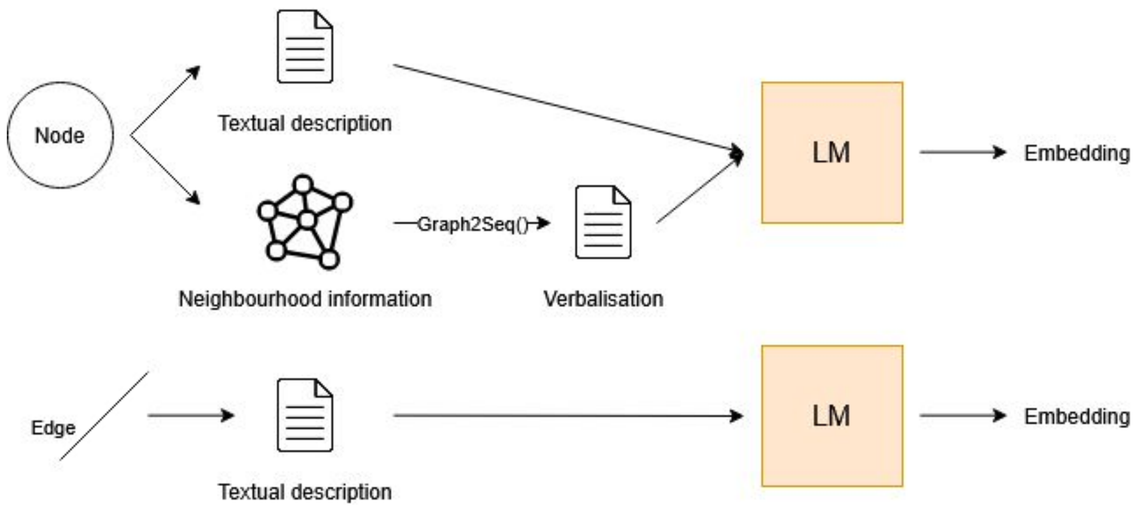


The **same** language model can process information (**textual descriptions and context**) from different graphs

Methodology

Leverage **biomedical fine-tunes** of pre-trained encoder-only language models (BLURB benchmark) as feature extractors.

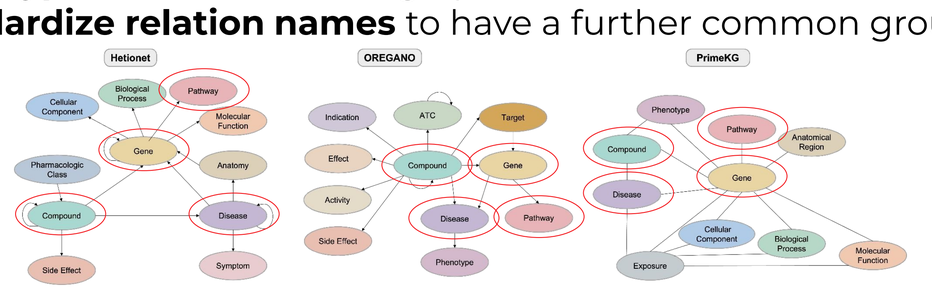
Node and **edge textual information** combined with verbalised **structural information**.



Attempted Modifications

Standardize KGs in order to have a singular shared schema - **same types of entities** in all graphs.

Standardize relation names to have a further common ground.



Preliminary Results

Cross-task:

Original Task	Original MRR	New Task	New MRR
DTI	0.197	*DR*	0.205
DR	0.433	*DTI*	0.383

Cross-dataset:

Original Graph	Original MRR	New Graph	New MRR
Hetionet	0.433	*Oregano*	0.130
Oregano	0.229	*Hetionet*	~0.0

Next Steps

Our assumptions were wrong - so we need to alter the agent so it is more aware of semantic similarity between entities.

Injecting **similarity information** into the **loss function** or the **decision making process** of the agent itself stand as the next ideas to test.

Acknowledgements: This work was supported by the LASIGE Research Unit, ref. UIDB/00408/2025. It was also partially supported by the CancerScan project which received funding from the European Union's Horizon Europe Research and Innovation Action (EIC Pathfinder Open) under grant agreement No. 101186829. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Innovation Council and SMEs Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

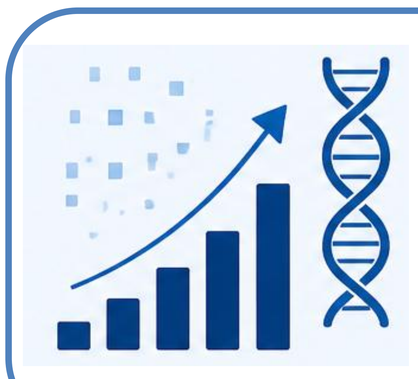
QUICK-DNA Quantum Innovation for Computational Knowledge in DNA

Helena Maki[1], André Souto[1], Margarida Carvalho[2]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] BioISI, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE health and biomedical informatics

Motivations



- Genomic data is growing faster than computation abilities.
- Genome alignment is a major bottleneck in bioinformatics.
- Quantum computing has the potential to provide quadratic speedup

Goal

Overcome difficulties presented by indels in quantum NGS alignment.

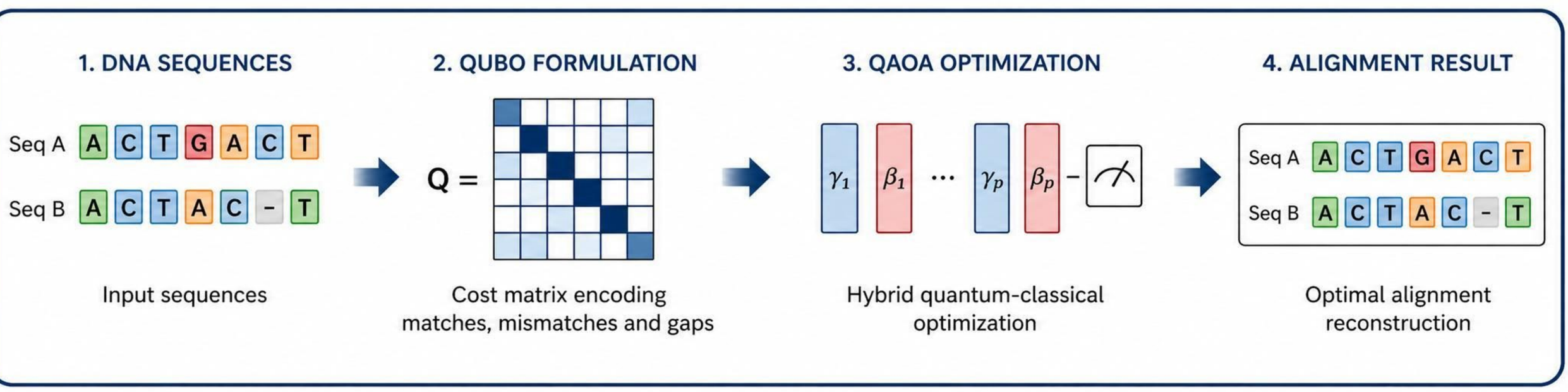
Problem

Quantum algorithms for NGS alignment struggle with gaps as they cause one part of the sequence to be out of phase with the rest.

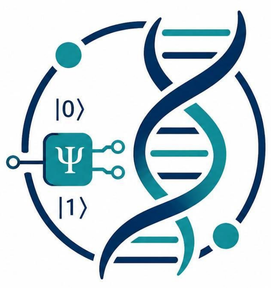
Core Idea

Gap is isolated to one half of the partition allowing the other half to align without issues.

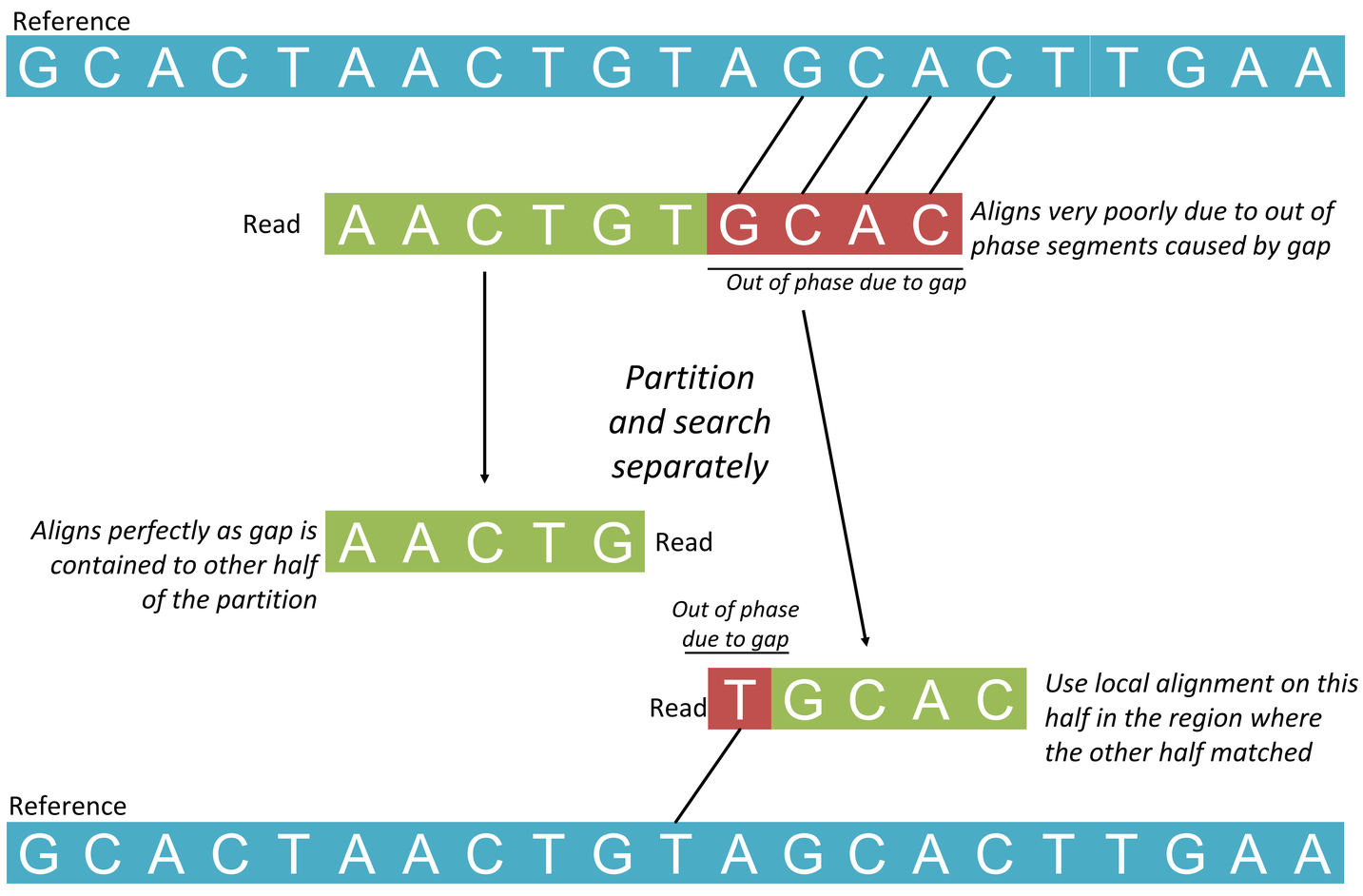
Pipeline



Contributions



- QUBO formulation for alignment
- Implementation of Grover-based quantum pattern matching.
- Recursive partitioning and re-querying allows for gaps to be detected.



Code and results

Acknowledgements: This work was supported by the internal seed project Quick-DNA and the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

Radiobiological Models in Radiotherapy – Use of AI for automatic data collection and integration into the 4Radiobiology Platform

Inês Esteves[1], Brígida Ferreira[2], Daniela Godinho[2], Márcia Barros[1]

[1]LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2]IBEB, Institute of Biophysics and Biomedical Engineering, FCUL, Lisboa, Portugal

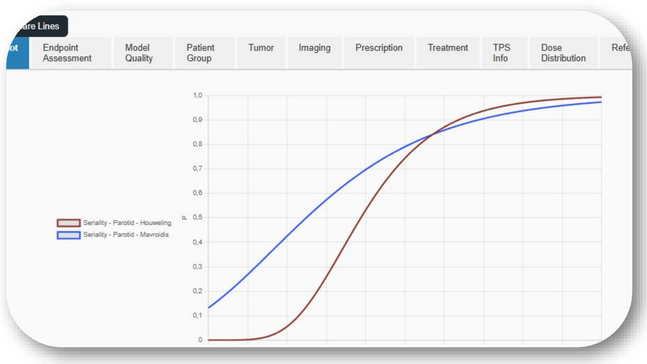
LASIGE health and biomedical informatics

Motivation

- Radiobiological modelling demands deep quality knowledge, facilitated by an available online library;
- Online library = Manual, time-consuming, and error-prone data collection process currently used in the 4Radiobiology.

Problems

- Manual and time-consuming data collection;
- Heterogeneity and lack of standardization in published studies and key parameters;
- Growing volume of dose-response models.



Methodology

Articles and Data Extraction

Article Classification Pipeline → Relevant RT Articles → Information Extraction Pipeline

1. Text Normalization
2. Named Entity Recognition (NER)
3. Table Extraction
4. Combinations Relation Extraction
5. Structured Output

Data Storage Update

Store Parameters Extracted in DB → Update 4Radiobiology website

4Radiobiology Optimization

Optimization of 4Radiobiology website

Solution

Automated, AI-driven pipeline for article classification and data extraction integrated in 4Radiobiology.

Radiobiological Articles → **pdf file** → **Text Conversion** PyPDF2 + OCR + PDFplumber → **txt file** → **Extraction Combinations** BioBERT(QA) + LLaMA + Regex → **csv file** → **Parameter Extraction Pipeline** Regex + BioBERT + LLaMa → **csv file** → **Normalization + Deduplication** → **Evaluation** Pipeline vs. manually extracted values → **Results so far: 81.30% Efficacy**

Conclusion

Manual radiobiology collection is no longer sustainable → Deliver more reliable, consistent data in 4Radiobiology Platform.

Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI: https://doi.org/10.54499/UID/00408/2025.

Learning from True Negatives in Scientific Open-World Graphs

Laura Balbi, Cátia Pesquita

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

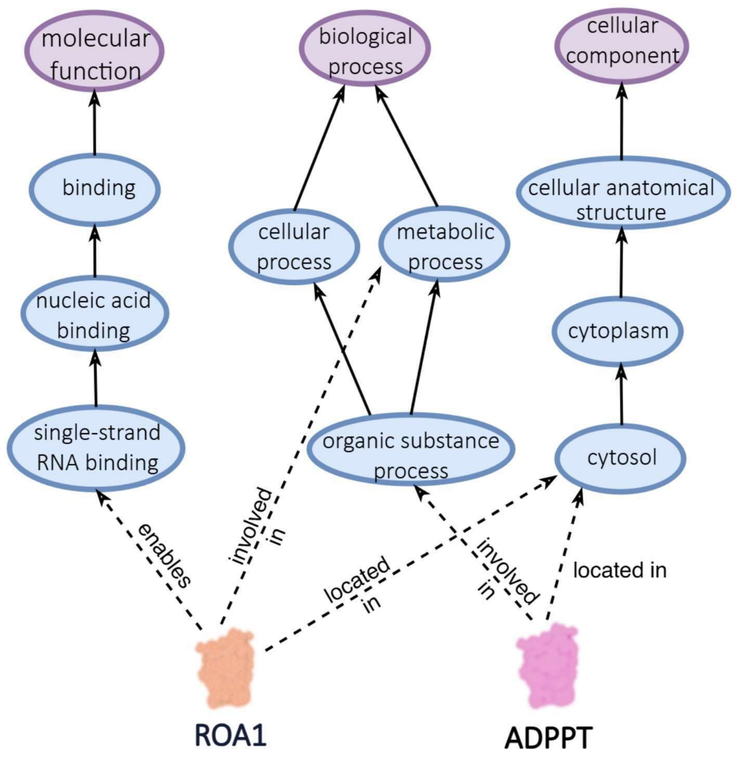
Scan here!

LASIGE health and biomedical informatics

Background

Knowledge Graphs (KGs) capture relational facts under an open-world assumption, where missing statements reflect incompleteness rather than falsehood. Still, KGs are dominated by positives and often omit negative knowledge (verified false statements).

KG representation learning often uses training negative generation strategies that assume a closed world – absence equals falsehood.



Example biomedical KG with protein annotations to Gene Ontology classes.

Motivation

Negation can be a meaningful supervision signal: to learn what an entity is and what it is not.

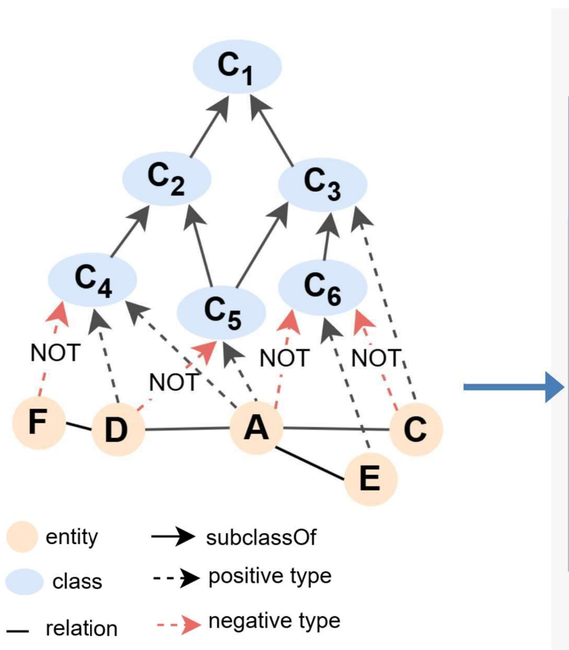
Sampling negatives from missing statements can create false negatives, semantic noise during training – and weaken the notion of negation.

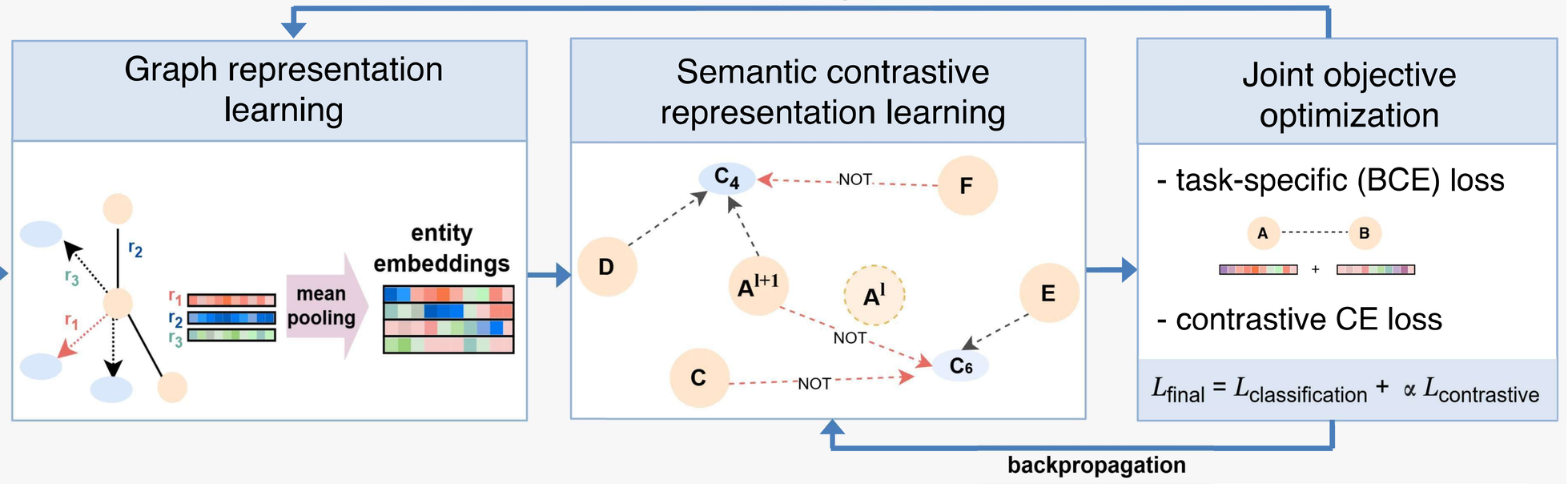
Objective

Investigate the impact of including and explicitly learning from verified negative evidence on KGRL.

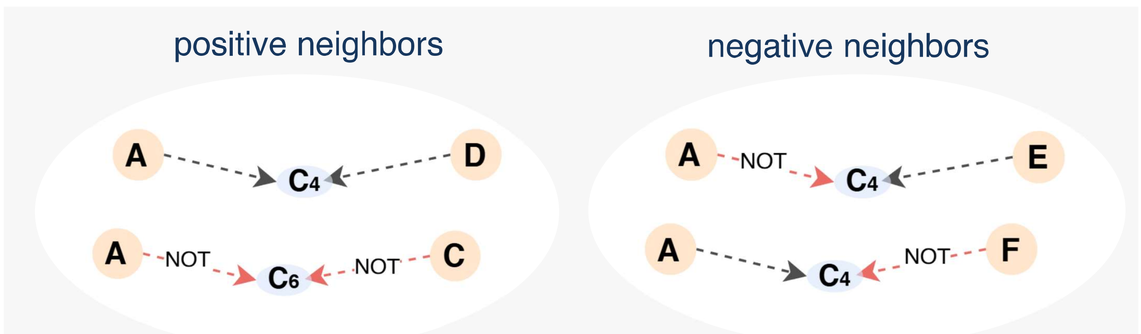
Method

NegK-GNN learns entity representations by explicitly using verified negative statements as training negatives and supervision signals, through a joint objective that combines a task-specific loss with a semantic contrastive loss.



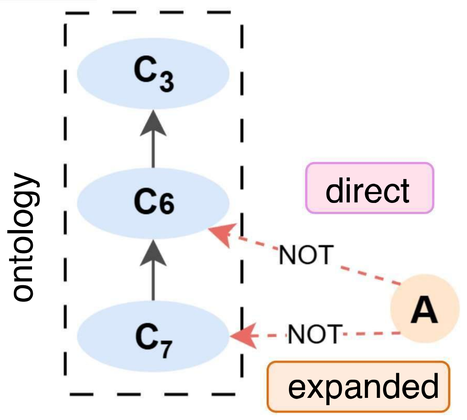


Contrastive sampling



Ontology-guided expansion

- negative statement expansion
- subclass reasoning – negation propagates downward.



Evaluation

Benchmark on Human protein–protein interaction and Gene Ontology functional annotations

Instances	12,880
Classification Triples	107,649
Positive Statements	74,901
Negative Statements	79,324

Task: Triple classification — to predict whether a candidate triple is true/false.

Results

Table 1. Mean results (%) for 10 runs.

Data	Model	Accuracy	F1	Prec (+)	Rec (+)	Roc Auc
Human	GCN	47.14	51.80	49.81	40.69	69.32
	Refined-GAE	48.76	47.84	48.37	35.84	47.29
	Signed GNN	58.90	55.26	66.43	43.38	**70.11**
	RGCN	56.38	52.82	63.72	29.14	65.50
	NegK-GCN	**61.77**	**61.06**	66.34	**48.54**	67.75
	NegK-RGCN	59.52	56.75	**67.95**	34.98	68.01

Acknowledgements: This work was partially supported by the CancerScan project through grant agreement No. 101186829, through the KAtY project through grant agreement No. 101017453, and was supported by FCT through research fellowship 2024.01208.BD, and the LASIGE Research Unit, UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Knowledge in Conflict: Towards Learning with Contradictions using Large Language Models

Leandro Duarte, Laura Balbi, Cátia Pesquita

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

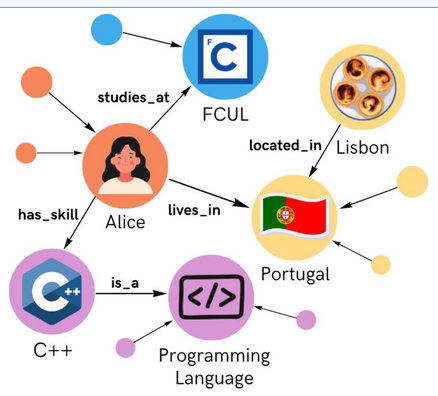
Motivation

Real-world knowledge is inherently contradictory, shaped by temporal change, contextual differences, and competing perspectives.

While *Large Language Models* demonstrate strong reasoning abilities, they often *behave inconsistently* when faced with *conflicting information*, especially when contradictions are *implicit* and require background knowledge to be identified.

Knowledge Graphs provide a *structured and semantic* representation of knowledge that makes these contradictions *explicit* through relations, hierarchies, and logical constraints.

By leveraging Knowledge Graphs as a *source of real-world contradictions*, we can systematically *evaluate how LLMs reason over conflicting information* beyond *synthetic* or purely textual *benchmarks*.



Objectives

1. Build a *benchmark* composed of sets of *explicit and implicit* contradiction datasets that cover two distinct domains (common-knowledge and biomedical).
2. *Test* a number of Large Language Models on their *zero-shot skills* on a contradiction detection task.

Taxonomy of Contradictions

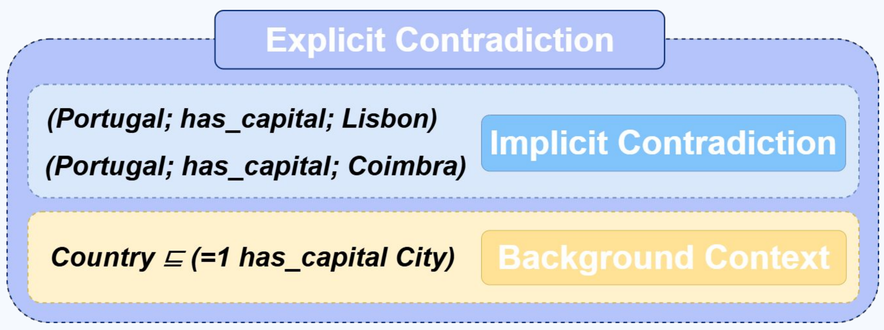
Contradictions primarily divide in *explicit and implicit.*

Explicit

Composed of a set of *contradictory statements* and a *background context statement* necessary to identify the contradiction.

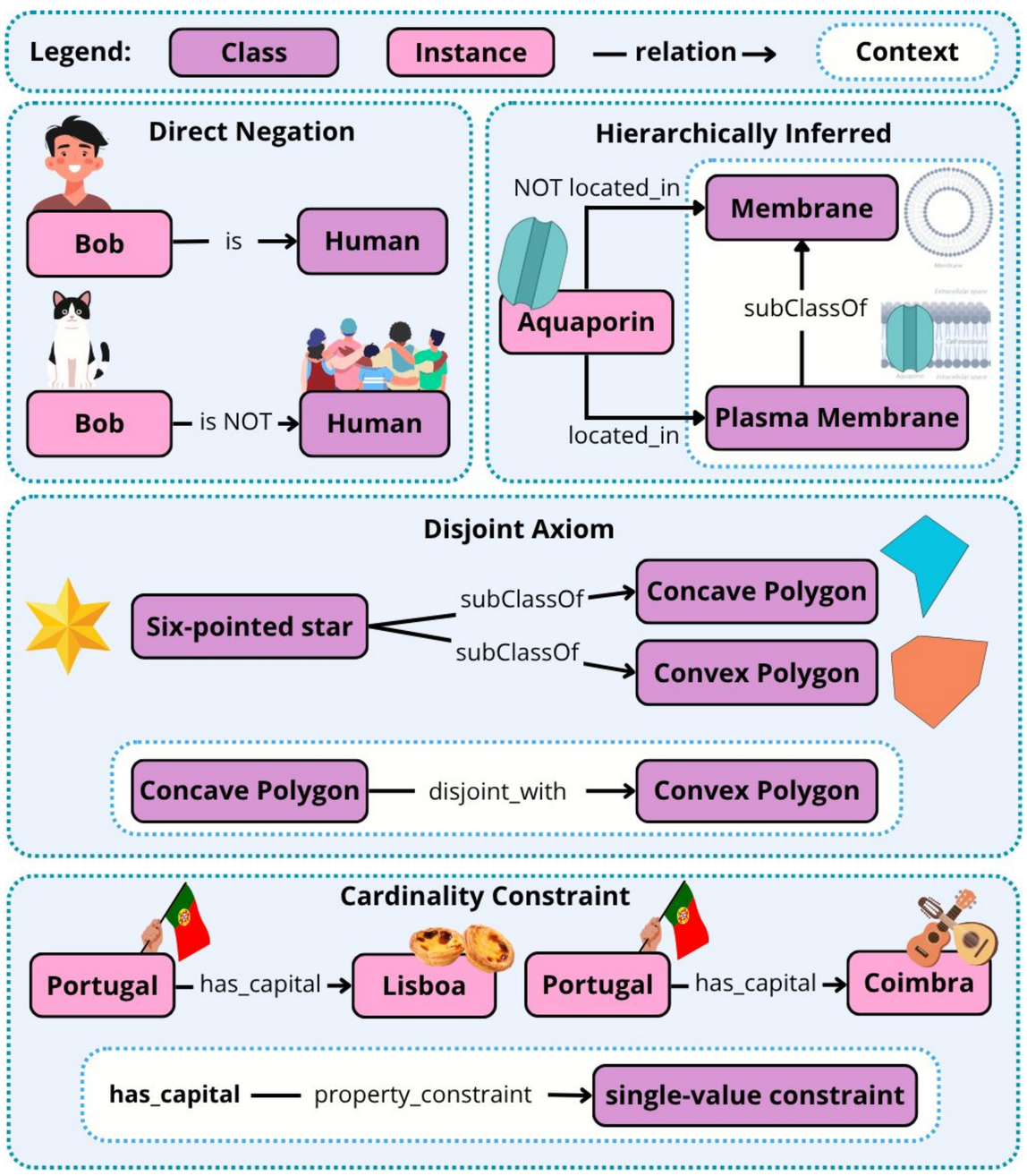
Implicit

Composed *only* of a *set of contradictory statements* that *can't be identified* without the necessary background knowledge.

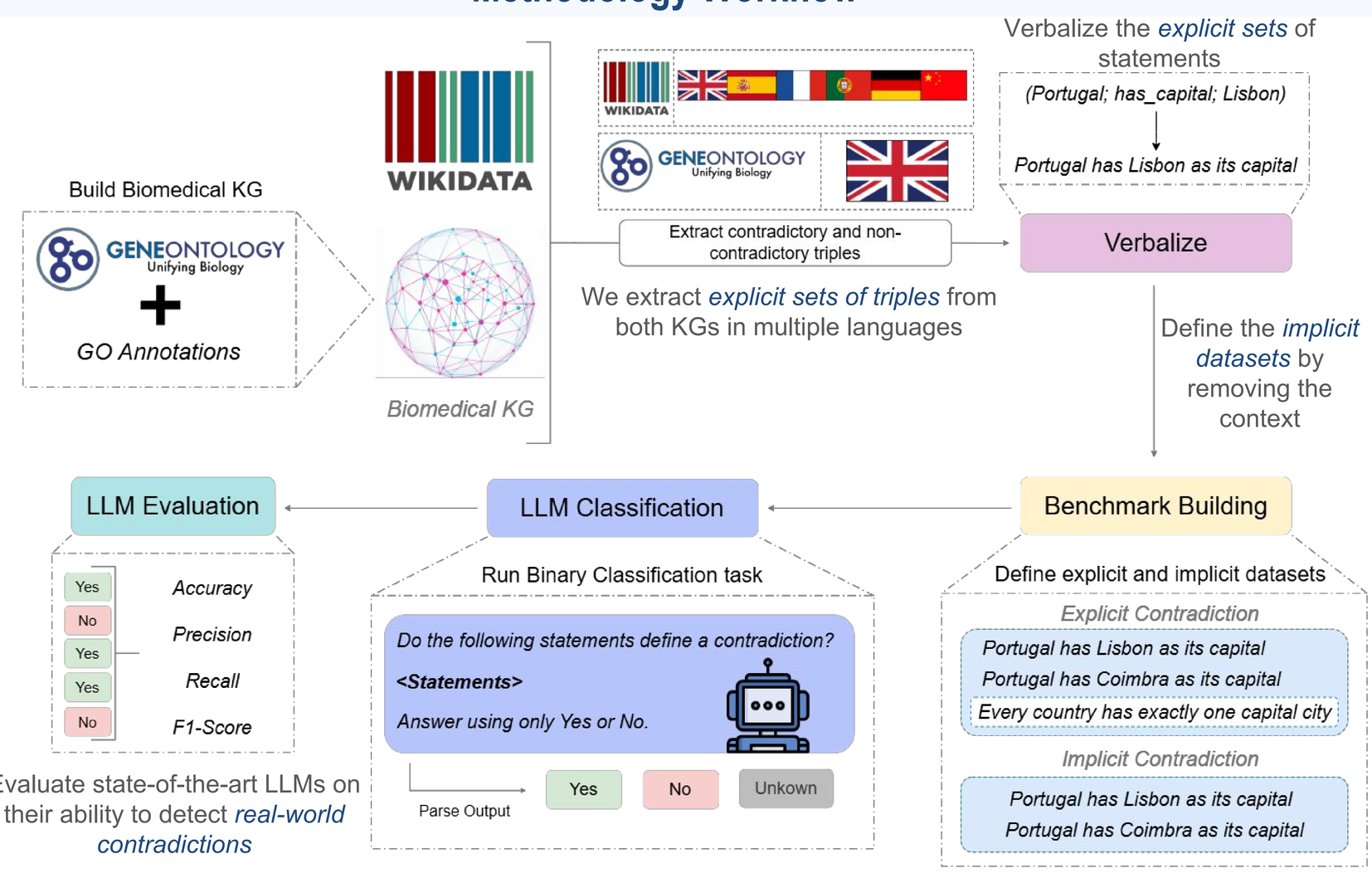


Types of Contradictions in Knowledge Graphs

We study four different types of contradictions defining their *explicit and implicit versions* when applied



Methodology Workflow



Primary Results and Future Work

The following results represent the number of contradictions, both in their *explicit and implicit versions*, that we found in each domain.

Domain	Contradiction Type	Count
Biomedical	Direct Logical	6.574
	Hierarchically Inferred	2.882
Common Knowledge	Disjoint Axiom (Class)	482.274
	Disjoint Axiom (Instance)	590.626
	Cardinality Constraint	2.346.706

Future work focuses on:

1. Defining relevant *state-of-the-art LLMs* for classification.
2. *Evaluating state-of-the-art LLMs* and discuss the obtained results.
3. Study whether *AI reasoning* differs from *Human reasoning*.

Acknowledgements: This work was supported by the LASIGE Research Unit, ref. UID/00408/2025. It was also partially supported by the CancerScan project which received funding from the European Union's Horizon Europe Research and Innovation Action (EIC Pathfinder Open) under grant agreement No. 101186829. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Innovation Council and SMEs Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

Predictive Models of Biological Control Agents for the Inhibition of Plant Pathogens

Leonor Fandinga[1], Ricardo Ramiro[2], Manisha Sirsat[2], Márcia Barros[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] InnovPlantProtect, Elvas, Portugal

LASIGE health and biomedical informatics

MOTIVATION

- Climate change influences the evolution and emergence of new plant pathogens
- Plant diseases cause significant crop losses and threaten food security
- Sustainable solutions are needed to reduce reliance on chemical pesticides

Chemical Pesticides

- Negative impact on environmental and human health

VS

Biological Control Agents (BCAs)

- Living organisms that suppress pathogens
- Sustainable alternatives

PROBLEM

- BCAs are effective but hard to identify
- Current development is costly and time-consuming
- Farmers struggle to identify effective products
- Researchers lack efficient data-mining tools to find effective BCAs

OBJECTIVE

- Developing a pipeline that uses machine learning to identify candidate BCAs from plant-associated microbiome data

METHODOLOGY

- Microbiome data from MGnify was processed into OTU tables filtered by kingdom, prevalence, and variance, then normalized (relative abundance) and CLR-transformed.
- Candidate neighbors were identified using kNN with cosine similarity via random projections, and interactions were inferred using LASSO, followed by bootstrap stability selection to retain robust signals.
- The resulting interactions were exported as a labeled dataset, used to train predictive models (RF, SVM, GNNs) for inhibitory relationships, and ultimately integrated into a Knowledge Graph (KG).

1 Data Extraction & Preparation

MGnify

- Pipeline version 5.0
- Plant Biome
- SSU Amplicon sequences

CSV

- Incomplete taxonomy
- Incomplete counts
- Missing OTU_id

- Keep only Bacteria
- Prevalence filtering
- Relative Abundance
- CLR transformation

Samples x OTUs

Microbiome Abundances

2 Interaction Inference

Per-OTU LASSO Regression (with coefficient thresholding)

Embedding-based k-Nearest Neighbor (k-NN) Candidate Selection

Bootstrap Stability Selection

Signed Edge dataset LASSO

Signed Edges: Positive (Agonism), Negative (Antagonism)

3 Model Training

Train Predictive Classification Models

- Random Forest (RF)
- Support Vector Machines (SVM)
- Graph Neural Networks (GNNs)

4 Validation

Predict & Generalize Inhibitory Relationships for New Pathogen-Microbe Pairs

Model Performance → F_1-Score, Recall, Precision

Model Result Validation → Compare to results obtained in a parallel MSc thesis project and in papers that predict and validate candidate BCAs

List of Predicted Inhibitions (example)

< Pathogen X, Microbe Y, Interaction >
< Pathogen X, Microbe W, Interaction >
< Pathogen X, Microbe Z, Interaction >
< …>

Recommender System integration

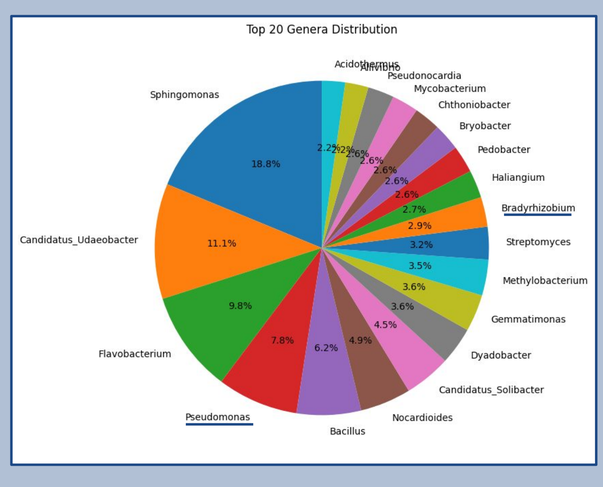
RESULTS

DATASET

20074730 RAW SAMPLES (Mgnify) → 311001 SAMPLES AFTER CLEANING (BACTERIA ONLY)

1887 UNIQUE SAMPLES

1246 UNIQUE OTU's




```
otu_i,otu_j,weight,freq,freq_pos,freq_neg,sign
OTU_0,OTU_2479,0.06525726,0.88,0.88,0.0,positive
OTU_0,OTU_1,0.09261859,0.92,0.92,0.0,positive
OTU_0,OTU_26,0.12757155,0.93,0.93,0.0,positive
OTU_0,OTU_2449,0.12342822,0.93,0.93,0.0,positive
OTU_0,OTU_9,0.24575149,0.97,0.97,0.0,positive
OTU_0,OTU_2498,0.3193416,0.98,0.98,0.0,positive
OTU_0,OTU_139,0.08632342,0.85,0.85,0.0,positive
OTU_0,OTU_2451,0.2819724,0.98,0.98,0.0,positive
OTU_1,OTU_2325,0.18602887,0.81,0.81,0.0,positive
OTU_1,OTU_139,0.46495235,1.0,1.0,0.0,positive
OTU_1,OTU_316,0.11005786,0.91,0.91,0.0,positive
OTU_1,OTU_2015,0.18161373,0.8,0.8,0.0,positive
OTU_1,OTU_689,0.18440732,0.85,0.85,0.0,positive
OTU_1,OTU_31,0.17627601,0.89,0.89,0.0,positive
OTU_3,OTU_13,0.1693638,1.0,1.0,0.0,positive
OTU_3,OTU_15,0.35618702,1.0,1.0,0.0,positive
OTU_3,OTU_23,0.48529285,1.0,1.0,0.0,positive
OTU_3,OTU_2478,-0.12255663,0.97,0.0,0.97,negative
OTU_3,OTU_73,0.17235158,1.0,1.0,0.0,positive
OTU_3,OTU_2444,0.14866324,0.98,0.98,0.0,positive
OTU_3,OTU_2443,0.15326364,0.97,0.97,0.0,positive
OTU_3,OTU_2055,0.1544678,0.85,0.85,0.0,positive
OTU_3,OTU_177,0.08321554,0.93,0.93,0.0,positive
OTU_3,OTU_632,0.11968076,0.98,0.98,0.0,positive
OTU_3,OTU_11,0.06022756,0.88,0.88,0.0,positive
OTU_3,OTU_2445,-0.093390316,0.97,0.0,0.97,negative
OTU_6,OTU_148,0.14380229,0.8,0.8,0.0,positive
OTU_9,OTU_41,0.14031038,0.93,0.93,0.0,positive
```

Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the project Recommender system for biological control agents against plant pathogens, ref. 2023.14580.PEX, DOI https://doi.org/10.54499/2023.14580.PEX, and under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025

ADDEx: Expert-Centered Explainable AI tool for Drug Development

Mariana Gouveia, Susana Nunes, Cátia Pesquita

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

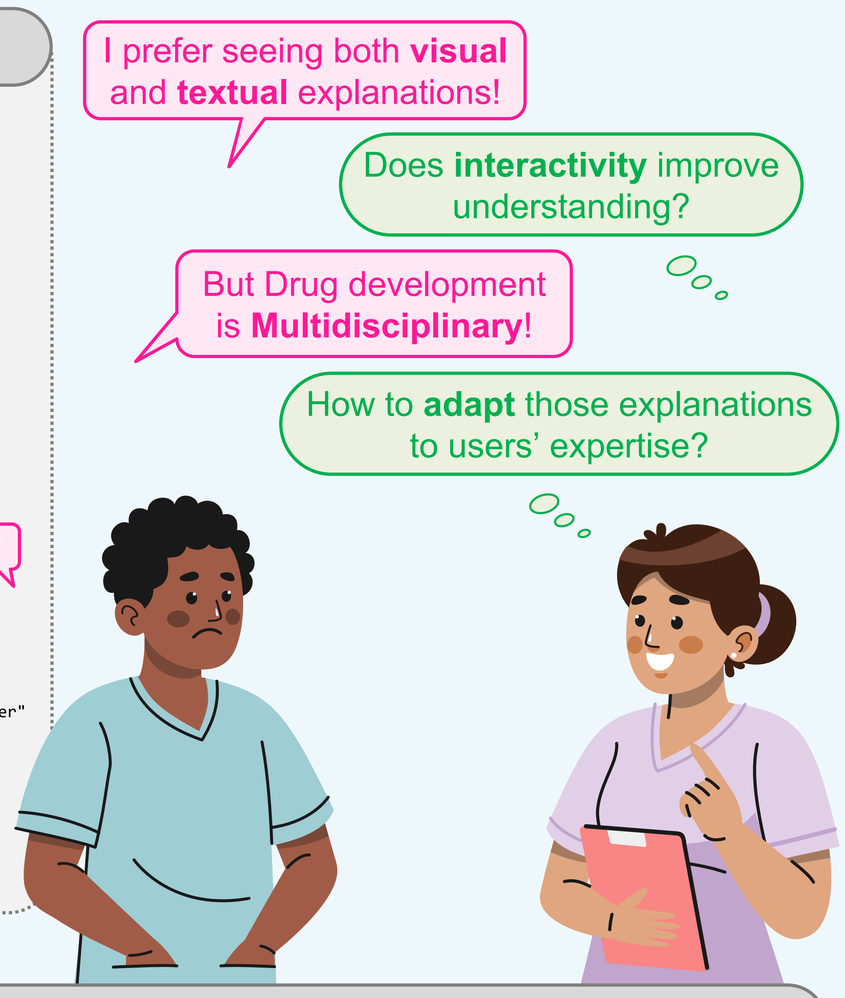
LASIGE health and biomedical informatics

Motivation

Drug Development	AI accelerates discovery	KG-based XAI adds transparency
✗ **10-15 years** per approved drug.	✓ **Identify promising drug candidates.**	✓ Mechanistic paths **explain predictions**.
✗ **$1–2 billion** per approved drug	✓ Enables **Drug repurposing.**	✓ Grounds predictions in **scientific Knowledge**.
✗ **< 90%** clinical trial failure rate.	✓ Predict **Drug-target interactions**.	✓ Supports expert **hypothesis validation**.
✗ **Most compounds fail before clinical validation.**	✗ Operates as a "**black box**", no mechanistic justification for experts.	✗ Static, non-personalized, hard to explore and interpret by domain experts.

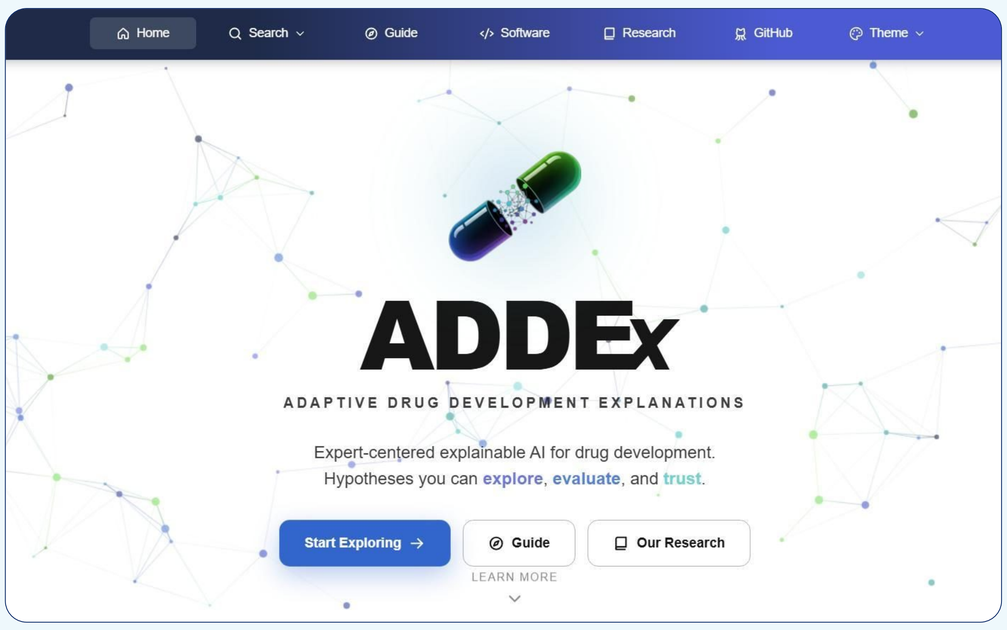
XAI Output


```
{
"id": "Compound::DB00175 - Disease::DOID:1936",
   "verbalization": "…may contribute to Atherosclerosis…",
   "paths": [{
        "score": {
                "ic_mean": 0.5756,
                "final_score": 0.6211,
                "agentic_score": 0.6667},
        "nodes": [
                {"id": "Pravastatin", "type": "Compound"},
                {"..."}],
        "edges": [
                {"source": "Pravastatin",
                "target": "Coronary Artery Disease",
                "label": "treats"}, {"..."}],
        "lowest_common_ancestors": {
                "Coronary Artery Disease, Atherosclerosis": [
                        "Non-Neoplastic Cardiovascular Disorder"
                ]},
        "id": "path_174"},
        {"..."}
   ]
}
```

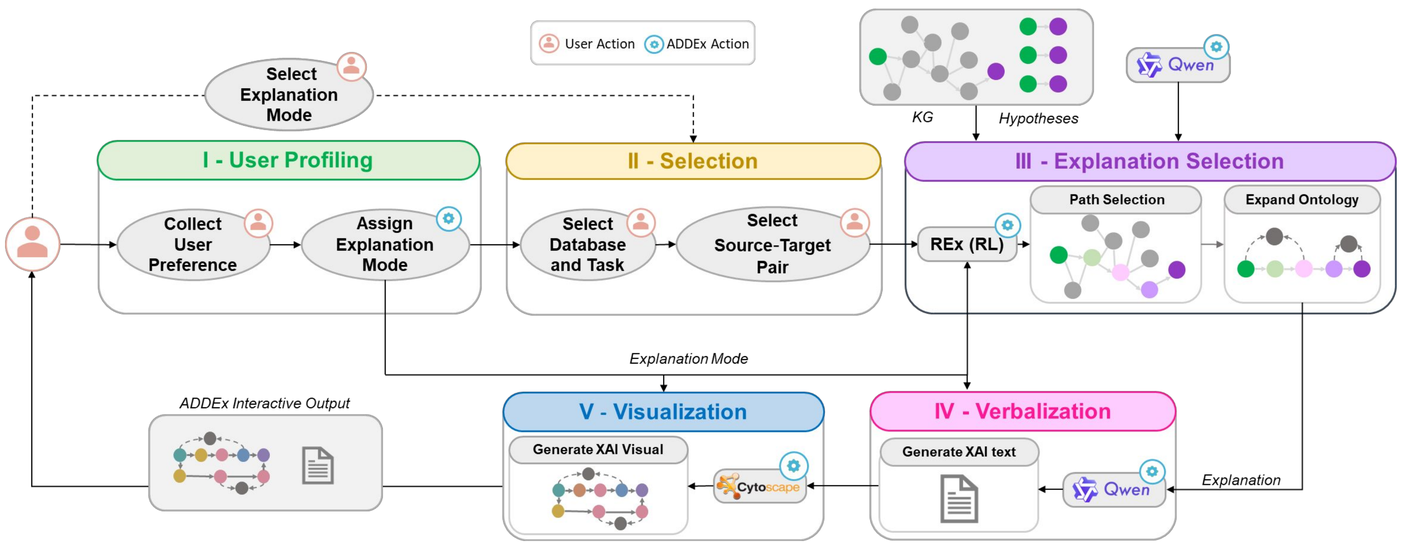



Proposal

Develop **ADDEx**, a personalized web platform that extends a pre-existing RL model (Reflexive REx) to deliver interactive, expertise tailored explanations through **Graph Visualization and Natural Language Verbalization**.



ADDEx workflow



ADDEx Functionalities

- **Adaptive Explanation Modes:** Persona-conditioned path selection and natural language verbalization.
- **Multi-KG & Multi-Task Coverage:** Support for drug repurposing and drug–target interaction tasks across Hetionet, PrimeKG, and OREGANO.
- **Interactive Graph Visualization:** Path highlighting, alternative explanation toggling, drag-based layouts, design customization, and ontological enrichment (LCA).
- **On-Demand Verbalization:** Natural language summaries dynamically updated according to user-selected paths.
- **Deep-Link Node Access:** Direct integration with DrugBank, NCBI Gene, UniProt, MONDO, ChEMBL, and PharmGKB.
- **Flexible Export:** Independent or combined export of explanations in TXT, SVG, PNG, JPEG, and PDF formats.
- **Result Caching:** Previously computed queries retrieved instantly from disk without recomputation.
- **Quick Search:** Manual query configuration with curated one-click example pairs.
- **Guide:** Step-by-step workflow guidance and FAQ integration.
- **Interface Personalization:** Light, dark, and high-contrast themes, with customizable focus and zoom components…

Future Work: User Studies, ADDEx Improvements, and Platform Hosting

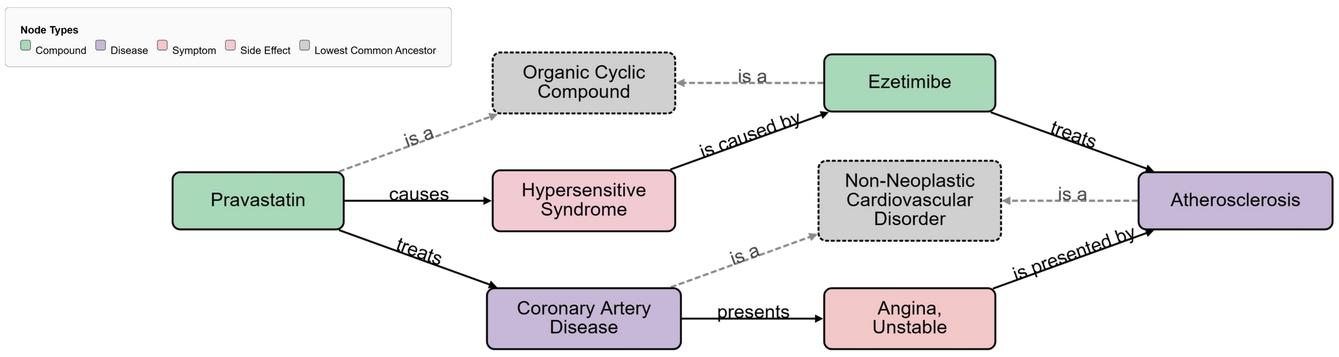


Image generated using ADDEx.

Acknowledgements: It was also partially supported by the CancerScan project which received funding from the European Union's Horizon Europe Research and Innovation Action (EIC Pathfinder Open) under grant agreement No. 101186829. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Innovation Council and SMEs Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.
LASIGE Research Unit, ref. UID/00408/2025

Context-Aware Parkinson's Disease Monitoring

Mariana Tomás[1], Diogo Branco[1], André Santos[1], João Ferreira[1], Filipa Pona-Ferreira[2], Joaquim Ferreira[2], Tiago Guerreiro[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] CNS – Campus Neurológico, Lisboa, Portugal

LASIGE health and biomedical informatics

Background

- Parkinson's disease (PD) is a progressive, chronic neurological disorder and the second most common neurodegenerative condition.
- Continuous monitoring in free-living environments is essential to understand daily symptom fluctuations.
- Current monitoring lacks real-time context. Capturing physical environment and social density is crucial to bridge the gap between clinical assessments and daily-life data.

Objectives and hypothesis

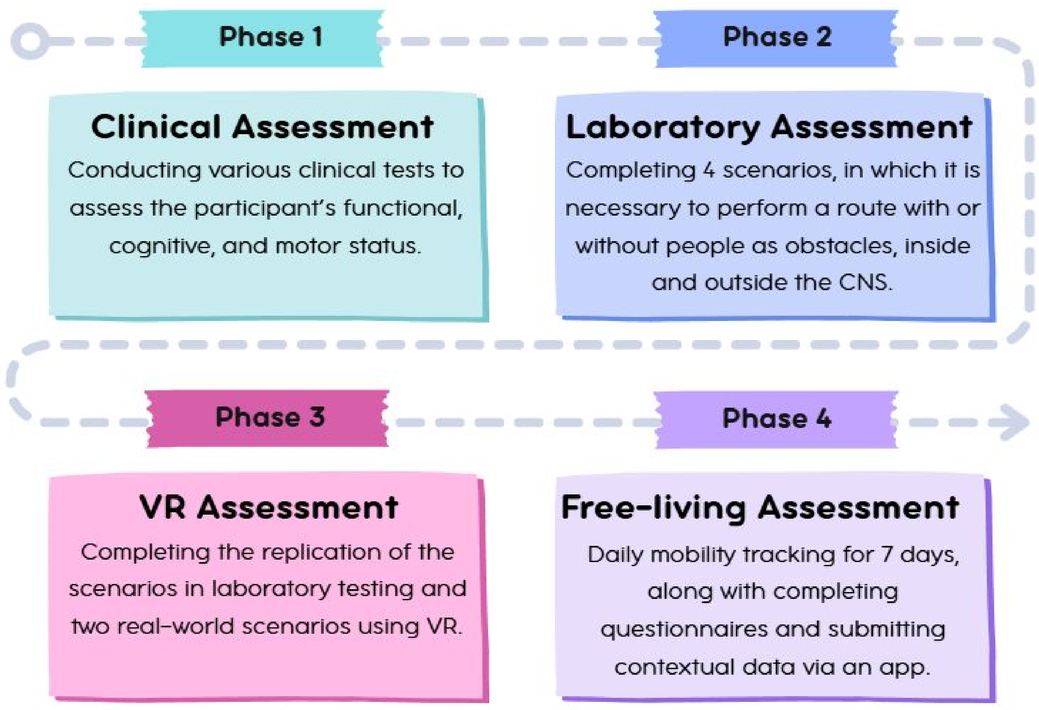
- The main objective is to establish whether integrating contextual data improves the clinical interpretation and relevance of gait data collected from real-world conditions.
- We hypothesize that complex environments, characterized by high social density or irregular flooring, induce greater variability in stride time than simple environments.

Research Questions

1. To what extent does integrating contextual information with free-living sensor data improve its interpretability and alignment with clinical assessments in Parkinson's disease?
2. How do different physical environments influence motor activity of people with Parkinson's disease in free-living conditions?
3. How does social density affect motor performance and mobility patterns in free-living contexts?
4. To what extent do VR-simulated everyday environments capture the same context-dependent changes in motor activity observed in real-world free-living settings?

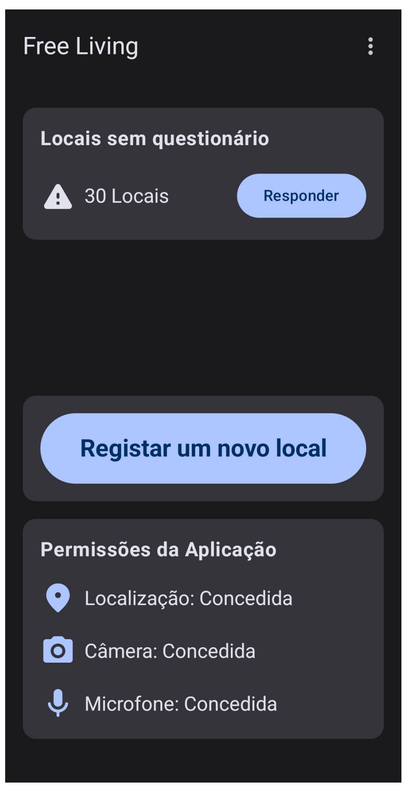
User study

Twenty patients diagnosed with idiopathic PD will be recruited from the Campus Neurológico (CNS) outpatient clinic. The following diagram explains the phases of the procedure:



A free-living app was developed for participants in the 4th phase of the study to collect contextual data, such as indoor/outdoor environments, social density, and how the person felt.

VR Assessment



Data colection and analysis

- **Data collection** – Extract raw data from the AX3/6 sensors via OmGui;
- **Processing, gait feature extraction** - Sci-Kit Digital Health package (for the lombar raw data) – and **integration of gait metrics and contextual data**;
- **Statistical Analysis** – Free-living vs. Clinical environments, VR validation, contextual data influence and contex data relation with clinical assessment.

AX3, sensor from Axivity that collects accelemeter data

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025..

CP-DA: Conformal Prediction–Guided Data Augmentation

Turning Uncertainty into Reliable Knowledge for Drug Discovery

Maryam Lotfi[1], Francisco M Couto[1], Marcia Barros[1,]

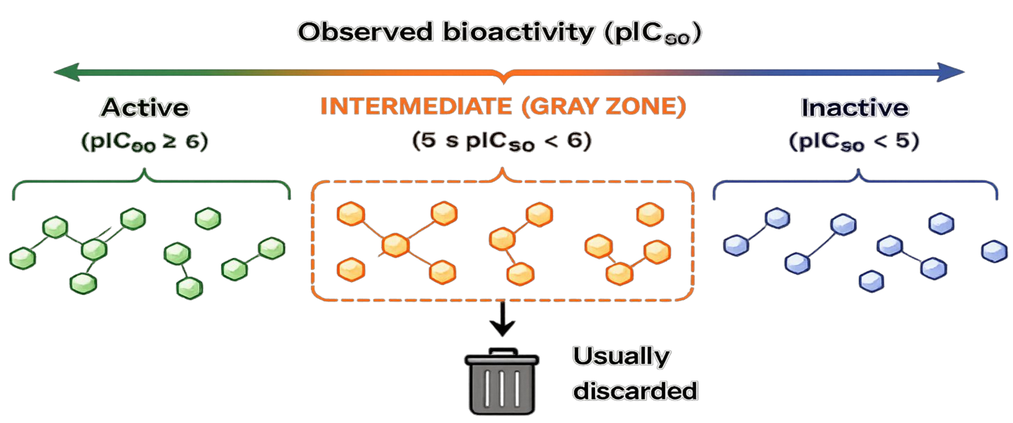
[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

health and biomedical informatics

1 THE PROBLEM

Intermediate compounds are often removed during preprocessing

- Loss of information
- Reduced diversity
- Weaker generalization
- Higher cost in data-scarce settings

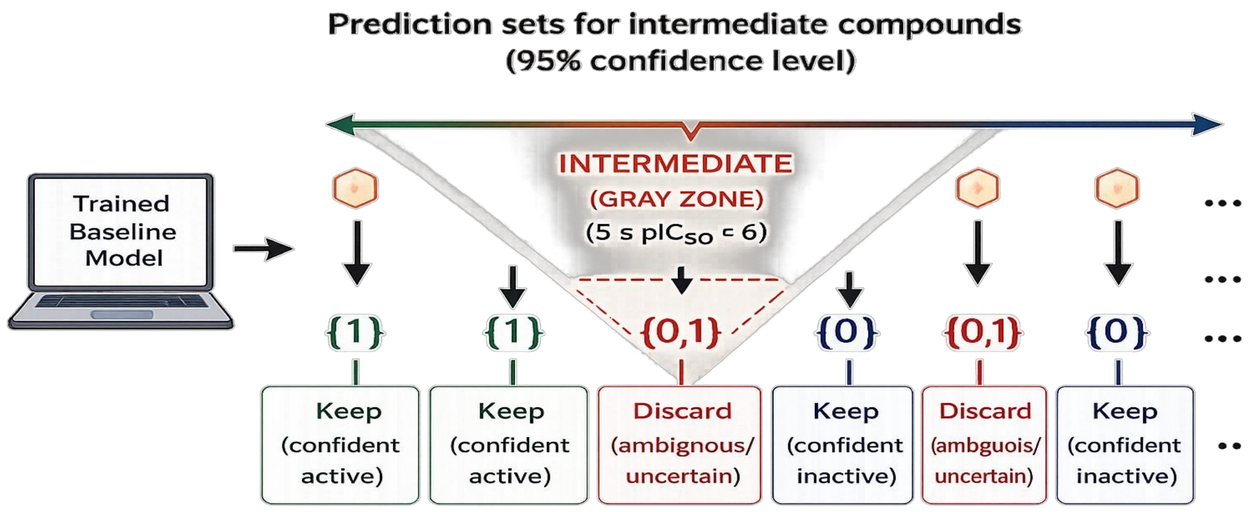


GOAL

Use conformal prediction to identify reliable intermediate compounds and improve the model.

2 MEASURE UNCERTAINTY (Conformal Prediction)

- Train a baseline model
- Apply conformal prediction to intermediate compounds (95% confidence).

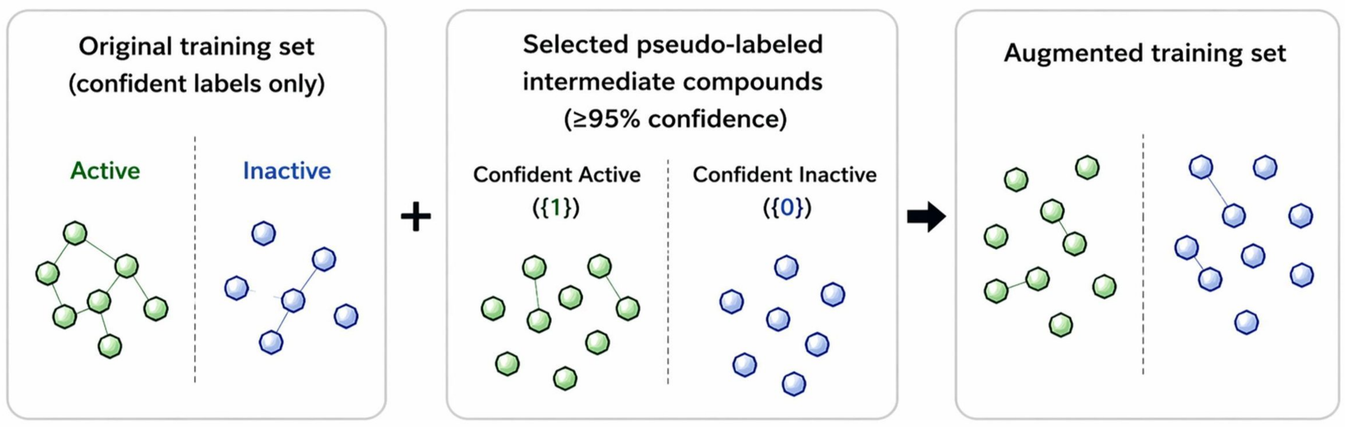


WHY CONFORMAL PREDICTION

- Provides valid confidence measures
- Works with different models
- No distributional assumptions

3 CP-DA SELECT AND AUGMENT WITH CONFIDENCE

- Keep singleton predictions ({1}, {0}) at 95% confidence
- Discard {0,1} sets

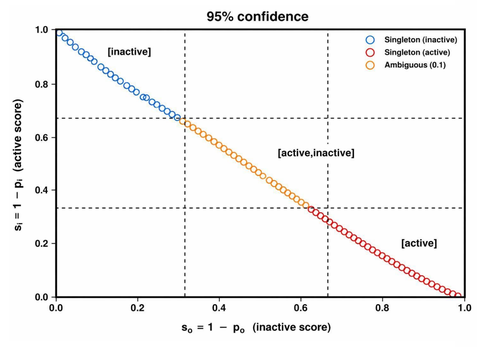


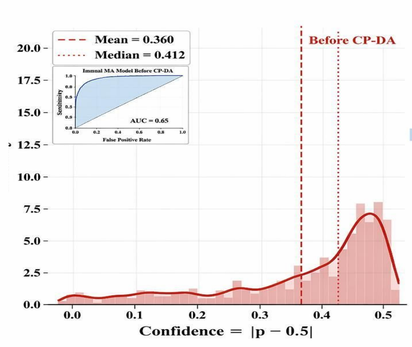
BENEFITS

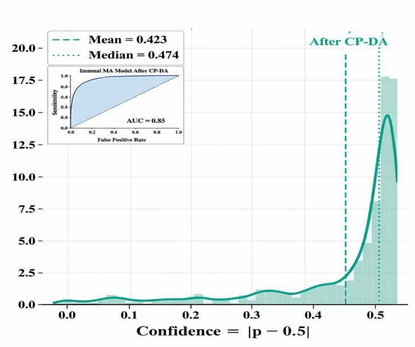
- Uses informative data
- Avoids reinforcing noise
- Maintains statistical validity
- Provides reliable pseudo-labels

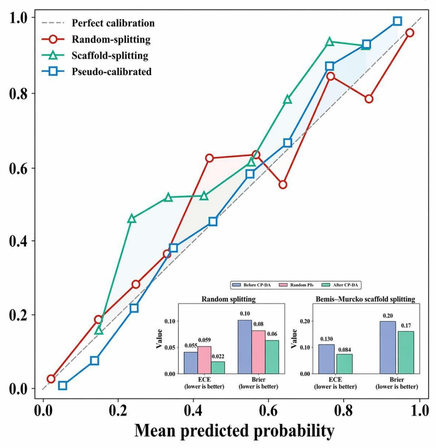
4 RESULTS: BETTER RELIABILITY WITHOUT SACRIFICING VALIDITY

- Improved performance (AUC, F1)
- Better calibration (lower ECE)
- Lower Brier score
- Higher singleton rate
- Coverage ≈ 95% (statistical validity preserved)

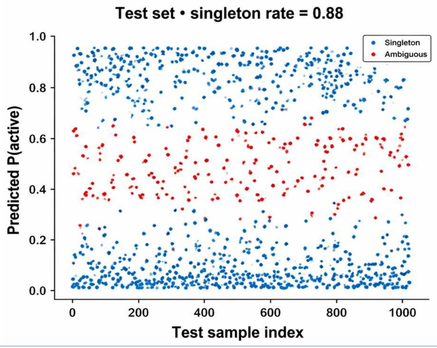




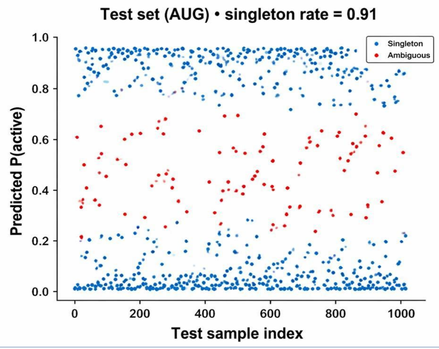




Test set • singleton rate = 0.88



Test set (AUG) • singleton rate = 0.91



Split	Setting	Accuracy	Precision	Recall	Macro-F1	AUC-ROC
Random	Before CP-DA	0.8928	0.888	0.877	0.882	0.955
Random	After CP-DA	**0.908**	**0.903**	**0.894**	**0.898**	**0.963**
Scaffold	Before CP-DA	0.699	0.784	0.677	0.658	0.829
Scaffold	After CP-DA	**0.743**	**0.806**	**0.726**	**0.719**	**0.834**
Random	Random PL	0.897	0.891	0.882	0.886	0.956

KEY MESSAGE

Uncertainty is not noise—
it is a signal.
We use it to learn better
and discover more.

CP-DA IN A NUTSHELL

Train on confident data → Measure uncertainty → Select confident → Augment → Improve reliability

IMPACT

- More reliable virtual screening
- Faster identification of repurposing candidates
- Reduce cost of false positives & false negatives
- Greater confidence in AI-assisted early-stage drug discovery

Acknowledgements: This work was supported by FCT through project <Hybrid Recommender Systems for Protein-Ligand Interaction Predictions >2023.03164.BD , and the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

U LISBOA · Ciências ULisboa · fct Fundação para a Ciência e a Tecnologia · LASIGE

Across the Universe of Autobiographical Media Memories in Space and Time

Miguel Silva, Yubing Chen, Teresa Chambel

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE health and biomedical informatics

Motivation

media: movies, music, etc.
present in our lives
as sources of information, entertainment, communication, education and culture

often part of
remarkable **memories**
link **moments**, **places** & **emotions**

enable connections of personal experiences and shared content

autobiographical
play crucial role for **identity** sense of self, **social bonding** and **continuity** across memories

Background

media & **memory**

autobiographical
collective & cultural **memory**
space, **time** & **emotions**

Interactive **storytelling visualization** & **navigation**
hypermedia & hypervideo

Related Work

media platforms
As Music Goes By
personal Information systems
social media & video storytelling
Family Stories
georeferenced 360º hypervideo
Sight Surfers

Future Work

user-centered **evaluation**

advance **link awareness** and immersive **context design**

add **user** generated **content** & **trajectories** in **memories**

consider **larger datasets** & advanced **recommendation**

Integrate **new media** types

Keeping privacy & significant ties in larger scale.

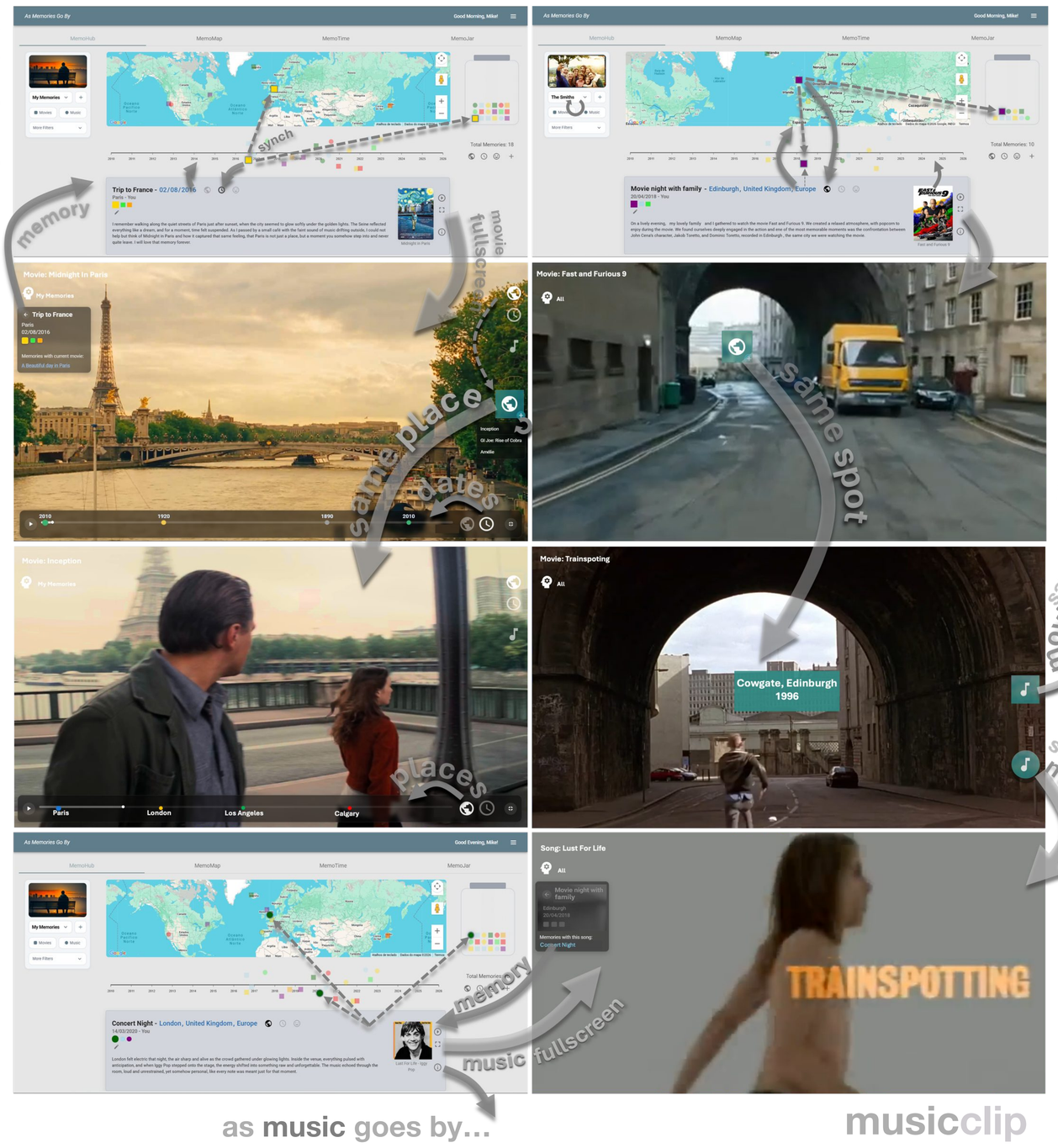
*"fundamental to experience, identify and **recall** life events"*

time . space . emotions

As Memories Go By

memohub

movies

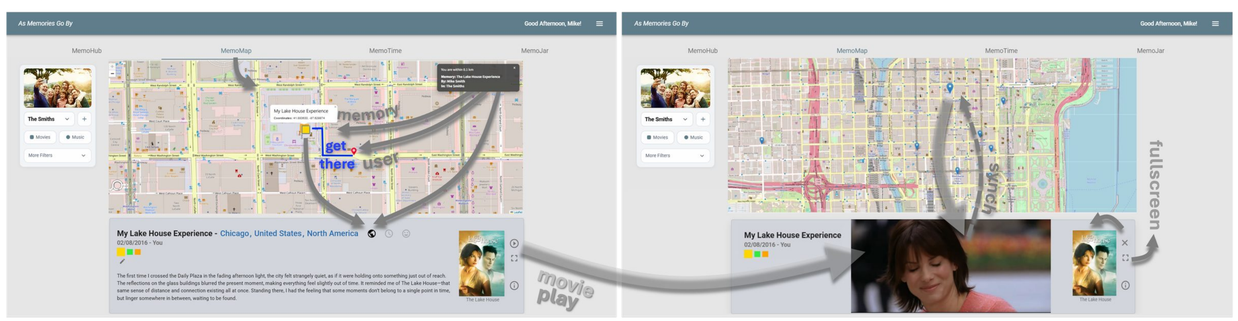


as **music** goes by...

musicclip

memory
My Lake House Experience

movie
The Lake House



notifications about **memories**: based on user **temporal** context (e.g. this day) or **location** nearby (in Fig) + **how to get there!**
+ access to memory, synchronizing view.

Media Memories

registered and **shared** in memory spaces: my, family, friends, etc.

1 Contextualization Navigation time . space . emotions

1 memo views
2 **2** media **fullscreen**
3 media **play** synch views

1. MemoHub view

context by:
filters & **synchronization** with **time**line, **map** & **emo**jar

aggregated summaries

memories' access & navigation:
text + ■ **movies** & ● **music**
shape: media type; color: main emotion

memory cards: detailed info

2. Across Media Memo

in **fullscreen**: immersive

context "In which **memories**?"
dates & places: in video **timeline**

navigation across **same music** or **time** or **place**
a) scene place or b) **specific spot**

to **another movie**
or music in **videoclip**

users **choose** which ones active

extended navigation: linking to As Music Goes By

3 3. Spatial Movie Navigation

movie **plays in synch** with map view

narrative trajectory
following the movie storyline through locations

spatial trajectory
based on proximity between locations
improved for physical navigation

Immersive meaningful connection with movie.

the paper

Acknowledgements: This work was partially supported by FCT through the AWESOME project, ref. PTDC/CCIINF/29234/2017 and the LASIGE Research Unit, ref. UID/ 00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Experiencing Movies with Emotional Awareness

Nélio Pereira, Bernardo Pereira, Teresa Chambel

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE health and biomedical informatics

Motivation

- Watching movies is a **deeply emotional experience**, yet these emotions remain shadowed;
- Current streaming platforms focus on content delivery but **do not reflect the viewer's emotional state;**
- Making **emotions visible** could enhance **awareness**, **engagement**, and **wellbeing**.

Related Work

Affective Computing

- Physiological emotional detection devices calculate and infer emotion and provide emotion feedback;

Model of Emotions

- Dimensional & Categorical;

Related applications

- Media4Wellbeing;
- Emojar.

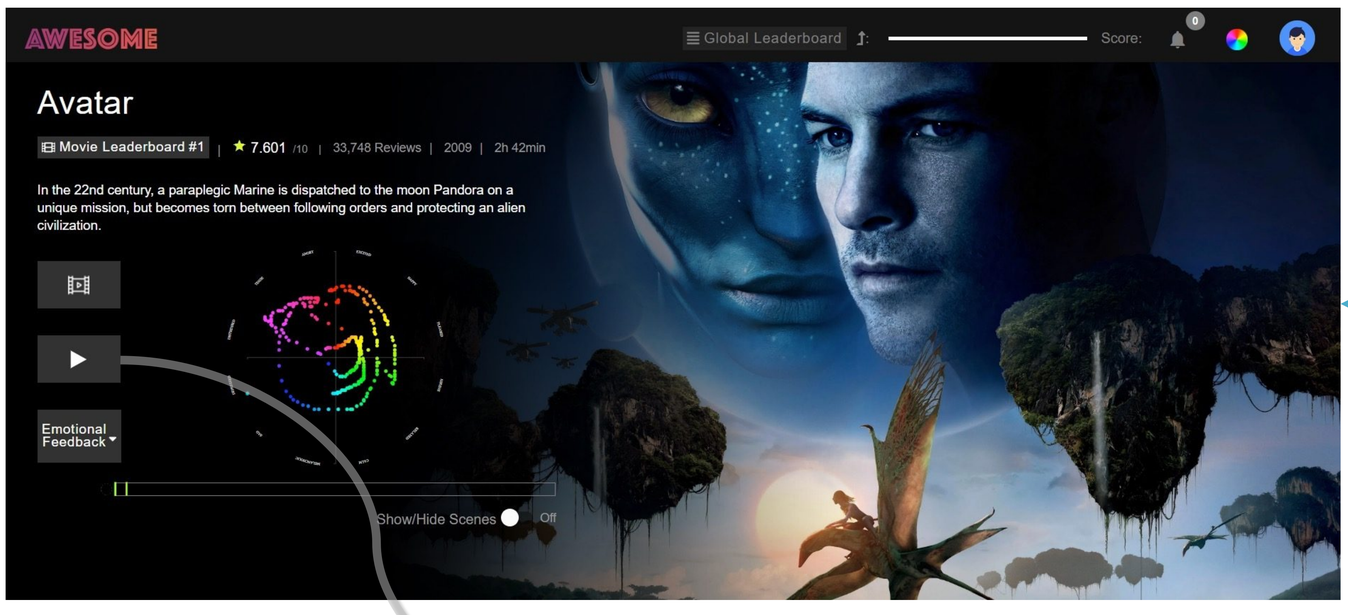
Visualization

- Shows **emotional transitions;**
- **Emotional patterns** (through Emowheel);
- Emoji overlay showing **current emotion**;
- Maps emotions to color;
- Perceived emotion through **peripheral vision**;
- Heartbeat animation;
- Zoom in/zoom out effect **based on heart rhythm;**
- **Emotional color around video** (Emobar/Emoglow);
- Simple, intuitive, glanceable;
- **Minimizes distraction**;
- Option to select emoji color (selection between emotion color or app accent color).

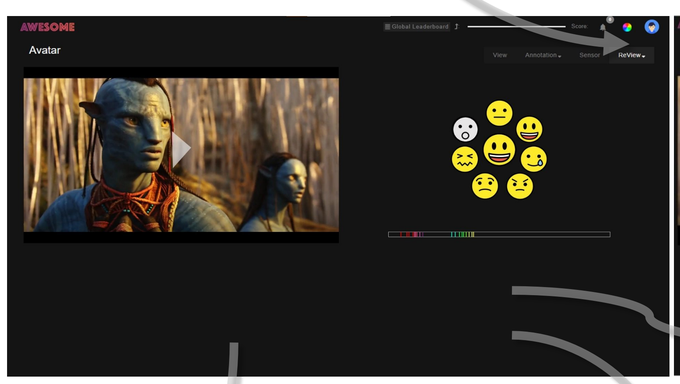
Contributions

- **Emotional Feedback**;
- Improved **customization and interaction**;
- On/Off toggles that allow **combination of widgets**;
- Detection sensors **integration** in web app.

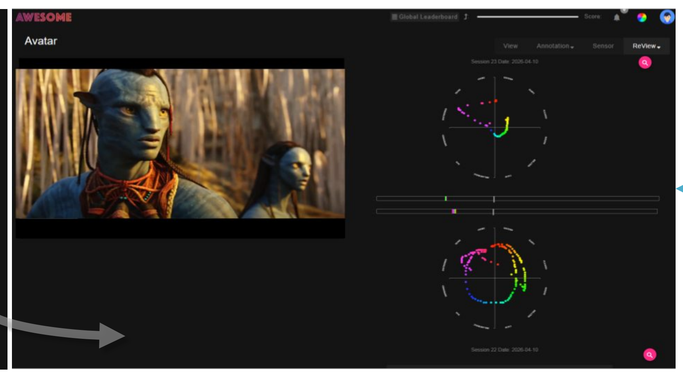
Movie Details (with emowheel from a past session)



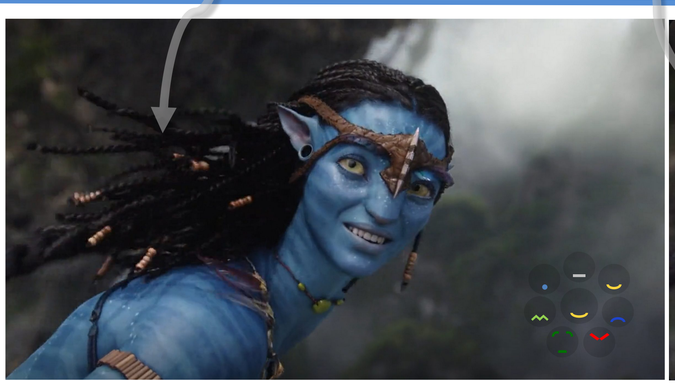
Emojiwheel (embedded)



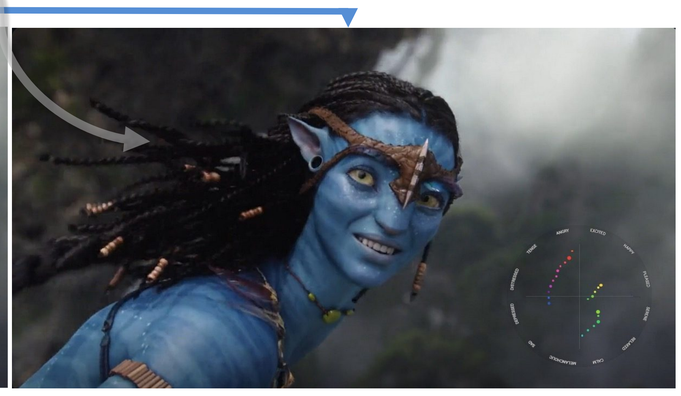
Emotional Comparison



Emojiwheel



Emowheel



Emoglow (embedded)

Emoglow

Emobar & Emoji

Heartbeat

Emotion Awareness

- **Interactive web app** designed and developed to access movies based on emotional impact;
- Focus on the **emotional representation** of movies, through physiological sensors while preserving immersion;
- Provides the ability to **watch movies and get a sense of the emotions felt** during a movie session.
- Uses webcam, MUSE Headband and Mi Band to detect the signals.

Comparison

- Timeline of emotional responses;
- Wheel of emotions depicting the emotions felt during a session;
- Comparison between two sessions;
- Allows for **reflection** after watching a movie;
- Selection between **emowheel, heartrate, emojiwheel.**

Sound & Haptic

- **Heartbeat sound** based on BPM;
- **Minimal visual attention** required;
- Wearable vibration (**heartbeat feedback**);
- Intensity **adapts** to **emotion** (inferred);
- Extends experience **beyond the screen.**

Future Work

- Methodologies to represent emotion;
- Further research on UI widget placement;
- Further research on immersion enhancement.

Acknowledgements: This work was partially supported by FCT through the AWESOME project, ref. PTDC/CCIINF/29234/2017 and the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025 .

EmOcean: Development of a mobile application to manage stress and emotions

Tiago Rodrigues[1], Ana Paula Afonso[1] , David Neto[2], Rita Sebastião[2]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] ISPA - Instituto Universitário de Ciências Psicológicas, Sociais e da Vida

LASIGE health and biomedical informatics

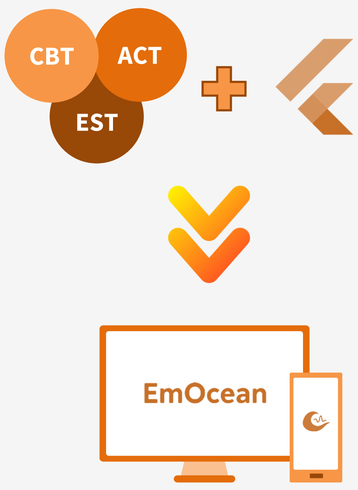
4 in 10 adults report **high stress**, impacting their mental health and relationships

MOTIVATION

- **Stress** is linked to severe health conditions, placing a huge burden on individuals and healthcare systems.
- Many existing **digital solutions** lack foundation in evidence-based psychological models, minimal cultural-linguistic adaptation, and fail to engage users long-term.

GOALS

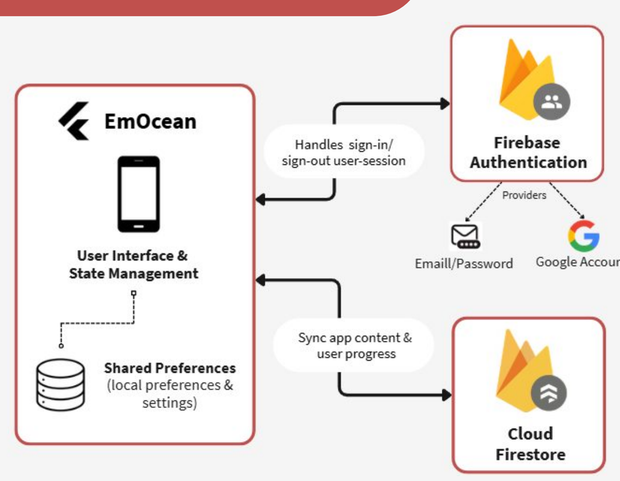
- Develop a mobile application that supports adults' stress management and emotional regulation through the guiding metaphor of navigation on an **emotional map**.
- Using a transdiagnostic approach that integrates Cognitive-Behavioral Therapy **(CBT)**, Acceptance and Commitment Therapy **(ACT)**, and Emotional Schema Therapy **(EST)** to target core psychological processes.



METHODOLOGY

Iterative & User-Centered Approach

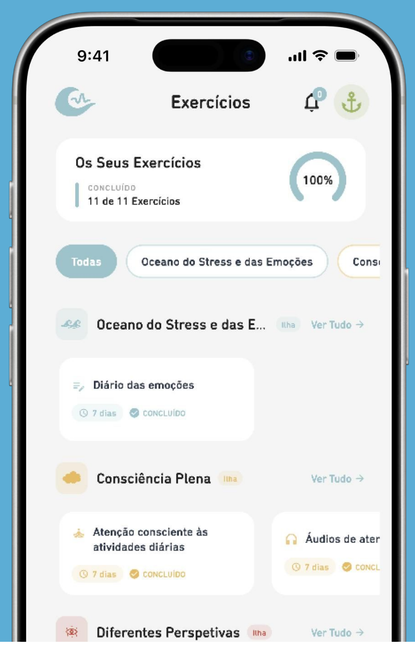
ARCHITECTURE



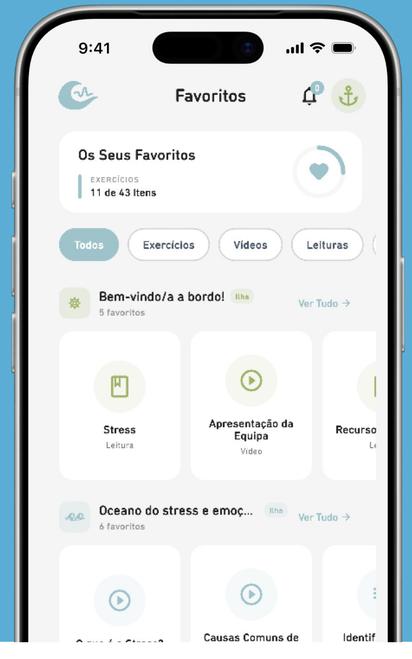
HOME

JOURNEY

EXERCISES



FAVORITES



Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE | Ispa Instituto Universitário

Semantic Similarity through Human-in-the-loop in Knowledge Graph Embeddings

Verdiana Schena[1,2]

Tutors:

Prof. Simona Colucci[1], Prof. Floriano Scioscia[1], Prof. Catia Pesquita[2]

[1] Polytechnic University of Bari, Bari, Italy

[2] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE health and biomedical informatics

Knowledge Graph Embeddings

Vectors mapping entities and relations from a Knowledge Graph (KG) into low-dimensional vector spaces, aiming to capture their semantic relationships.

- **Applications**: Link prediction, recommendation, knowledge base completion, entity similarity, and conceptual clustering;
- **Evaluation metrics**: Mean Reciprocal Rank (MRR), HITS@K;
- **Dataset**: DrugBank (focus on drug entities).

Research Question 1

Semantic Similarity in Knowledge Graph Embeddings (KGE)

Similar (drug) entities do not always correspond to similar KGEs.

Issues:

- ❑ Entity similarity definition is ambiguous and context-dependent;
- ❑ Embeddings similarity is usually assessed using numerical metrics (e.g., cosine similarity).

State of the Art findings:

- ❑ Link prediction metrics are *poor indicators* of semantic coherence;
- ❑ Entities are similar if they share predicates in the KG.

Research Question 2

Human-in-the-loop in Machine Learning

The *human-in-the-loop* approach enables human intervention within the AI's decision-making process, thereby enabling more effective execution of specific tasks.

Issues:

- ❑ Measuring and evaluating objective and subjective benefits.

State of the Art findings:

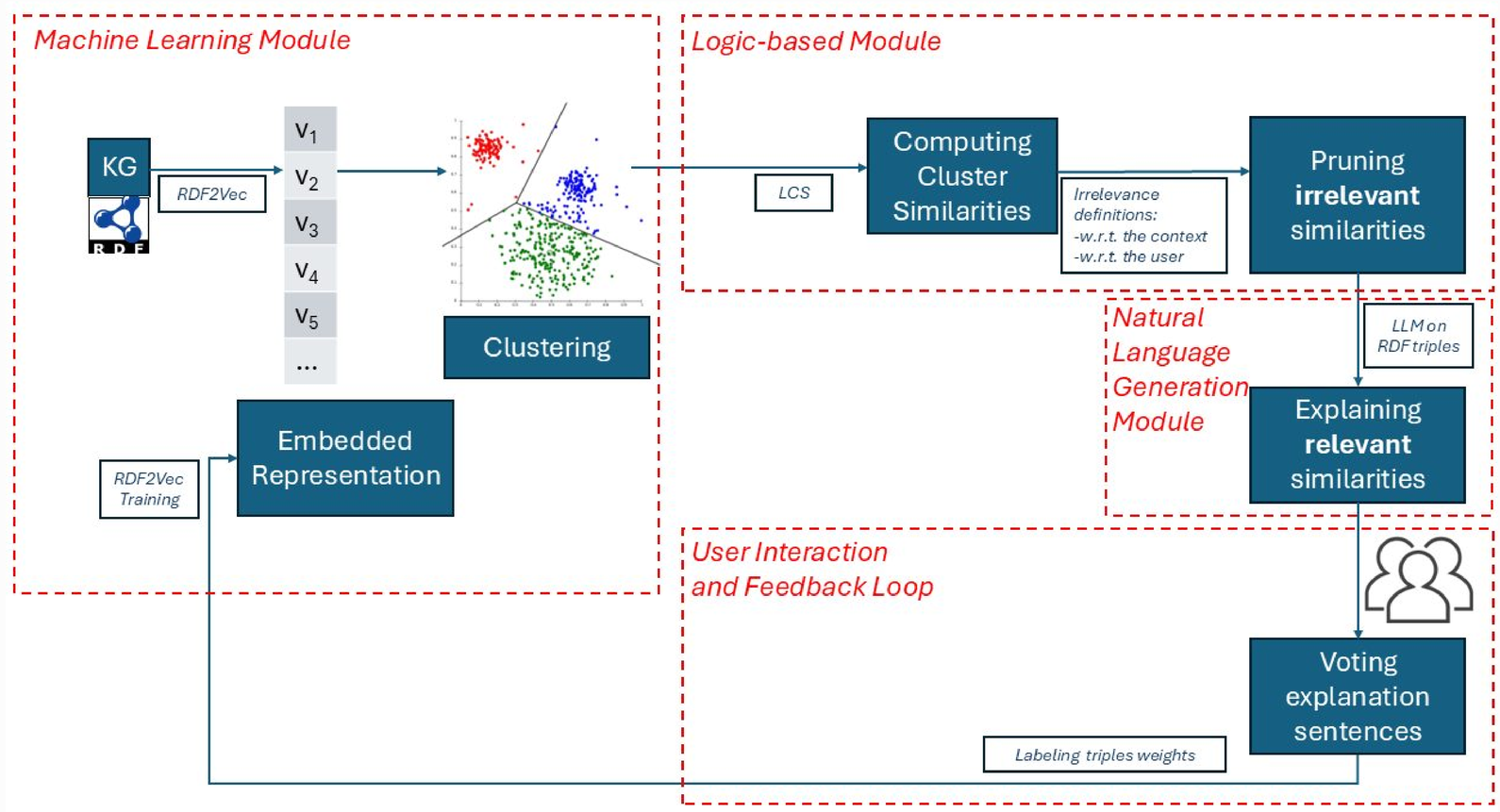
- ❑ Combined approach outperforms fully automated or manual methods.

Proposed Approach

LISE: a Logic-based Interactive Similarity Explainer

Post-hoc and method-agnostic approach that explains RDF similarity and refines KGEs for user-driven clustering.

1 **Machine Learning Module:** generates and clusters KGEs, without explaining shared entity characteristics.



2 **Logic-based Module:** generates an RDF KG (LCS) for each cluster, capturing shared features, then filters it to produce a Common Subsumer (CS).

3 **Natural Language Generation Module:** produces natural language explanations of the CSs.

4 **User Interaction and Feedback Loop Module:** collects user ratings to refine embeddings and clustering.

Ongoing Work

- ❑ Evaluation of the impact of using user ratings on embedding refinement (experiments conducted on DrugBank);
- ❑ Evaluation of the correlation between semantic similarity and geometric similarity in embeddings.

Future Work

- ❑ Refinement of embeddings using alternative models and accounting for node specificity in the knowledge graph;
- ❑ Evaluation of the possible correlation between the size of the LCS (number of shared information among resources) and geometric cohesion.

Acknowledgements:
Supported by FCT through national funds via project UID/00408/2025 (DOI: 10.54499/UID/00408/2025).
Supported by Distretto Tecnologico Aerospaziale S.c.a.r.l.
The candidate thanks Prof. Francesco M. Donini for useful discussions and collaboration on the research project.

Development and optimization of web applications to promote academic success

Alexandra Teixeira[1], Catarina Carvalho[2], Carlos Duarte[1]

1 LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
2 GAPsi, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

1. Problem

University transition demands autonomy, yet many students lack self-regulation and time management skills.

Structured schools → universities require higher autonomy and planning.

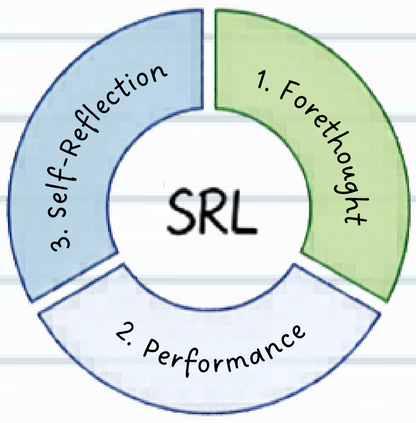
F-

Often manifested as:
→ Procrastination
→ Severe Stress & Anxiety
→ Low Motivation
→ Potential dropout

2. Theoretical foundation

Social Cognitive Theory (Albert Bandura)

The conviction that one can successfully perform a behavior.



Self-Regulated Learning (SRL)

Based on Zimmerman's cyclical model. divides learning into 3 interactive phases that influence each other continuously.

3. Solution

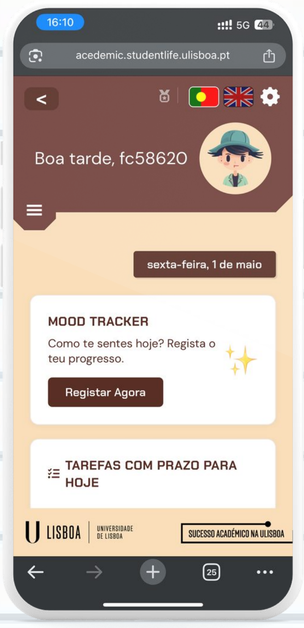
Acedemic Tracker

Acedemic Challenge

Scan to access

Study & Time Management

Two integrated web apps based on Zimmerman's SRL model and Social Cognitive Theory to overcome behavioral and psychological barriers.



A platform that provides planning, focus, and monitoring tools.

Gamified 21-day challenges

Builds motivation and academic self-efficacy through gamified daily challenges of different levels.

4. Future Work

- → Integration with a virtual study room app
- → Research on scalability security challenges
- → Redesign of Acedemic Tracker
- → Research on the feasibility of implementing automatic timetable import
- → Addition of gamification features

5. Validation Methodology

(1) Usability Tests

Identification and resolution of UI/UX barriers using RITE method

(2) Longitudinal Study

Quasi-experimental (pre/post-test) during the semester

Scales used:
- GSE (General Self-Efficacy Scale) by Araújo, M. and Moura, 2011
- SRLI-R (Self-Regulated Learning Inventory – Revised) by Barros and Veiga Simão, 2017

This work was supported by the Portuguese Recovery and Resilience Plan (PRR) and by the European Union NextGenerationEU funds, under the PULSARA project. This work was also supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025 (DOI https://doi.org/10.54499/UID/00408/2025)

Designing Support for Engagement Strategy Selection in Digital Health Interventions

Ana Rita Mendonça[1], Isabella Cadoni[2], Rúben Gouveia[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] University of Twente, The Netherlands

LASIGE inclusive human-computer interaction

Digital Health Interventions (DHIs) are increasingly used to support **health-related behavior change**. Their effectiveness depends on **sustained user engagement**. When engagement is low, users are less exposed to intervention content, **limiting its potential impact**.

However, **designing for engagement remains challenging**. Designers must make decisions based on **dispersed evidence**, with **limited guidance** across contexts.

What is missing are **tools to support these decisions**.

What **support** do designers and researchers need when **selecting engagement strategies** for digital health interventions?

14 designers and researchers

Semi-structured interviews

Imagined a **toolkit** for engagement design

Design considerations for engagement support tools

1. Contextual and Non-Prescriptive Support

   Support strategy selection based on the **intervention's goals, target population, and context of use**.

   Adapt guidance to designers' **roles and levels of experience** (e.g., more support for novices, less for experts).

2. Reflective Design Support

   Identify **risks** (e.g., narrowing the design space too early).

   Challenge habitual choices (e.g., surface assumptions behind choosing a familiar approach).

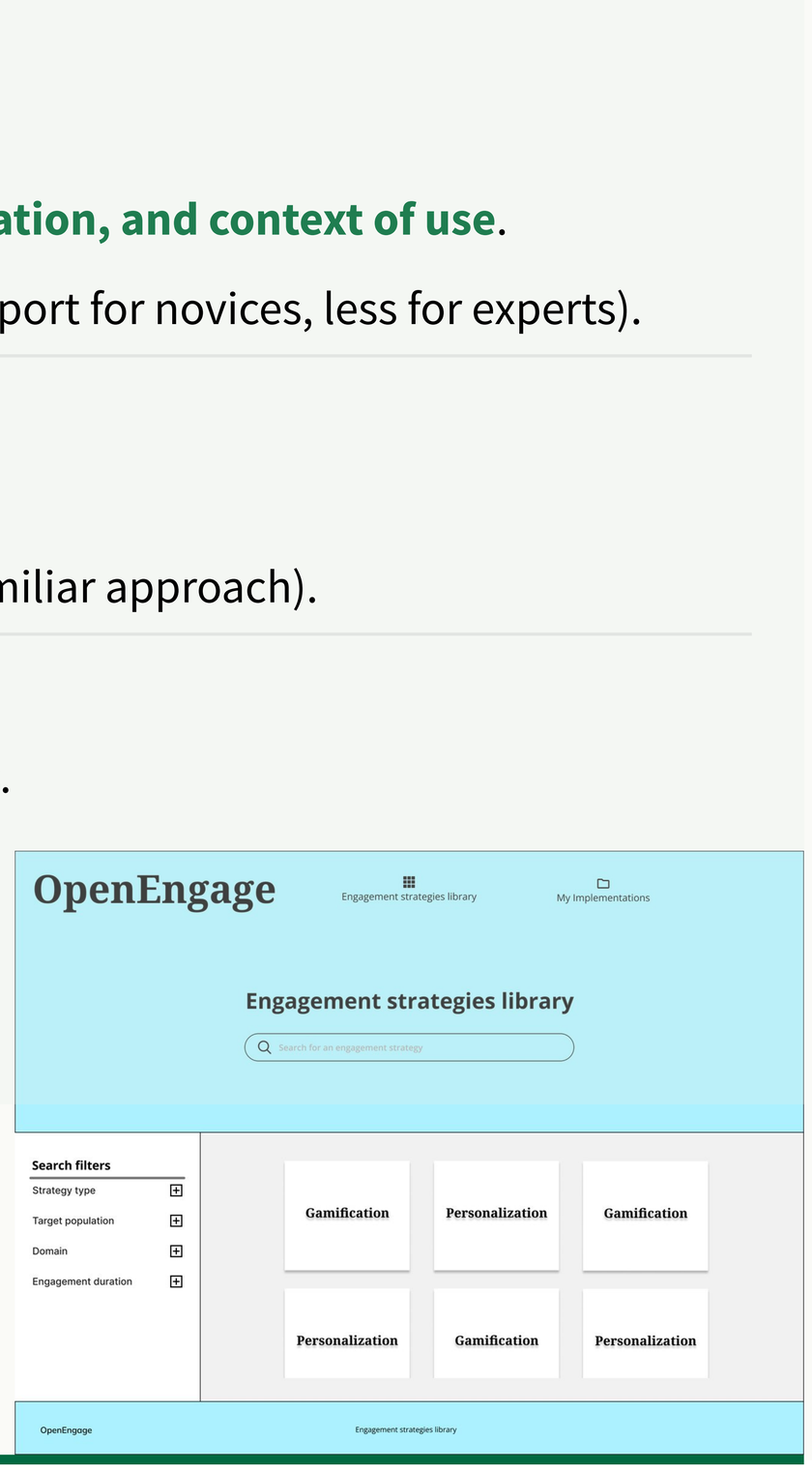
3. Learning from Prior Work

   Show how engagement strategies were implemented in **prior interventions**.

   Enable **comparison** across strategies.

4. Living Knowledge Resource

   Provide **up-to-date** research evidence.



Building on these findings, we are designing **OpenEngage**, a knowledge-sharing platform designed to support designers in selecting engagement strategies

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025 (DOI: https://doi.org/10.54499/UID/00408/2025).

U LISBOA · Ciências ULisboa · fct Fundação para a Ciência e a Tecnologia · LASIGE · UNIVERSITY OF TWENTE. · Funded by the European Union · erc European Research Council Established by the European Commission · ENGAGE

Reputation and Reciprocity in Child-Robot Interaction

Bárbara Rosa[1], Isabel Neto[1], Tiago Guerreiro[1], Hugo Simão[1], Alexandre Pires[2]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] SIAS, Faculty of Science, University of Amsterdam, Netherlands

inclusive human-computer interaction

MOTIVATION

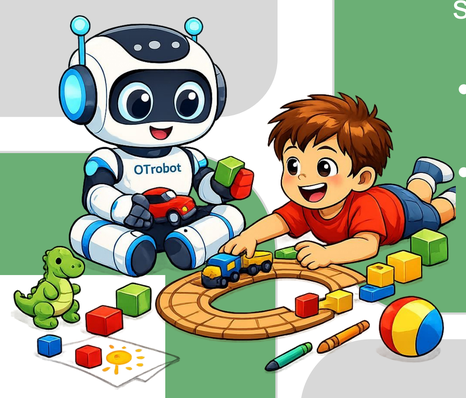
Social robots influence children's behaviour, but we still do not understand how children cooperate with robots in social dilemmas.

In particular, the role of gossip-based reputation remains unexplored.

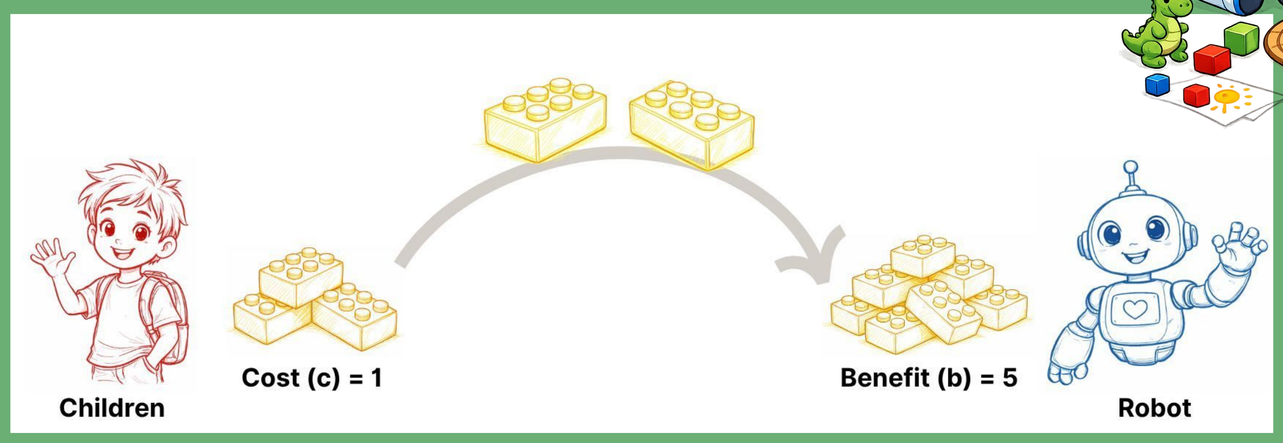
But how do children decide to cooperate with robots?

We study how children cooperate with a social robot by combining:

- **Indirect Reciprocity (Reputation via Gossip)**
- **Direct Reciprocity (Repeated Interaction)**



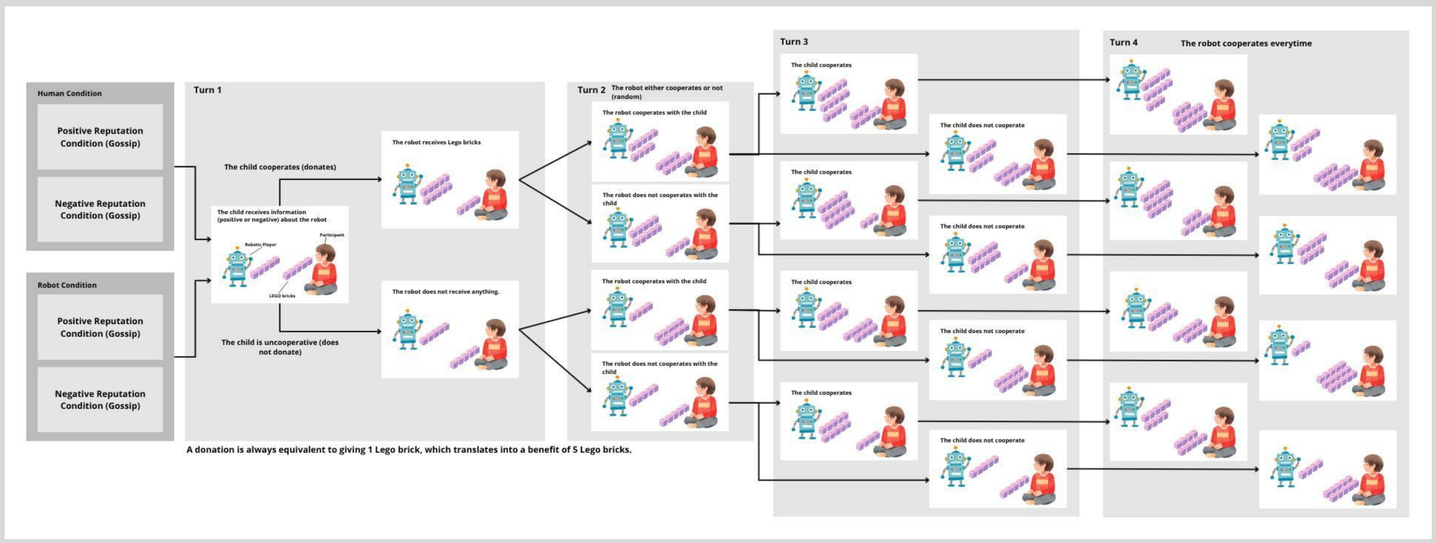
INTERACTION (Donation Game)



KEY INSIGHTS

- Children apply social norms to robots;
- Reciprocity influences cooperation in HRI;
- Reputation affects initial trust;
- Gossip remains unexplored in child–robot interaction.

SESSION INTERACTION DESIGN



REPUTATION MANIPULATION (Human vs Robot)

Positive Reputation Condition

"This robot has played with other children before, and they said he was friendly."

Negative Reputation Condition

"This robot has played with other children, they said he was not very friendly."

MEASURES

1. Cooperation Behaviour: Children's binary decision (cooperate vs do not cooperate);
2. Perceived Reputation: Measured after each round, through a short question about the robot - "How much do you want to be friends with this robot?";
3. Questionnaires: Familiarity with robots (pre-interaction) and enjoyment/usability (post-interaction);

EXPECTED CONTRIBUTIONS

1. Empirical insights into cooperation in child–robot interaction;
2. Understanding how children use reputation and reciprocity with robots;
3. Methodological contribution: adapting social dilemmas for children;
4. Design implications of educational social robots.

NEXT STEPS

1. Conduct the study in school settings with child participants;
2. Collect and analyse behavioural data on cooperation and reputation;
3. Evaluate the impact of gossip and reciprocity on children's decisions;
4. Develop and refine the theoretical model based on empirical results.

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

KeepsakeBox: Development of a Platform to Support Reminiscence Activities and Cognitive Stimulation for Older Adults

Bernardo Taciano[1], Daniel Gonzalez[1], Daniel Luís[1], Jesse Bráulio[1], Rodrigo Taciano[1], Tiago Rocha[1], Filipa Ferreira-Brito[1], Manuel J. Fonseca[1], Vânia Mendonça[1], Soraia M. Alarcão[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

Context

- In 2021, 57 million people lived with dementia worldwide, with ~10 million new cases each year
- Reminiscence therapy triggers personal memories, reinforces identity and facilitates communication between Person with Dementia (PwD) and caregivers
- Cognitive stimulation allows actively engagement of cognitive functions in older adults
- Formal caregivers manage multiple patients; informal caregivers have limited time and resources

Problem

- Scrapbook supports multimedia reminiscence but lacks shared caregiving and profile management
- KeepsakeBox supports shared caregiving but lacks cognitive stimulation
- Both operate independently on different tech stacks, hindering continuity of care
- Neither supports independent older adults engaging with activities on their own

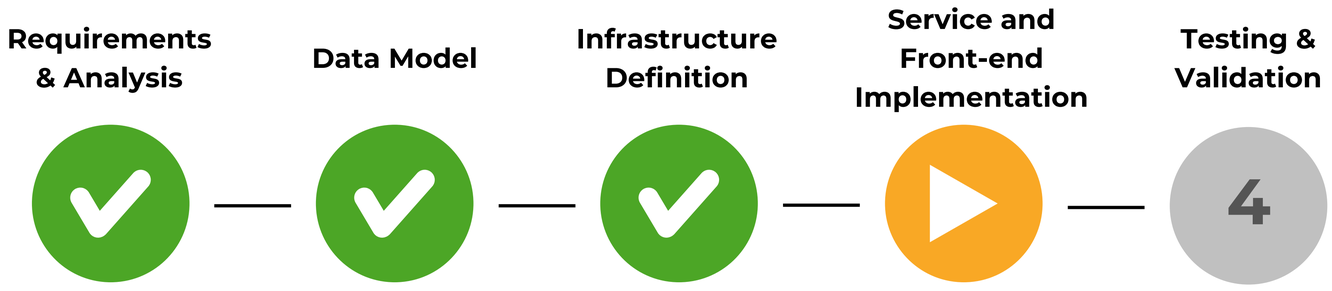
Proposal

Merge Scrapbook and KeepsakeBox into a single platform for **reminiscence therapy** and **cognitive stimulation** for caregivers and independent users, supporting:

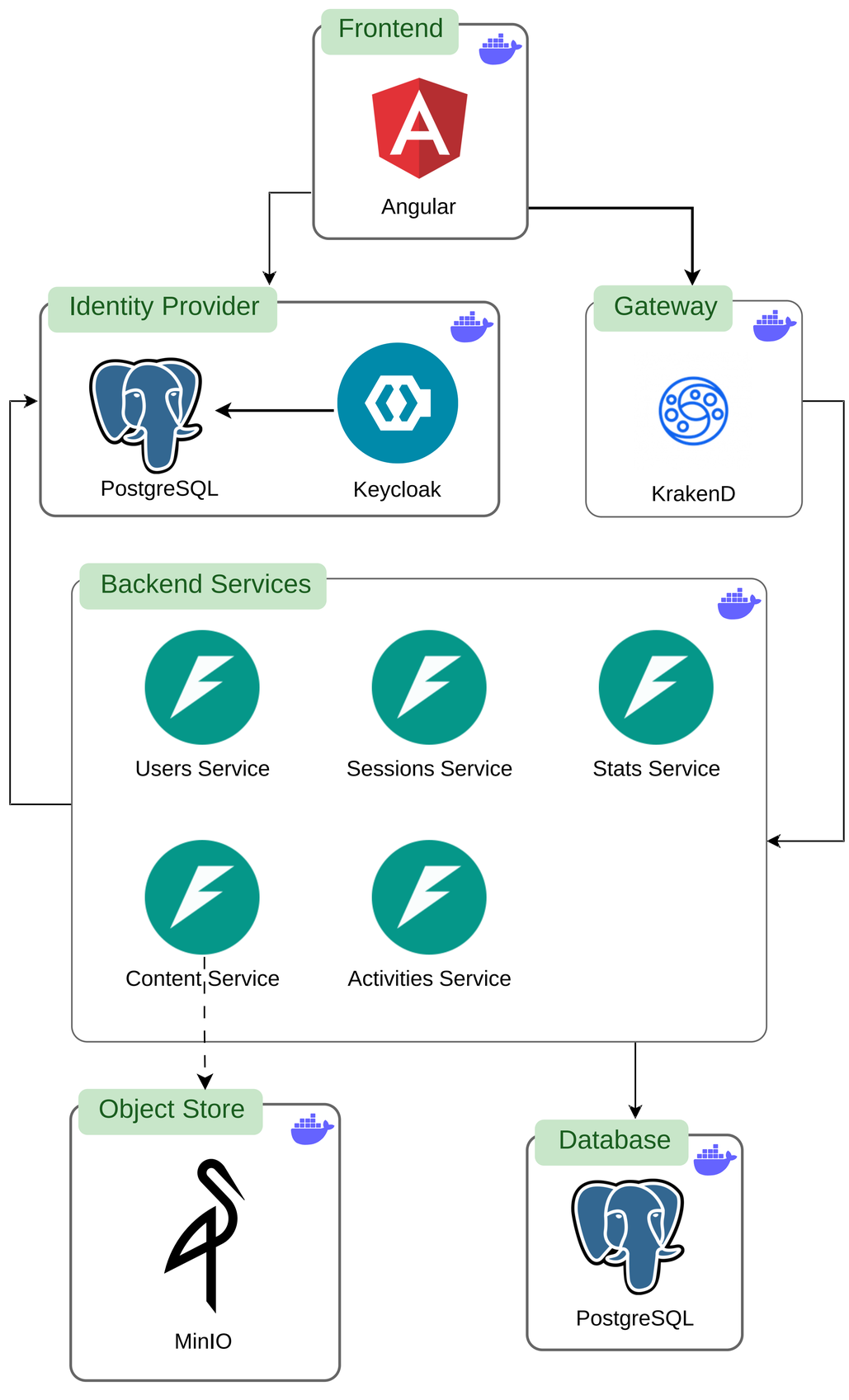
- Reminiscence sessions based on the PwD's biographical profile
- Cognitive stimulation
- Personalized multimedia content management
- Shared caregiving
- Session statistics

Services

Users	User and profile management with Keycloak integration
Sessions	Creation and management of reminiscence therapy sessions
Activities	Cognitive stimulation session management
Content	Multimedia content management via MinIO
Stats	PwD progression analysis and emotional trend visualization



Architecture



Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025

Modular Gaming: Exploring Contextual Play

Bruna Santos, David Gonçalves, Inês Gil, João Guerreiro, André Rodrigues

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

inclusive human-computer interaction

MOTIVATION

Digital Games

- A way to spend time together
- Part of everyday life

Everyday constraints

Contexts change

However...

Games assume fixed play context

Understanding how games can adapt to different contexts has the potential to support more **inclusive** and **socially rich** play.

OBJECTIVES

Explore how a game can support social play across multiple gaming contexts

- How does **modular, context-adaptive gameplay shape players' perceptions** of social presence, collaboration and engagement?
- How do game adaptations to context **influence player experience?**
- What **design challenges** arise when adapting gameplay to context?

APPROACH

IDENTIFYING GAMING CONTEXTS

Households offer rich contextual diversity within a single social setting, so we focus on a domestic environment.

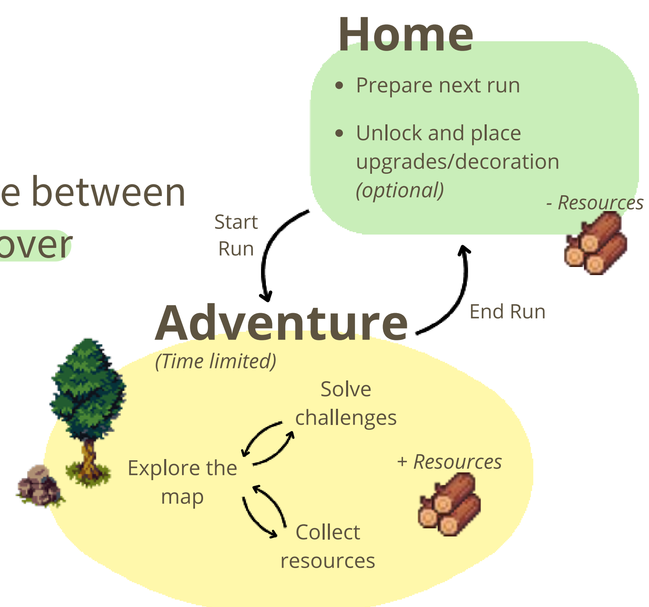
Three context-defining dimensions:

- Synchronicity — **Synchronous vs Async**
- Player Location — **Nearby vs Distant**
- Device Configuration — **Shared or Separate devices PC or Mobile**

Spectating · **Shared Screen** · **Nearby Mobile + Mobile** · **Nearby PC + Mobile** · **Distant PC + PC** · **Distant PC + Mobile** · **Alone**

GAME DESIGN

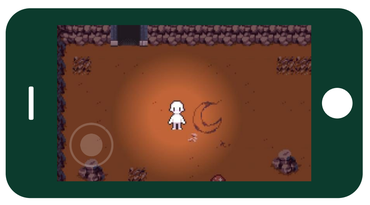
A **cooperative game** with a shared world. Players alternate between a shared space where persistent changes reflect progress over time, and exploratory adventure runs where players solve challenges and collect resources.



Context-adaptive design

The core loop remains the same, but the game adapts to the context.

- **Environment complexity:** Mobile vs PC
  - Game environment changes to adapt to device type
  - *E.g. smaller, simplified maps on the smartphone to support easier navigation*
- **Co-op challenges are altered** depending on context
  - Challenges are designed as modules that can be added or removed from the game
  - *E.g. playing on different screens introduces asymmetric information puzzles*
- Changes in **interdependence levels**
  - Adventure challenges require more or less interdependence depending on context
  - *E.g. higher interdependence when players are co-located*



USER STUDY

Qualitative Approach

30 participants, in pairs

Controlled laboratory setting

Exposure to the different context conditions

Data Collection

- Post-play interviews to gather qualitative insights
- Questionnaires (miniPXI) after the exposure to each context condition

Acknowledgements: This work was supported by the LASIGE Research Unit, ref. UID/00408/2025 (https://doi.org/10.54499/UID/00408/2025).

U LISBOA · Ciências ULisboa · fct Fundação para a Ciência e a Tecnologia · LASIGE

Opening the Conversation

Methodological Commitments for Inclusive AAC Research with People with Cerebral Palsy

Daniel Freitas[1,3], Carlos Duarte[1], Jaime Ribeiro[2]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] aTOPLab, CiTechcare, Politécnico de Leiria, Portugal
[3] Fundação MEO, Portugal

LASIGE inclusive human--computer interaction

MOTIVATION

People with Cerebral Palsy (CP) and Complex Communication Needs (CCN) rely on AAC yet are **systematically excluded from AAC research.**

AAC technologies appear to succeed in basic conversation and lab setting but fail in real-world communication, leading to **AAC abandonment** and increased **social exclusion for people with CP**.

Exclusion reflects a methodological accessibility **mismatch between traditional HCI methods and the communicative realities of people with CCN**, as these methods assume fluent speech and rapid interaction.

GOAL

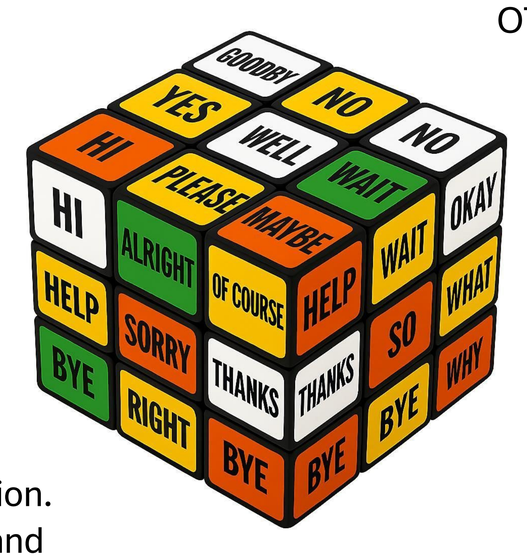
A Methodological Accessibility Framework. This work focus on identifying **methodological commitments** that ensure **research participation itself is accessible** to people with CP and CCN, **before** design decisions are made.

MULTIPLE FACETS OF THE PROBLEM, DIVERSE COMBINATIONS OF USERS NEEDS

Users require temporal flexibility
Methodologies should be adapted for slow, effortful, and variable communication skills.

Multimodal Communication
Treat gaze, gestures, vocalizations, typing, and behavior as valid data.

Proxy Support ≠ Substitution
Prioritize direct user participation. Use proxies to contextualize, and not substitution.



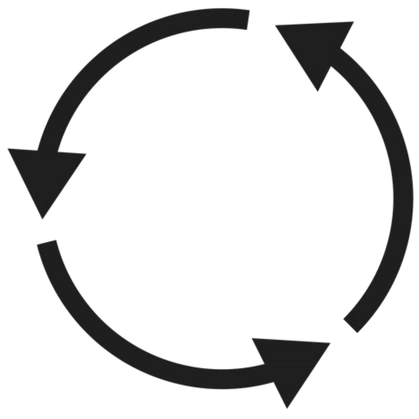
Multidisciplinary teams
Integrate Speech-Language Therapists & OTs and clinicians into research teams.

Small-N & Case-Based Research
Depth and longitudinal insight over breadth.

Community Partnership
Work with CP communities; value lived experience as expertise.

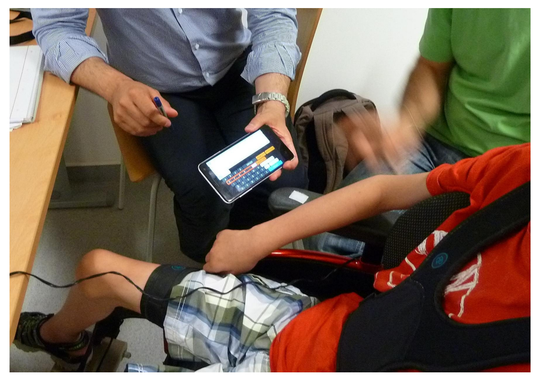
MULTI-STUDY APPROACH

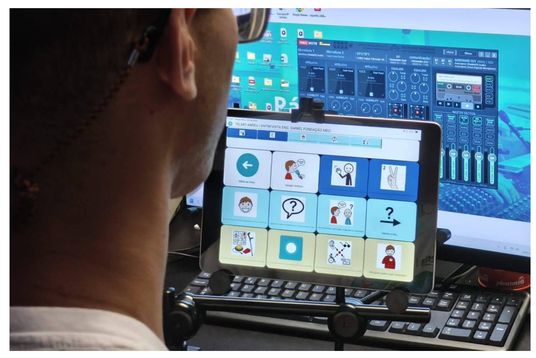
Scoping Review
A scoping review of inclusive AAC research with people with Cerebral Palsy (CP), mapping how user involvement is currently conceptualized and practiced, and identifying methodological gaps that motivate the proposed commitments.

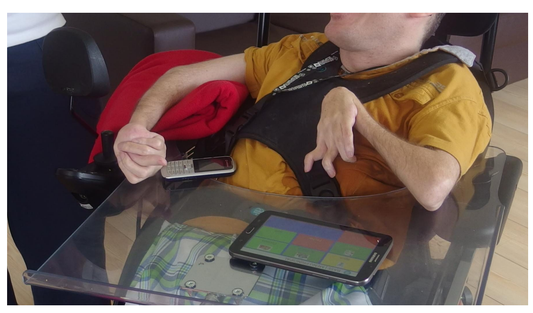


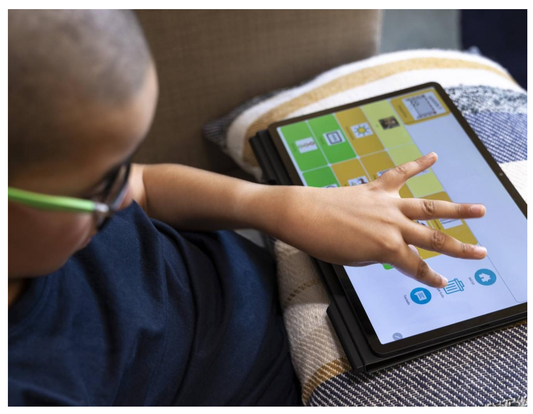
Best Practices in AAC User Research
Interviews with researchers and AAC users with CP to examine how AAC research is conducted in real-world contexts, assessing feasibility and methodological barriers.

Real-World validation application
within an AAC design and development process to evaluate their effectiveness across diverse communicative contexts and support iterative refinement.









This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025 (DOI:https://doi.org/10.54499/UID/00408/2025).

Auth4ALL: Co-design of Context-aware Adaptive Mobile Authentication for Older Adults

Duarte P. Gonçalves[1], Henrique Rosado[1], Tiago A. Santos[1], Vânia Mendonça[1], Soraia M. Alarcão[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

Problem

- Ageing brings cognitive and motor challenges
- Traditional authentication methods (e.g., PIN, Fingerprint) become hard to use
- Security mechanisms often compromise usability
- Current authentication systems are not designed for ***all*** users

Objectives

- Design and evaluate an adaptive context-aware authentication system
- Improve balance between security and usability
- Promote digital inclusion for older adults

Proposed Solution

A mobile authentication system that dynamically adapts based on:

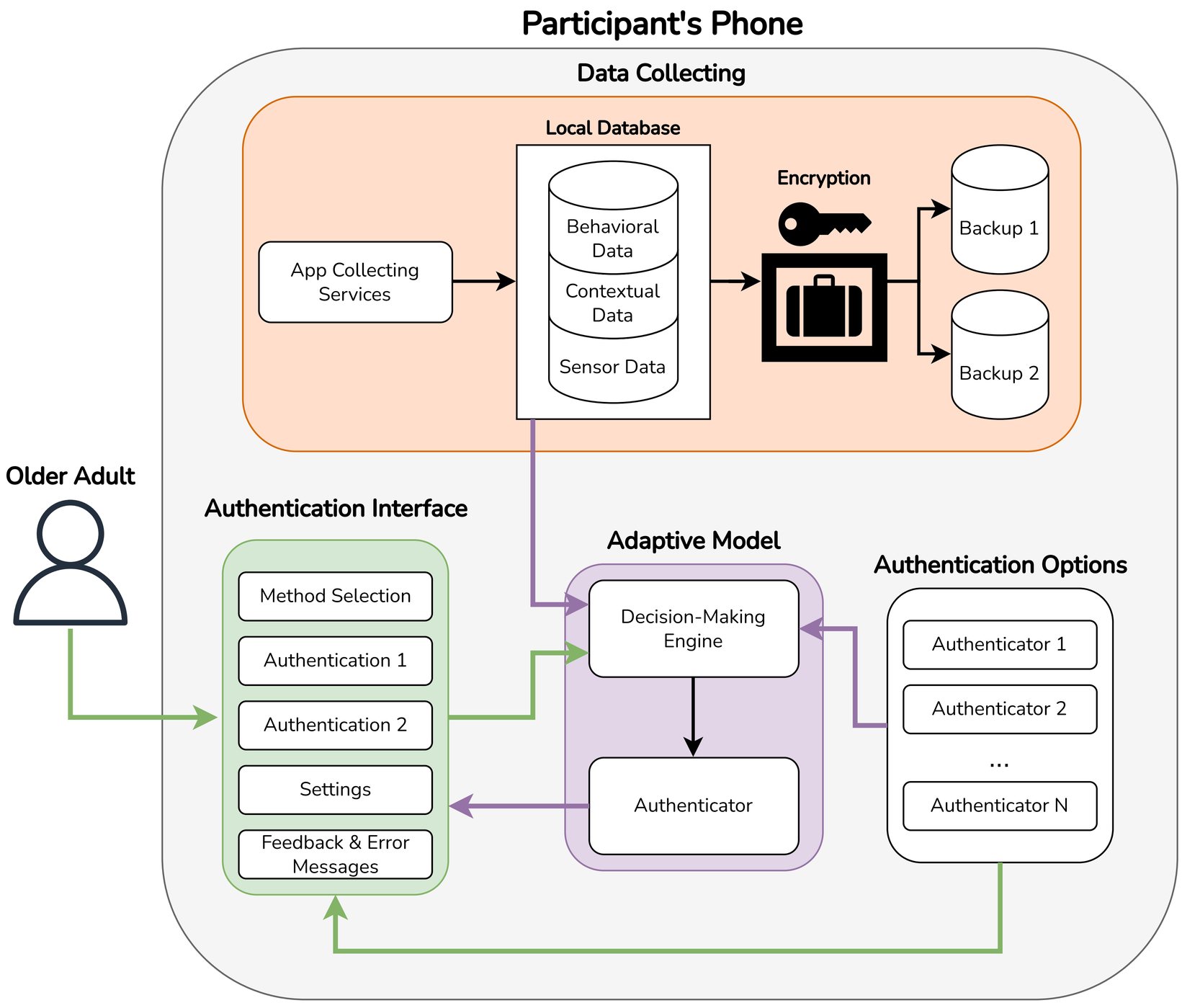
- User's behavior and preferences
- Context (e.g., location)

Example:

- At home (Low risk) → Weaker or no explicit authentication
- In public (High risk) → Stronger authentication (e.g., 2FA)

Work in Progress

- Interviews with target audience
- Co-design of authentication interface
- Development of data collection application
- Synthetic dataset generation
- Development of adaptive authentication model



Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., through project Auth4ALL, ref. 2024.14974.PEX (https://doi.org/10.54499/2024.14974.PEX), and the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Virtual TV **Remote Control** for Users With **Cerebral Palsy**

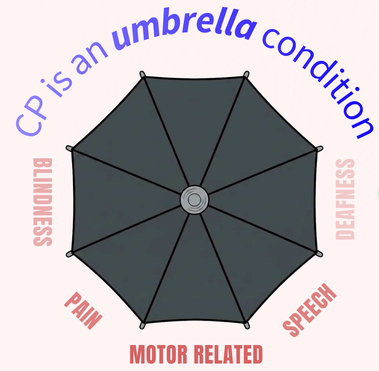
Keywords: *Augmentative and Alternative Communication (AAC), Cerebral Palsy (CP), Television (TV), Text-To-Speech (TTS)*

Guilherme Gouveia, Daniel Freitas, Carlos Duarte

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

PROBLEM



TV accessibility for people with Cerebral Palsy (CP)

MAY BE IMPRECISE OR UNINTENTIONALLY INCORRECT

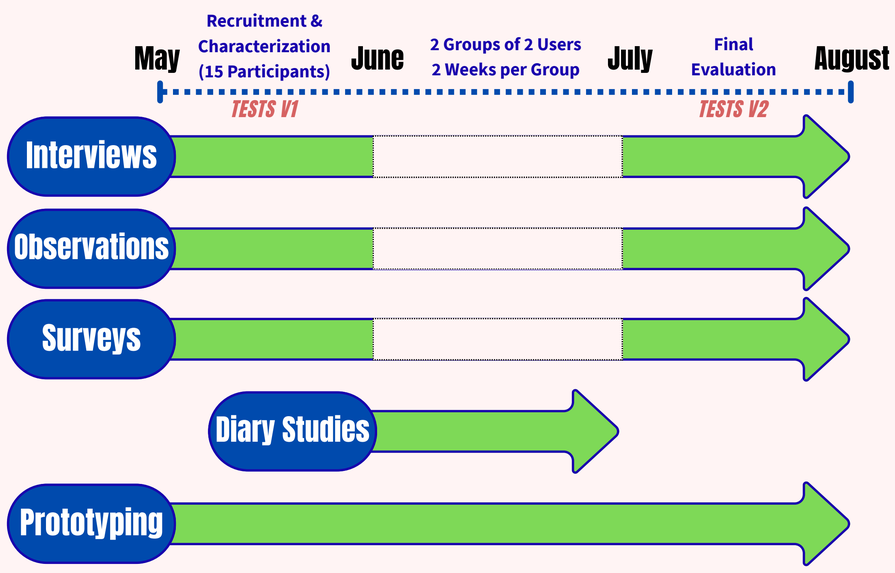
"It would be fantastic to be able to control the TV with your own mind"

Wi-Fi (LAN)

OBJECTIVE

Tackle the **specific challenges of CP users** by designing a **functional** virtual remote control that enables them to interact with the TV more **autonomously**.

METHODOLOGY



We will recruit 15 participants with CP who frequently watch TV at home:

- 10 from the ***Centro de Paralisia Cerebral Calouste Gulbenkian***
- 5 are from personal networks

USB-C

PROTOTYPES

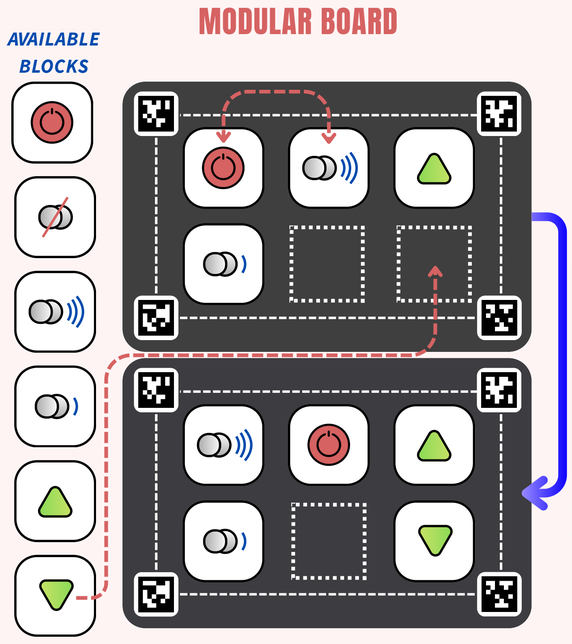
Mobile App

- Customizable & Modular
- *TTS* Feedback
- Scalable (*ADB* Protocol)
- Runs "Anywhere" (*Flutter* Framework)
- Different Forms of Interaction:
  - Touch
  - Scanning
  - Lines
  - External Button

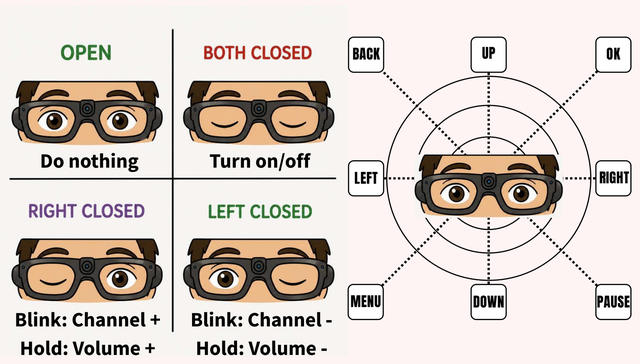
Eye / Gaze Controller

- Customizable & Modular
- *TTS* Feedback
- Scalable (*ADB* Protocol)
- Can Run on Phone (*Python*)
- Uses Pupil Labs Neon Glasses
  - Requires *Neon Companion App*
- Different Forms of Interaction:
  - Modular Board
  - No Board, only eye movement

WITH MODULAR BOARD



WITHOUT MODULAR BOARD



Acknowledgements:
This work was supported by Faculdade de Ciências da Universidade de Lisboa (FCUL) through scholarship ref.\BI-MESTRANDO-SPID_CIENCIAS_CD (376871) and FCT through the LASIGE Research Unit, ref. UID/00408/2025 (DOI:https://doi.org/10.54499/UID/00408/2025).

U LISBOA · Ciências ULisboa · fct Fundação para a Ciência e a Tecnologia · LASIGE

WebAR Experiences for Museums
“A Verdade e a J-u-s-t-i-ç-a”

Henrique Matos[1], Beatriz Carmo[1], Hugo Ferreira[2], Ana Fonseca[3]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] IBEB, Departamento de Física, Faculdade de Ciências, Universidade de Lisboa, Portugal
[3] Independent Artist, Portugal

LASIGE inclusive human-computer interaction

1 Motivation

Create ***innovative*** & ***gamified*** cultural experiences

Mitigate the need to **download** apps on-site

WebXR supports *only* **Android**, neglecting iOS’s potential

2 Objectives

Seamless experience cross-OS

Use AR to improve learnability & enjoyment

Gamify to enhance engagement

} WebAR experience for iOS & Android

3 Requirements

Easy access to the experience

iOS & Android Cross-compatibility

No app downloads (cannot be native)

High performance

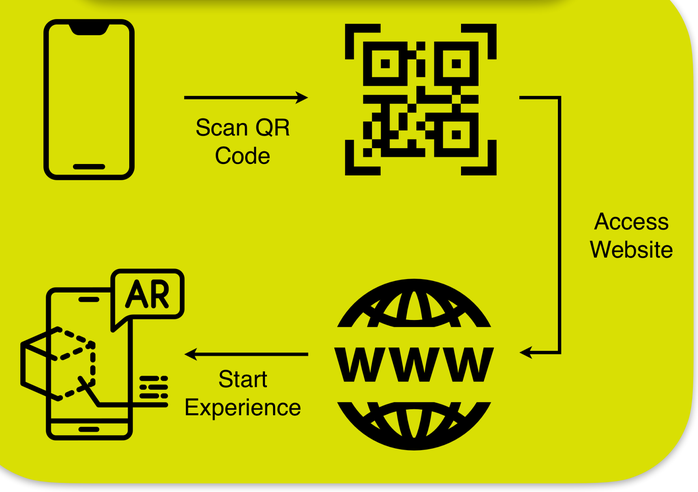
Use free existing Software

Solution?

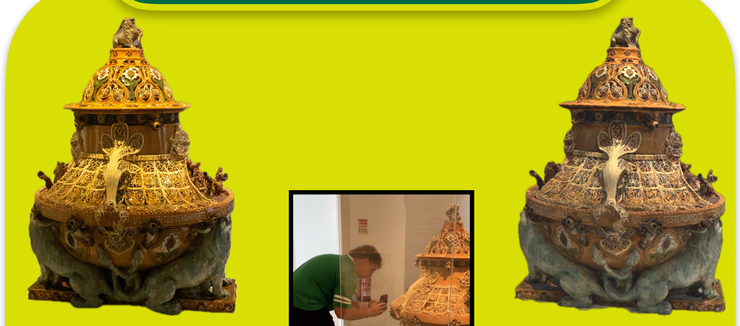
Progressive Web App!

4 Methodologies

Desired Interaction Pipeline



Photogrammetry Process



Technologies

WebAR Frameworks

- ARKit
- ARCore
- MindAR

- RealityScan

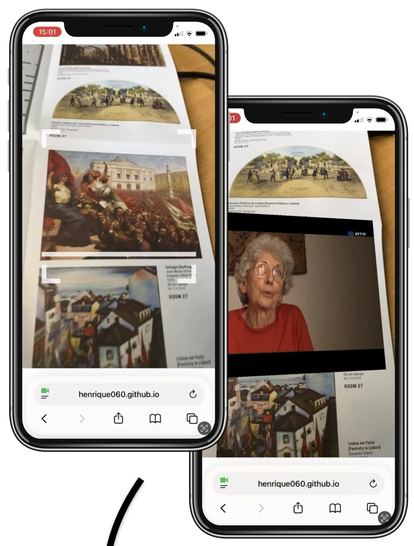
Photogrammetry

5 Experiences

A VERDADE e a J-U-S-T-I-Ç-A

Marker-based AR

Great for **tethering experiences** into a specific area



Scanning flyer and placing a video on top, automatically

Marker-less AR

Useful for **moving around** & **inside** object

Bringing ‘Terreiro do Paço’ into the museum grounds

Placing monument on the museum garden

6 Conclusions

High quality photography is required to deal with complex objects

WebAR is possible on iOS through Computer Vision-enabled tools

While high-quality AR is possible with MindAR, WebXR support would enhance the WebAR scene

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE | MUSEU DE LISBOA

FROM BOARD TO SCREEN AND BACK: EXPLORING CONTINUOUS TRANSMEDIAL PLAY

Inês Gil, David Gonçalves, João Guerreiro, André Rodrigues

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

What is **CONTINUITY?**

How players perceive they are **continuing a game** over **multiple play sessions**

What is **TRANSMEDIAL PLAY?**

Where elements such as **game theme** or universe **cover more than one media**

Hybrid games use **both analog and digital mediums** in **a single play session**, and to **some players** it can lead to negative experiences. To leverage the affordances of each medium in **separate sessions**, we explored

CONTINUOUS TRANSMEDIAL PLAY

USER STUDY

- What can continuous transmedial play **contribute** to (or **hinder**) player **engagement**?
- How can **narrative** and **progression** support a continuous player experience?
- How is **group play** affected?

16 8 7 GROUPS

FRIENDS & FAMILY

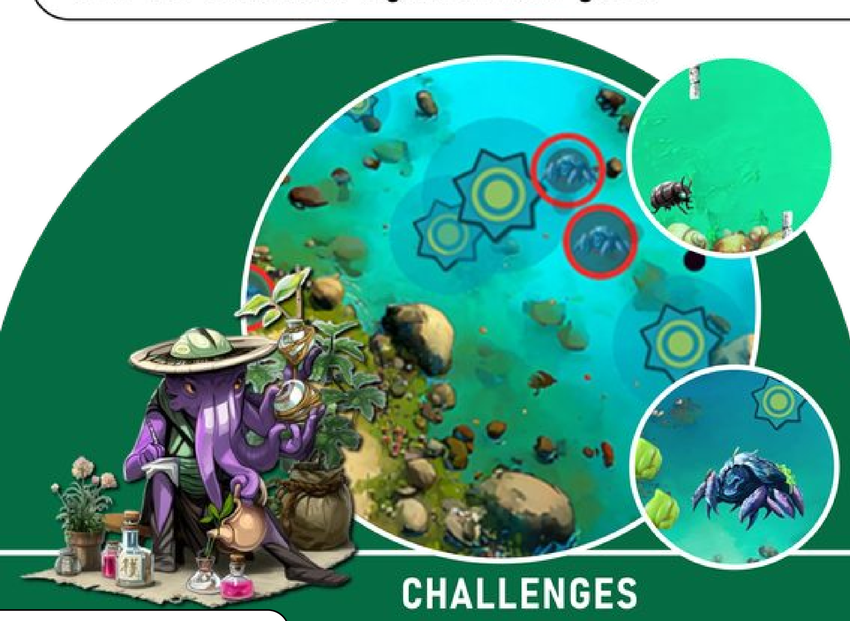
1. Played **a chapter** of the **board game**
2. Played the **digital game** for **1-2 weeks**
3. Played **another chapter** of the board game, incorporating **changes** from the digital game, folowed by an **interview**

We expanded a board game to include digital play sessions, to do this we created a digital mobile game

1 PHYSICAL SESSION

2 DIGITAL SESSION

3 PHYSICAL SESSION



CHALLENGES

NARRATIVE

ARENA

MARKETS

FINDINGS

CONNECTING THE MEDIUMS

Participants suggested ways the mediums could be more connected, such as **changes** in:

NARRATIVE OR GAMEPLAY

For players that enjoy the **additional narrative** the value of the **digital medium** was **enough.**

But others wished their actions had more

IMPACT IN THE GAMEPLAY

"Meeting a new character in physical sessions could open a new space to train in digital sessions!" (P4)

OR

"Maybe we let some enemies escape in the physical session and now we have to defeat them here [digital session]" (P8)

BALANCING THE EXPERIENCE

Players can also reach **different levels of progression** depending on:

AVAILABILITY AND PREFERENCES

And in some cases players will play a lot and others will not be **able to play at all** and end up **feeling the**

⚠ FEAR OF MISSING OUT

Being able to **share rewards between the members of a group** could be the key, but it can also lead to a sense of

⚠ LOSS OF OWNERSHIP

A SOLUTION TO THIS could be to **gain knowledge** instead!

"I read that enemy has a weakness to fire!" (P8)

BRIDGING THE GAP

The digital medium was seen as a way to

SUSTAIN MOMENTUM

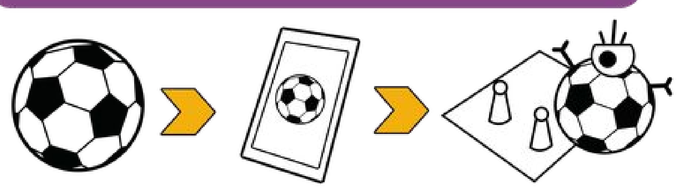
between physical sessions by **revisiting the narrative**, and helping players **remember the rules, for example:**

KEEPING TRACK OF TROPHIES

Players **discussed** the experience with other members of the group, even outside the game

FUTURE WORK

Could further expand transmediality to include activities outside the game, that influence the game and helps the group being aware of each other's activity



This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025 and through scholarship ref.: 1406.2 – 2 BI EM LASIGE

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

ATIVA-ER: Designing an Emotion Regulation Mobile App for and with Neurodivergent Adults

Isabel D. Trancoso[1], **Sofia Costa**[1], Filipa Correia[2], Vânia Mendonça[1], Vivian G. Motti[3], Soraia M. Alarcão[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] ITI-LARSyS, Instituto Superior Técnico, Lisboa, Portugal
[3] Human-Centric Design Lab, George Mason University, Virginia, United States

LASIGE inclusive human-computer interaction

MOTIVATION

- **Mobile applications** can provide **emotion regulation** support to **neurodivergent adults**
- However, the **lack of a user-centric design** hinders their acceptance, usability and effectiveness

GOAL

- To develop an emotion regulation mobile application, **customizable** to the individual preferences and needs, and **adaptable** to the textual and physiological emotional expression of **neurodivergent adults**

METHODOLOGY

Two Scoping Reviews

Literature Review

1) How can artificial agents promote emotion regulation in neurodivergent individuals?

2) What emotion detection systems have been developed for the neurodivergent population?

Online Survey & Interviews

Neurodivergent Adults' Preferences

3) Do neurodivergent adults use mobile apps to regulate their emotions, and what features and functionalities do they prefer?

Co-Design Sessions

Design Space

4) How can emotion regulation exercises (e.g., journaling, mood tracking) be digitized into a mobile app and customized according to the needs of neurodivergent adults?

Dataset & Models Development

Emotion Detection Framework

5) Can models trained on the textual and physiological data from neurodivergent adults accurately detect their emotions?

User & Expert Evaluation

Mobile App Prototype

6) To what extent is the mobile app prototype designed for neurodivergent adults suitable to support their emotion regulation needs?

8-month ABC | ACB Design

Longitudinal Evaluation

7) How do the customization and adaptability of our mobile app increase the usability, acceptance by neurodivergent users?

8) Can our mobile app prototype enhance the emotion regulation skills of neurodivergent users?

Acknowledgements: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025

Social Play Between Deaf and Hard of Hearing Children and Hearing Peers: Learning from Children and School Ecosystems

Jing Zhao[12], Isabel Neto[2], Michaela Okosi[3], Paulo Vaz de Carvalho[4], Hugo Nicolau[1]

1 ITI/LARSyS, Instituto Superior Técnico, Universidade de Lisboa, Portugal
2 LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
3 Gallaudet University
4 Catholic University of Portugal

LASIGE inclusive human--computer interaction

MOTIVATION

Is communication really the barrier to social play?

METHODOLOGY

- Drawing and Interviewing with 19 Children
- Interviewing with 6 Educators
- Observation for 26 Days

DESIGN SPACE

CONTEXTUALIZED SOCIAL PLAY TECHNOLOGY

A SHARED PEER CULTURE

Bidirectional Social Norms Exchange and Co-creation

Communication Accomadation

BUILD CHILD AGENCY

Actively Join or Invite

Lead the Play

Assert Preferences

Independent Negotiation

Bilingual Support

Bicultural Awareness

Collaborative infrastructure

FOSTER SCHOOL CULTURE

Acknowledgements: This work was supported by FCT through project 2022.06596.PTDC, 10.54499/LA/P/0083/2020, UID/50009/2025, 2024. 15729.PEX, and through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025, and by the European project DCitizens: Fostering Digital Civics Research and Innovation in Lisbon (GA 101079116

RobotsVote:
Do Robots Influence Your Vote?

João Pereira[1], Isabel Neto[1], Hugo Simão[1], Alexandre Pires[2], Tiago Guerreiro[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] SIAS, Faculty of Science , University Of Amsterdam, Netherlands

LASIGE inclusive human--computer interaction

1 Human opinions are strongly shaped through **social interaction**.

People may be influenced by those around them, through **group pressure**, **persuasion** or **new information,** for example.

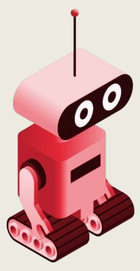
2 In the **political** field, these social processes are **fundamental.**

Empirical evidence suggests that individuals often update their **beliefs**, shifting slightly toward the information provided even if it challenges their **prior affiliations**.
This constant interaction and influence eventually leads to a **shift** in public opinion, one step at a time, for example, through each interaction on social media.

3 The landscape of political persuasion is being reshaped by **virtual agents.**

Recent research indicates that AI agents can be as **persuasive** as humans in **political** contexts and possess the capacity to **influence** user opinions significantly.
These virtual agents can be **chatbots**, such as ChatGPT or Gemini, or **bots** on social media.

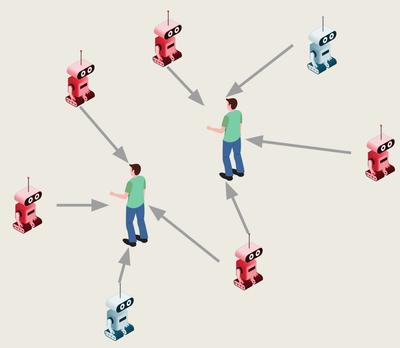
4 The influence of AI extends into the realm of **physical robotics**.



Through **non-verbal communication**, **gestures**, and **social cues**, robots engage users in ways that go beyond text.

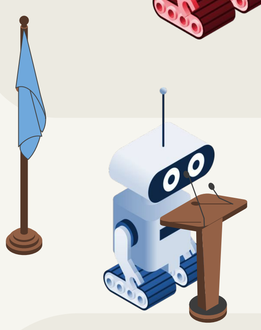
The integration into daily environments allows for the emergence of social mechanisms like **reciprocity** and **conformity** that are typically found in human-to-human interactions.

5 **What we know so far...**

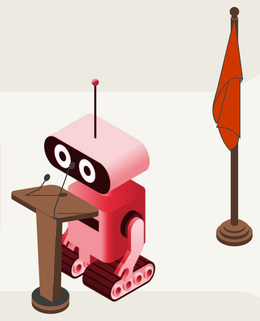


Research in HRI shows that robots can influence humans, mainly through **informational conformity**, when someone uses the answers of others as a source of information for their own judgment, especially in ambiguous situations.

However, most existing studies focus on analytical or perceptual tasks, such as visual perception, perception of feelings, and arithmetic operations, rather than contexts involving attitude and beliefs changes.



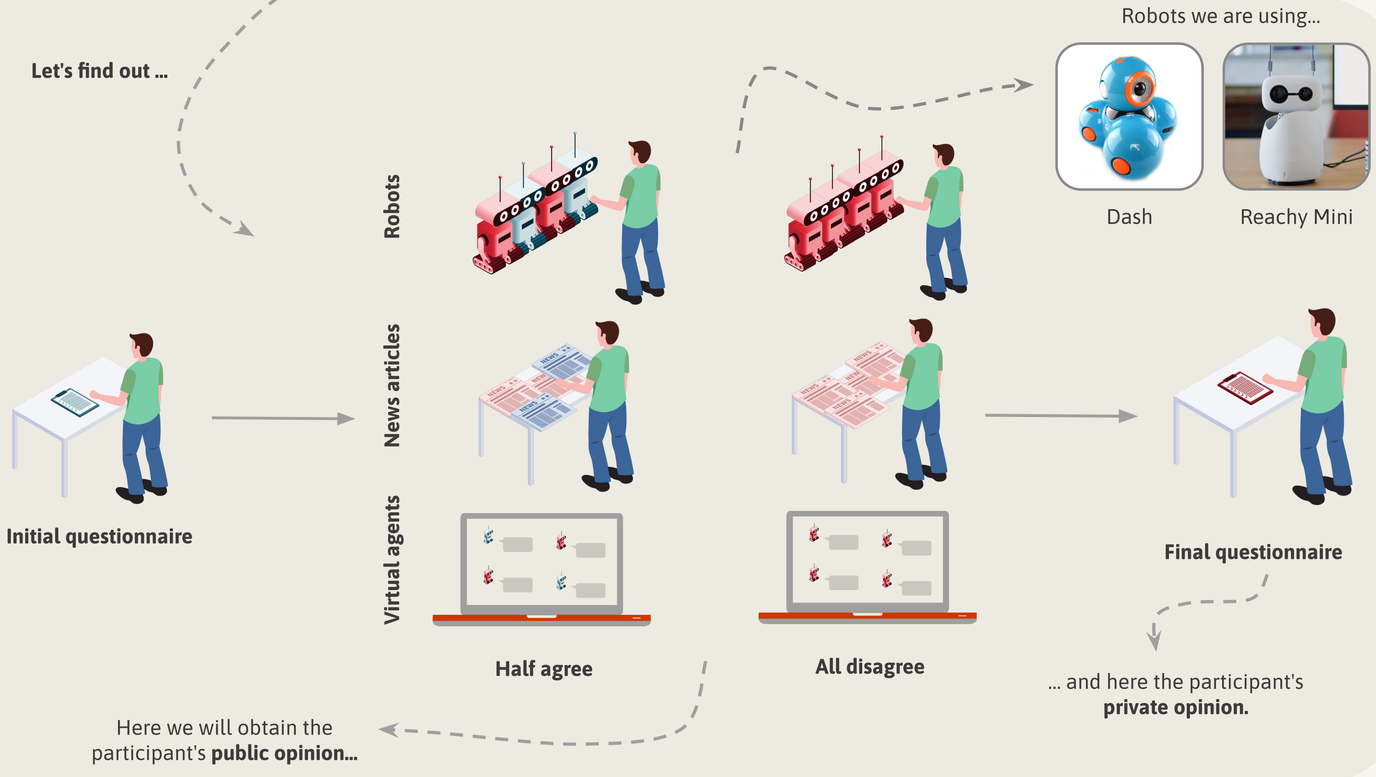
However, it remains unclear whether social robots can influence human political attitudes.
Can robots actually change what we think about politics?



6 **User Study**

The study will be conducted with young Portuguese people, with an expected sample size of around 150. The experiment consists in discussion rounds between 1 participant and 4 agents.

1. 1 of the 2 topics, socioeconomic or robot-related, is assigned to each participant, a design between subject.
2. Each participant will answer an initial questionnaire with statements related to their condition.
3. After completing the questionnaire, participants go through 6 separate conditions: **all robots disagree, half robots agree, all virtual agents disagree, half virtual agents agree, all articles disagree and half articles agree**.
4. In each round, a different statement is discussed; all participants give their opinion, leaving the last word to the participant.
5. After the discussion rounds, participants complete another questionnaire with the same statements as the initial questionnaire



Research Questions

Robots

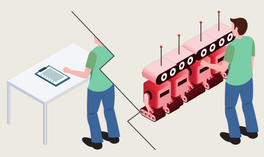
1. Do humans conform to a group of robots when discussing social, political and robot-related issues?
2. Is the shift in public opinion reflected in private opinion?
3. Which condition(economic, social, or robot-related) is there greater conformity?

Robots vs Virtual agents vs news

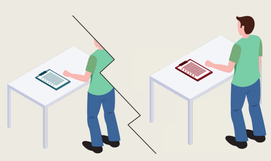
4. Do humans conform more with groups of robots, virtual agents, or news articles?
5. Is the shift in public opinion more reflected in private opinion with groups of robots, virtual agents, or news articles?

7 **What we will measure...**

Difference in the **public opinion** and the response to the same question in the **initial questionnaire**.



Difference between the **two questionnaires,** private opinion before and after the experiment.

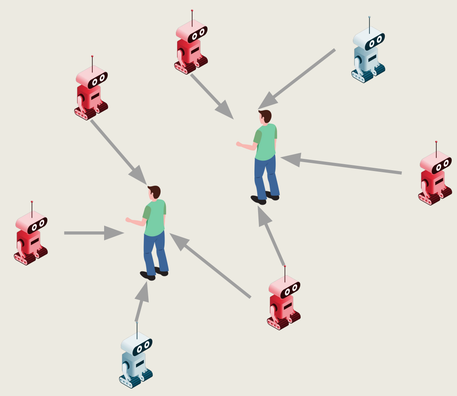


8 **Last but not least... Modeling opinion dynamics...**

Using the data from the user study, we will develop models to simulate **dynamics** of opinion change under robot social influence.

The model output will estimate the **evolution** of public and private opinion over time, based on the **biased voter model** and **expressed private voter model.**

With this, we will model the voter behavior using evolutionary models to estimate the impact on the population when these robots, or agents of influence, hold more left-leaning or more right-leaning views, for example.



What's next...

We will soon begin a user study, with a target of 150 participants, which will last about a month and a half. This study will take place at the ULisboa rectory and at Caleidoscopio

During that time, we will be able to start analyzing the data.

After that, we will develop opinion dynamics models based on the data obtained.

Acknowledgements: This work was supported by FCT through LASIGE Research Unit, ref. UID/00408/2025 (https://doi.org/10.54499/UID/00408/2025) and ref. 2024.15729.PEX.

Exploring the Effect of Communication Timing on Social Dynamics in Cooperative Games

Manuel Cardoso, Isabel Neto, André Rodrigues, David Gonçalves

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

inclusive human-computer interaction

BACKGROUND

In multiplayer games, especially cooperative ones, **communication is a key aspect**. It allows players to coordinate actions, solve problems together and connect. Communication can be designed to happen at specific timings, which can affect player experience:

Walkie-talkies:
Battery runs out, incapacitating players' communication

Emergency meetings:
Players have a specific period of communication

GOAL

We aim to inform the design of more engaging and intentional communication systems in cooperative games. We pose the following research questions:

- How does communication timing affect feelings toward the other players?
- How does communication timing influence the qualities and content of the communication itself?

APPROACH

We developed BallTogether, a two player cooperative puzzle game, in order to analyse how different communication modes affect players' social dynamics: **Free**, **Pre-round**, **Spaced Intervals** and **Resource-based**.

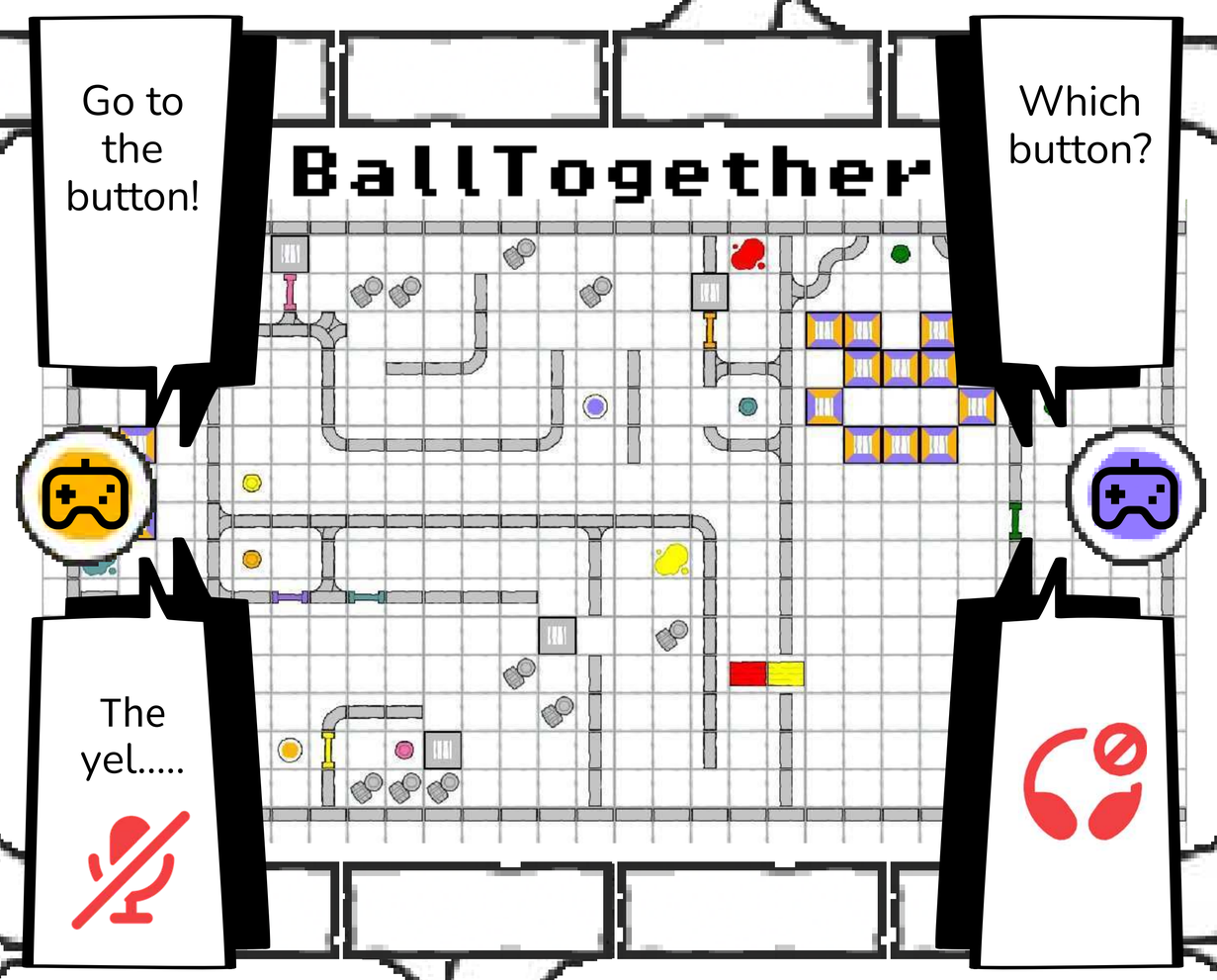
COMMUNICATION MODES

Free: no restrictions

Pre-round: plan your strategy before moving

Spaced Intervals: communication toggles between on and off

Resource-based: spend resources to speak



Player 1
Player 2

We also developed a Discord Bot that works in parallel with the game to enforce the communication modes, by deafening the players.

Do players **trust** each other more when not able to communicate?

How does communication mode influence **connectedness?**

Does communication mode affect **team effectiveness**?

How does **communication** vary based on communication mode?

USER STUDY

20 pairs of participants will be recruited to play 4 levels of the game prototype, experiencing all comunication modes.

Sessions will include gameplay, questionnaires and a semi-structured interview.

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Understanding Player Experience in Multiplayer Games with Fluid Goal Structures

Martim Emauz, Isabel Neto, André Rodrigues, David Gonçalves

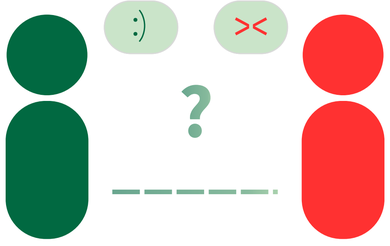
LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

inclusive human-computer interaction

Motivation

Multiplayer games are typically framed as either **cooperative** or **competitive** - but some allow players to shift between both

Games with **fluid goal structures,** where players shift between both, are underexplored



Context

Fluid goal structures — games where cooperation and competition are not fixed, but emerge from player decisions and game events

Bounded Generalized Reciprocity - players act prosocially toward others based on expectations of reciprocity, even with strangers

Goal

Characterize how players **perceive**, **behave**, and **react** when playing games with fluid goal structures, using a **multimodal approach**

Research Questions

RQ1 — What motivates **constructive or destructive behaviour** with others?

RQ2 — How do players deviate from **prosocial behaviours** predicted by the Bounded Generalised Reciprocity framework?

RQ3 — How do players **perceive other players** and newly formed teams?

RQ4 — How do **physiological indicators** for arousal, engagement, and tension change during gameplay?

User Study 1

Game: Arc Raiders

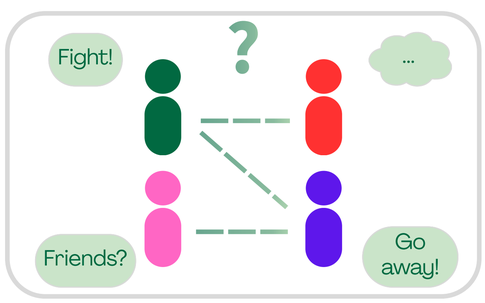
Fluid Goal Structure — Players enter short online matches and decide — cooperate, compete, or avoid others — They can fight, loot, explore or complete quests

12-20 participants

~1h30 per session

1 player per session

User Study 2



Using **relevant patterns highlighted in study 1**, we will design a multiplayer game with fluid goal structures — four players, one exit, shared enemies and intertwining goals

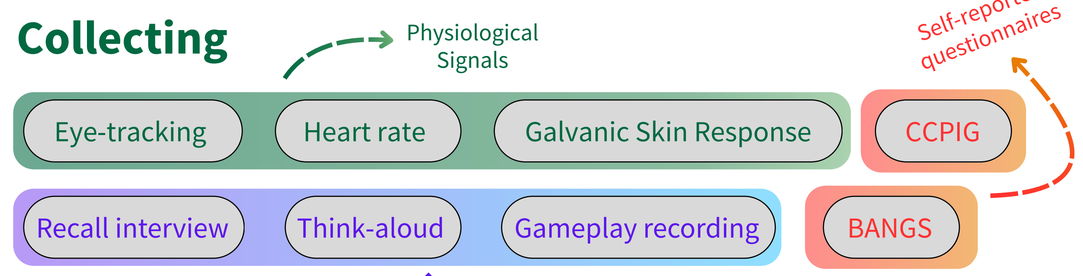
12-20 participants

Prototype game

4 players per session

Data Collection & Analysis

Collecting

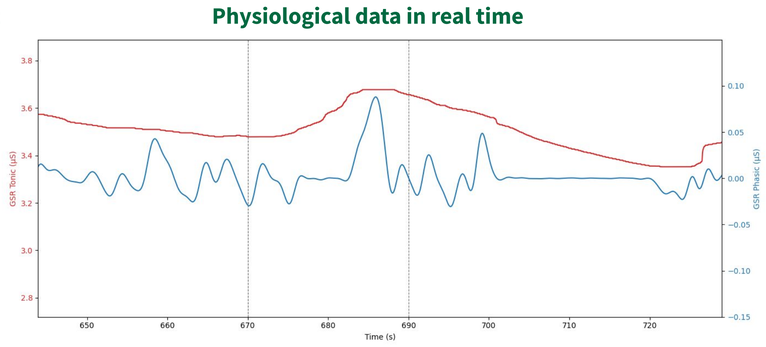


Understanding

- The player's behaviour, through reviewing the gameplay sessions
- Their perceptions, via PX questionnaires and recalling moments from gameplay
- Thematic analysis of the transcripts and data

Gazepoint GP3 & Biometrics Kit

Physiological data in real time



Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

EXPLORING GAME ANALYTICS IN BOARD GAMES:

An NFC-Based Approach for Tracking and Analyzing Card Game Playtesting Sessions

Miguel Martins, Inês Gil, David Gonçalves, João Guerreiro, André Rodrigues

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

THE PROBLEM

Playtesting is essencial in game design, but **data collection** is often:

- Manual and slow
- Incomplete or subjective
- Difficult to analyze

Designers lack structured feedback from playtesting sessions.

OUR SOLUTION: PLAYLYTICS

Playlytics is a mobile application that uses **NFC technology** to **automatically capture** events during board game **playtesting**.

Capture | **Trace** | **Analyze**

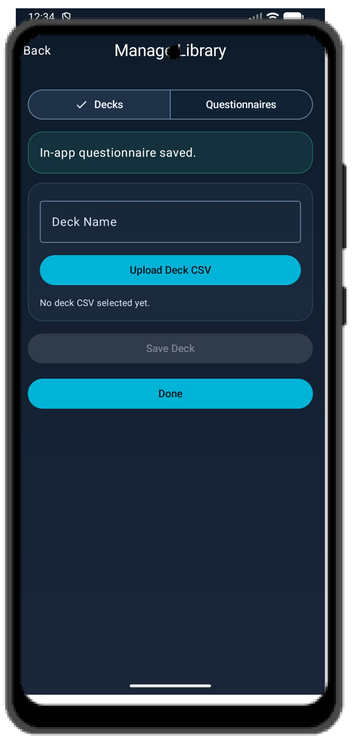
HOW WE TRACK CARDS

Each **card** is placed in an **NFC-tagged sleeve**.

HOW IT WORKS?

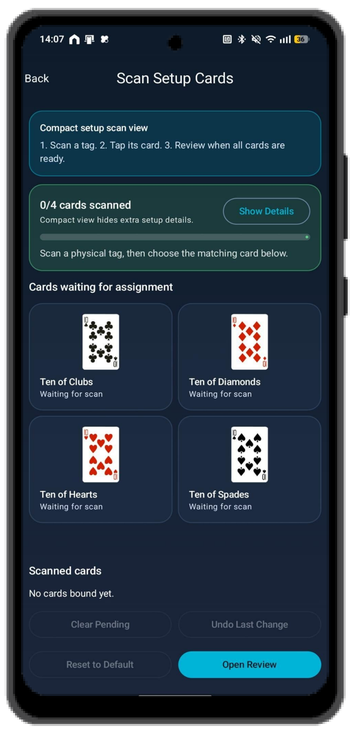
PRE PLAYTESTING SESSION

1 IMPORT DATA



Import decks and **questionnaires** from CSV files or linked forms.

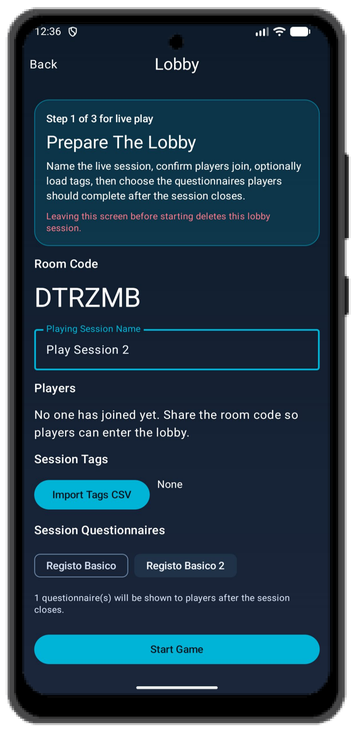
2 CREATE GAME SETUP



Configure the **game setup** and **assign** the cards to NFC sleeves.

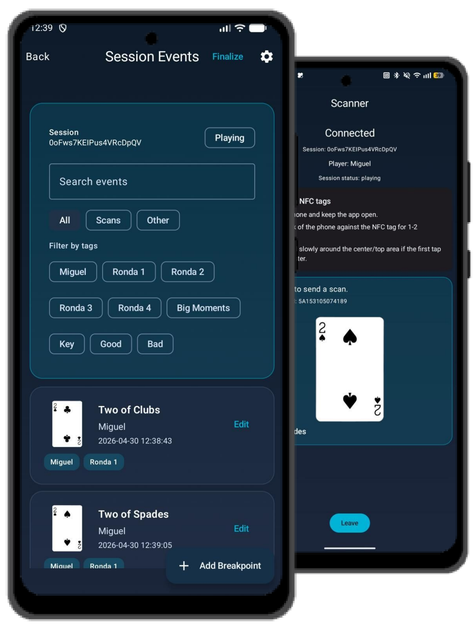
DURING PLAYTESTING SESSION

3 START SESSION



Create or join a **playtesting session.**

4 LOG KEY EVENTS



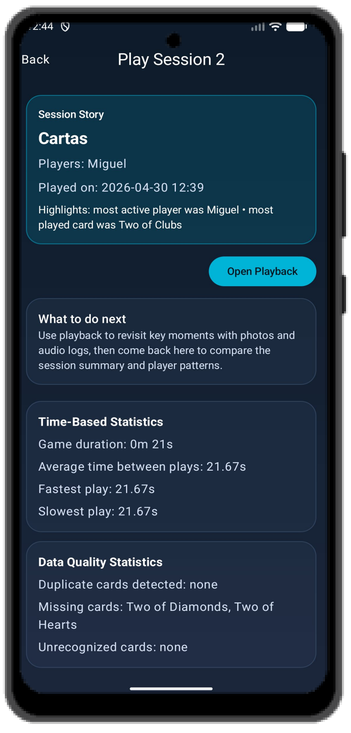
Log gameplay events, create **tags**, and add **breakpoints** during the **playtesting session**.

POST PLAYTESTING SESSION

5 FILL POST-GAME FORM

Collect feedback with a **post-game form**.

6 REVIEW SESSION



Review recorded events, explore the **session timeline**, and **analyze** key insights.

PRE PLAYTESTING SESSION | DURING PLAYTESTING SESSION | POST PLAYTESTING SESSION

RESEARCH QUESTIONS

RQ1: How does an app support board game playtesting before, during, and after a session?

RQ2: What impact does digital tracking have on the playtesting process for designers?

RQ3: How can digital data collection support game analytics in board game playtesting?

USER STUDY

6 designers

30–45 minute playtesting session

Procedure

1) Set up game
2) Sueca session
3) Complete form
4) Review data
5) Interview

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Harmonizing Health Data: Co-Designing Personalized Music Feedback for Accessible Health Trackers for Blind people

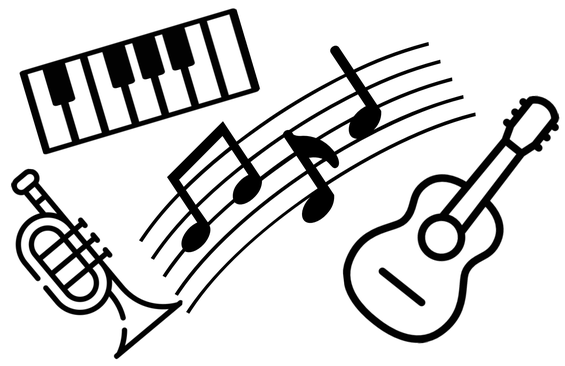
Pedro Costa[1], Eun Kyoung Choe[2], João Guerreiro[1], Rúben Gouveia[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] College of Information, University of Maryland, College Park, Maryland, USA

inclusive human-computer interaction

GOAL: Make health and physical activity tracking more accessible for visually impaired people through ambient sound feedback.

HOW? By co-creating personalized audio cues with users, replacing voice-based feedback with short musical cues that preserve privacy and reduce intrusiveness



MOTIVATION

- Mobile Health Apps rely on visual graphics that are inaccessible to Blind and Low Vision users.
- Ambient feedback comes as a great solution, however, generic designs, lacking aesthetic customization, can lead to the abandonment of the technology.
- The use of voice in public spaces can compromise privacy and also cause cognitive fatigue.

METHODOLOGY

Phase 1 – Co-design Sessions

- **Select Health Metric**: Participants choose what they want to track - Daily Step Goal, Sedentary Duration, or Sitting Breaks.
- **Express Preferences**: Users express musical preferences through vocal sketching (humming) and verbal descriptions.
- **Co-Creation of Sounds**: The team transforms these ideas into short melodies, with the support of musicians, audio editing, and Generative AI as a mediator.

Phase 2 & 3 - Prototyping & Field Study

- **App Integration**: The custom soundtracks are integrated into a mobile app.
- **In-the-wild Evaluation**: Participants use the prototype in their daily lives to evaluate its real-world impact on privacy, social comfort, and motivation.

Audio Samples

Acknowledgements: This work was supported by FCT through project Accessible Health Tracking with Ambient Feedback for Visually Impaired People, ref. 2024.13723.PEX (DOI: https://doi.org/10.54499/2024.13723.PEX), and the LASIGE Research Unit, ref. UID/00408/2025 (DOI: https://doi.org/10.54499/UID/00408/2025).

Inclusive by Design: Rethinking Endometriosis Tracking Tech for Trans and Gender-Diverse Users (TGD)

Sara Canhoto, Rúben Gouveia, Tiago Guerreiro

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

LASIGE inclusive human-computer interaction

What happens when a ‘women’s disease’ is tracked by people who are not women?

Endometriosis is framed as a “women’s disease”, but many who live with it are excluded by design

APPROACH

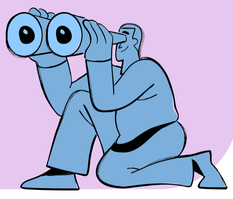
We conducted an online questionnaire (**n = 34**), recruiting participants via **Reddit and Facebook** communities focused on **endometriosis and TGD health**, to **explore tracking practices, barriers, and emotional experiences** with health-tracking tools

USE OF HEALTH TRACKING TOOLS

74% track digitally — but only 38% use apps.

The rest adapt their own systems: notes, spreadsheets, paper… **or stop tracking entirely.**

Users choose **flexibility and privacy over rigid, surveillant apps**.



MAIN BARRIERS

26% Reductionist Body Models
Bodies reduced to cycles and numbers

“The system assumes it’s an error rather than a different experience.”

19% Privacy & Surveillance
Tracking exposes identity, not just health

“My data being given to authorities to track me being NB.”

13% Gendered Design
Tools assume gender and fertility goals

“Apps assume I’m a woman and want fertility tips.”

11% Tracking Burden
Tracking demands more than users can give

“When I’m in pain, I just want one click.”

50% Disappointment

38% Frustration

29% Overwhelmed

Tracking is not neutral
It often creates emotional harm

DESIGN TENSIONS

STRUCTURED DATA VS LIVED EXPERIENCE
Tracking tools demand clean, standardized inputs, while users experience complex and fluctuating bodies

CONVENIENCE VS PRIVACY
Digital tools are easy to use, yet often avoided due to a lack of trust and data concerns

GENDERED SYSTEMS VS IDENTITY DIVERSITY
Apps assume binary, female-centered bodies, misaligning with TGD identities

QUANTIFICATION VS EXPRESSION
Tools prioritize measurable data, while users seek to capture emotions, context, and nuance

STABILITY VS CHANGE OVER TIME
Systems assume consistent patterns, while bodies and identities evolve (e.g., hormones, surgery, irregular cycles)

Check full paper

Acknowledgments: This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025, and the LASIGE Seed Project ref. IBD_LASIGE-SP-2025.04.

U LISBOA | Ciências ULisboa | fct Fundação para a Ciência e a Tecnologia | LASIGE

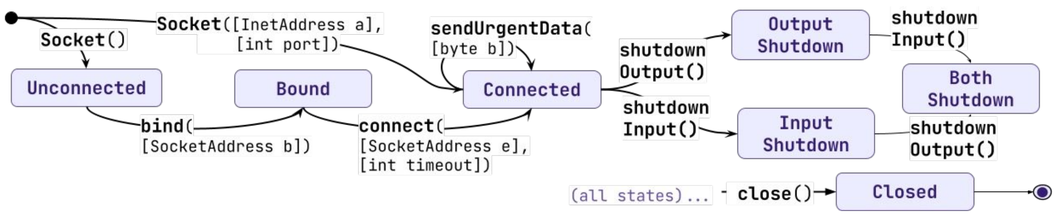
Barista: Synthesizing Typestate Specifications with LLM Agents

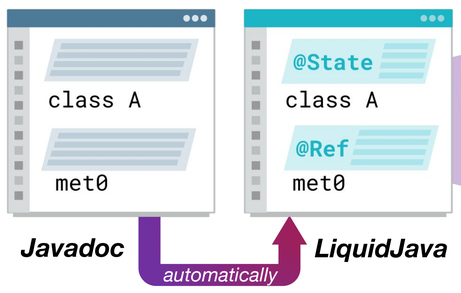
Catarina Gamboa[12], Paulo Canelas[12], Ricardo Costa[1],
Marcio Caetano[1], Jonathan Aldrich[2], Alcides Fonseca[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] S3D, Software and Societal Computing Department, Carnegie Mellon University

reliable software systems

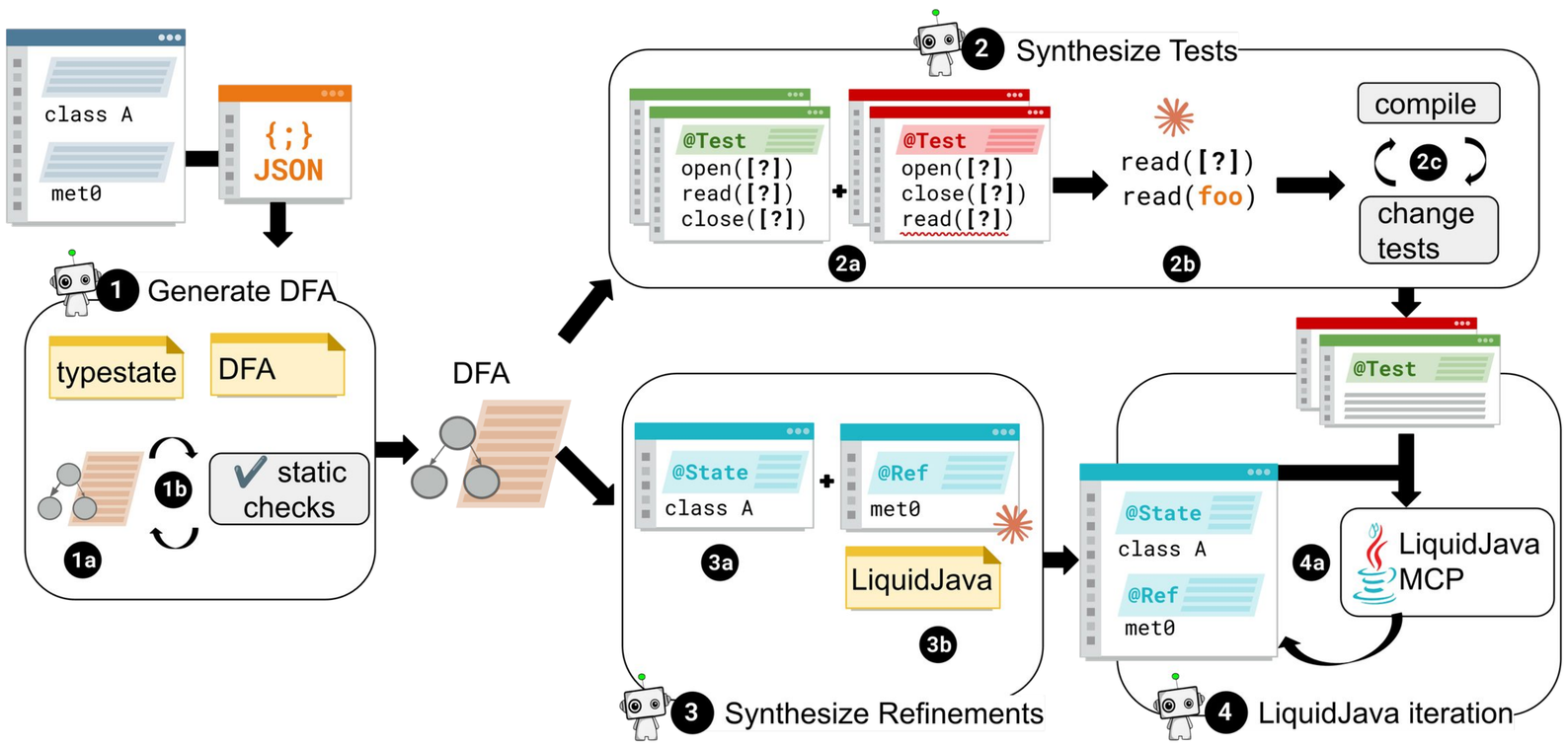
Object-oriented classes often encode implicit protocols that clients must follow, e.g **`send()`** is only possible if a socket was not already **`close()`**






```
import liquidjava.specification.*;
import java.net.InetAddress;
import java.net.SocketAddress;

@StateSet({"unconnected", "bound", "connected", "inputshutdown"
      , "outputshutdown", "bothshutdown", "closed"})
@RefinementAlias("ValidPort(int p) {p >= 0 && p <= 65535}")
@ExternalRefinementsFor("java.net.Socket")
public interface SocketRefinements {
    // Constructors
    @StateRefinement(to = "unconnected(this)")
    public void Socket();

    @StateRefinement(to = "connected(this)")
    public void Socket(InetAddress address,
                       @Refinement("ValidPort(_)") int port);

    // Connection methods
    @StateRefinement(from = "unconnected(this)",
                     to = "bound(this)")
    public void bind(SocketAddress bindpoint);

    @StateRefinement(from = "bound(this)",
                     to = "connected(this)")
    public void connect(SocketAddress endpoint,
                        @Refinement("_ >= 0") int timeout);

    @StateRefinement(from = "connected(this)")
    public void sendUrgentData(int data);

    // Shutdown methods, shutdownOutput is similar
    @StateRefinement(from = "connected(this)",
                     to = "inputshutdown(this)")
    @StateRefinement(from = "outputshutdown(this)",
                     to = "bothshutdown(this)")
    public void shutdownInput();

    @StateRefinement(to = "closed(this)")
    public void close();
}
```

BARISTA: Agentic workflow to synthesize LiquidJava specifications from documentation



dataset

35	5	5-89
JDK Classes	Domains	Methods in classes
e.g., Socket, ImageReadParam, BufferedReader	*Graphics, I/O, System, Network, Security, Data*	

Are the generated specifications accurate? Why do they diverge from experts? Is every agent needed?

RQ1 How accurate are the generated specifications?

Can the generated specifications detect protocol violations?
Where do generated and expert specifications diverge, and why?

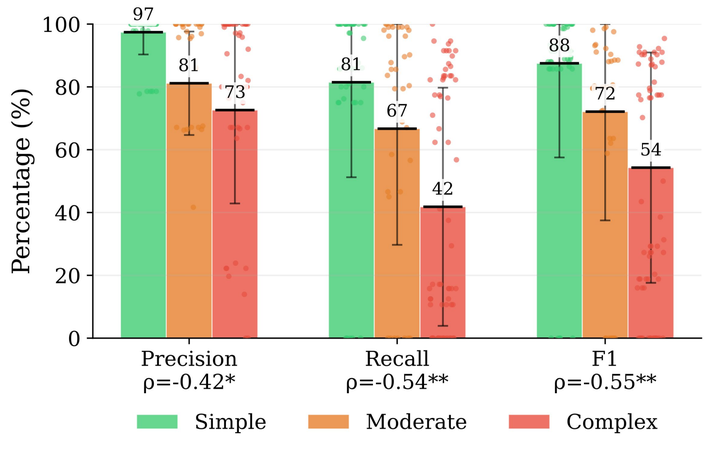
	Accuracy	Precision	Recall	F1
Barista	74.8%	87.0%	68.3%	76.5%

	Barista Detection		
	Always	*Partial*	*Never*
Expert ✅	23	5	1
Expert ❌	2	2	2

Failure root causes

- 7 Parameter-conditional transitions
- 6 Getter under-restriction
- 5 Beyond-doc extrapolation
- 5 Ambiguous documentation
- 2 Orthogonal state dimensions
- 2 Source code dependency
- 2 Cross-class contract opacity

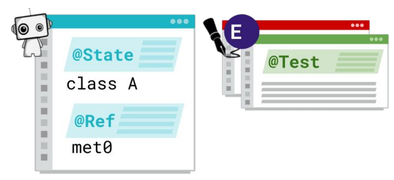
Information Gap / Modeling Gap



*Plot 1: Metric means and standard deviation per protocol complexity. Spearman rank correlations (ρ), * p<0.05, ** p<0.01.*

Expert Annotations

Quantitative

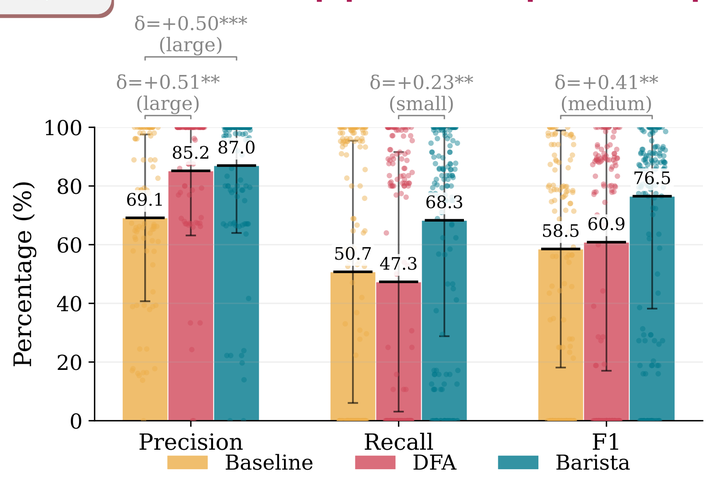


Qualitative

Answer RQ1

Barista achieves 87.0% precision and 68.3% recall. When divergences occur, Barista under-restricts rather than over-restricts, and 75% of specifications are usable directly or with minor edits.

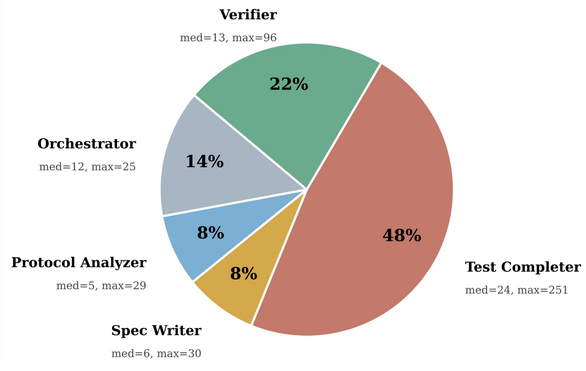
RQ2 Does each pipeline component improve specification quality?



*Plot 2: Ablation study. Brackets annotate significant improvements (Wilcoxon signed-rank, p<0.05) with Cliff's δ effect size; only non-negligible effects are shown. * p<0.05, ** p<0.01.*

Answer RQ2

DFA generation boosts precision; test-driven validation boosts recall. Barista: $2.20/class at 4.2× the speed of manual annotation.



Plot 3: Distribution of turns in each sub-agent

Discussion & Future Directions

1. Optimize for **precision over recall** to better support developer workflows with less false positives
2. More **expressive intermediate representations** can improve recall
3. **Better error messages** from the verifier can improve the refinement loop

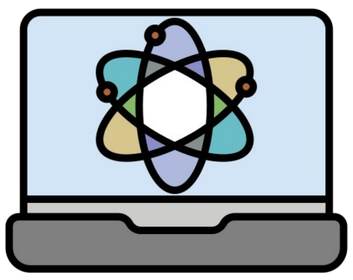
Acknowledgements:
This work was funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the project CMU-Portugal, ref. PRT/BD/154254/2021 and SFRH/BD/151469/2021, and under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025, which includes the LASIGE Seed project ref. LASIGE-SP-2025.07.
This work was also funded by the European Union - NextGenerationEU through the PRR - Recovery and Resilience Plan under the LASIGE Research Unit, ref. UID/PRR/00408/2025, DOI https://doi.org/10.54499/UID/PRR/00408/2025 and ref. UID/PRR2/00408/2025.

Framework For Non-Quantum Developers

Gabriel Stranoi[1,2], André Souto[1], Nabil Abdennadher[2], Tewfiq El-Maliki[2]

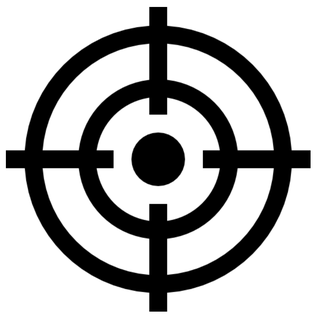
[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] University of Applied Sciences and Arts, Western Switzerland, Geneva

reliable software systems



Motivation

Quantum computing is advancing rapidly but remains inaccessible to most developers. Unlike classical computing, it lacks high-level abstraction layers that hide complexity.

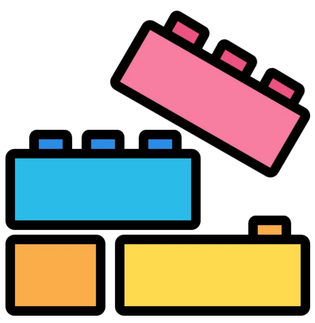


Goal

Design a framework that enables non-expert developers to use classical, quantum, and hybrid methods without deep quantum knowledge.

Problem

- Low-level circuit-centric development
- Fragmented ecosystem (Qiskit, Cirq, etc.)
- Complex hybrid classical–quantum workflows

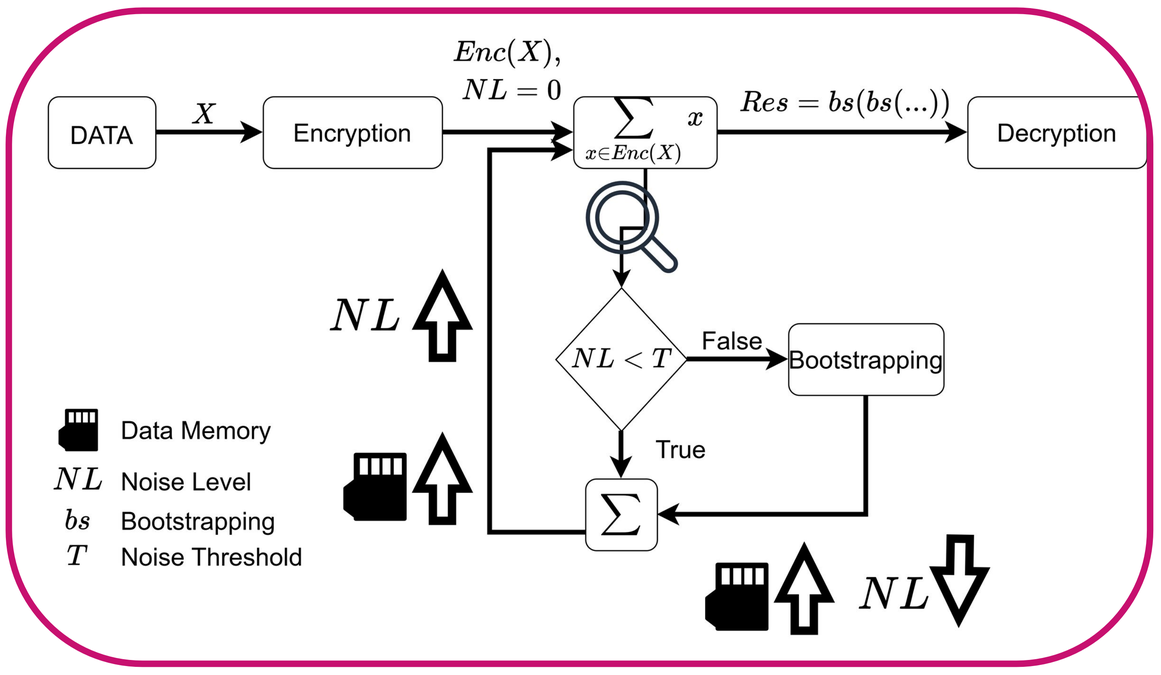


Components

- Abstract complexity
- Automate decisions
- Enable hybrid computation

Research Questions

RQ	Domain	Artifact Component	Validation Strategy
RQ 1	*Abstraction*	Unified High-Level API	Usability case studies (Novice vs. Expert scenarios).
RQ 2	*Data Encoding*	Adaptive Encoding Selector	Benchmarks against manual selection strategies.
RQ 3	*Security*	Unified Crypto-Interface (FHE, MPC, QOTP)	Security analysis and interoperability tests.
RQ 4	*Orchestration*	Backend-Agnostic Engine	Portability assessment across multiple providers.
RQ 5	*Architecture*	Hybrid Runtime Architecture	Performance benchmarks and architectural analysis.



PoC 1 HE - HE is not scalable, **1.** Memory growth during processing. **2.** Long execution time due to bootstrapping

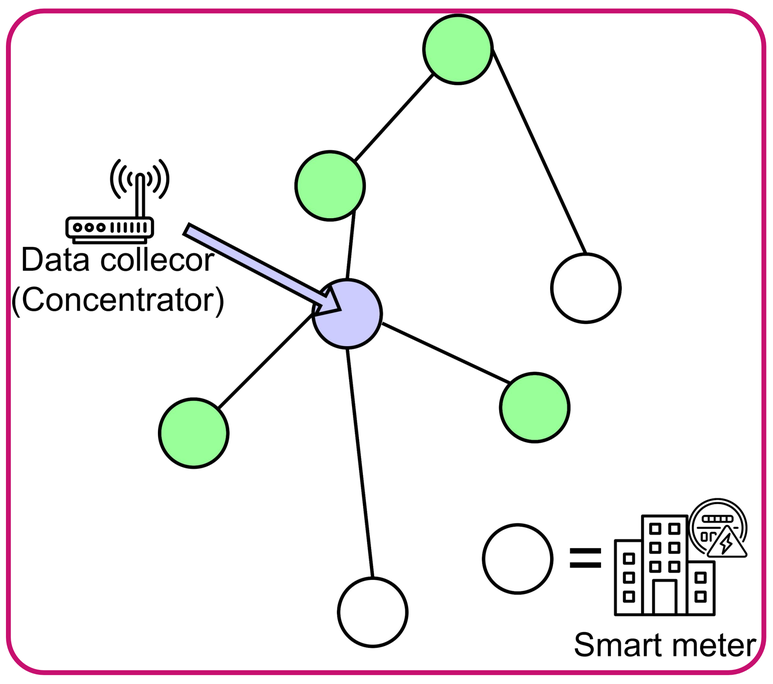
Highlights Proof of Concept

Worldwide Agro-Food Trading Statistics

Big compagnies needs global trading statistics without revealing sensitive data. The use of **homomorphic encryption** (HE) allows mathematical operation on encrypted data but lacks scalability due to the noise growth. Quantum computing could help solve this issue

Optimisation Problem in PLC network

Energy compagnies want to minimize concentrator placement to reduce costs. The Quantum computing is useful to explore all possibilities to find the optimal solution instead of suboptimal ones.



PoC 2 PLC - Power Line Communication network with smart meters and concentrator placement optimisation

Acknowledgements: This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025.

Developer Sentiment Analysis towards LLM-based Coding Assistants

A Multi-Platform Pipeline

Miguel Cut, Diana Costa, Wellington Oliveira

LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

reliable software systems

Pipeline

X

Stack Exchange

Reddit

GitHub

Data Collection

Aggregates diverse, platform-specific data via dedicated APIs and targeted searches into a unified, pipeline-ready dataset of developer thoughts about AI coding tools.

Data Preprocessing

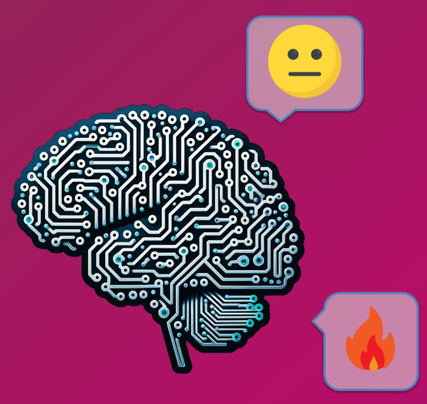
Cleans, normalizes, and keyword-filters raw conversations to isolate relevant data, while strictly preserving thread hierarchies and semantic context such as code and quotes for downstream analysis.

Data Analysis

Processes comments through multi-model sentiment analysis and hybrid topic modeling to comprehensively reveal both how developers feel about AI coding tools and the specific themes they are discussing.

Data Validation

Tracks temporal sentiment shifts linked to AI milestones, along with cross-platform comparisons, evaluating whether developer communities share a consensus on the use of AI coding tools.



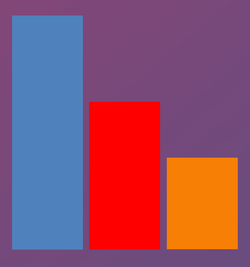
Results

How do developers feel about AI-generated code?

Copilot · Claude · ChatGPT · Cursor · Gemini

What do YOU think about it?
Share your thoughts:

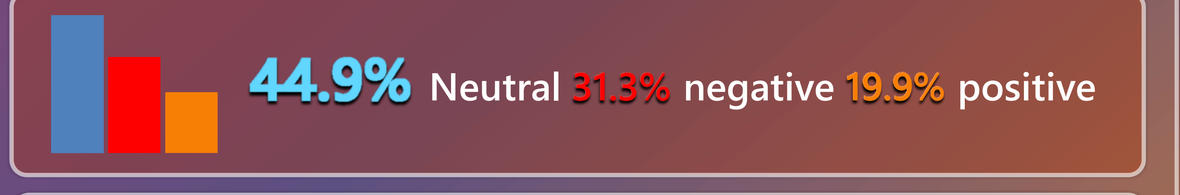
20,784 Items analyzed



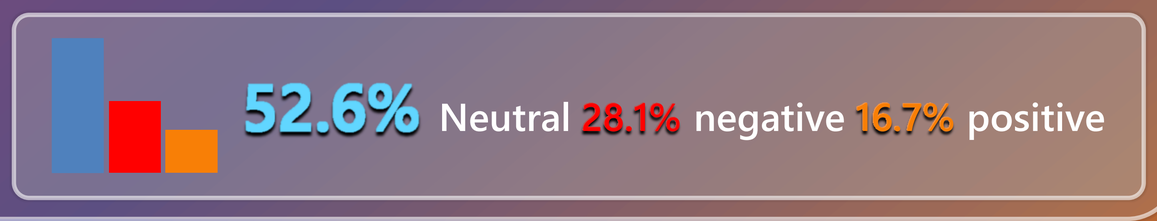
47.8% Neutral / **30.1%** negative **18.7%** positive

93% Sentiment accuracy

2 Platforms



44.9% Neutral **31.3%** negative **19.9%** positive



52.6% Neutral **28.1%** negative **16.7%** positive

Acknowledgements:
This work was supported by FCT through the LASIGE Research Unit, ref. UID/00408/2025, **DOI https://doi.org/10.54499/UID/00408/2025**.

Improving the Usability of LiquidJava

Ricardo Costa[1], Catarina Gamboa[1,2], Alcides Fonseca[1]

[1] LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
[2] S3D, Carnegie Mellon University, Pittsburgh, PA, USA

reliable software systems

What is LiquidJava?

LiquidJava is a type system for Java that uses **liquid types** and **typestates** to verify value properties and object protocols, catching common bugs at compile time.

```java
ArrayList<String> l = new ArrayList<>();
l.add("lasige");
l.get(0);
l.get(1);
```

Its **VS Code extension** automatically performs the verification and reports diagnostics in the editor.

```java
@ExternalRefinementsFor("java.util.ArrayList")
@Ghost("int size")
public interface ArrayListRefinements<E> {
  @StateRefinement(to="size(this) == 0")
  public void ArrayList();

  @StateRefinement(to="size(this) == size(old(this)) + 1")
  public boolean add(E elem);

  public E get(@Refinement("0 <= _ && _ < size(this)") int index);
}
```

It's Great, But...

Unhelpful diagnostics and **limited IDE support** negatively impact developer experience, limiting adoption in mainstream software development.

Why did the verification fail?

What's wrong?

```java
@Refinement("_ > 0")
public int half(@Refinement("_ > 0") int x) {
  return x / 2;
}
```

Poor Readability

Limited Context

Plain Text

```
Refinement Error:
Failed to check refinement at:

return x / 2

Type expected:(#ret_1 > 0)
Refinement found:true && #ret_1 == #x_0 / (2)
&& #x_0 == x && (x > 0)
...
```

Redundant Expressions

Internal Variables

No Explanation

To address the usability barriers of LiquidJava, we adopted a user-centered approach focused on **improving verification feedback** and **IDE support**

IDE Integration

Enhanced the **VS Code extension** with practical features that help developers read, write, and debug refinements in the editor.

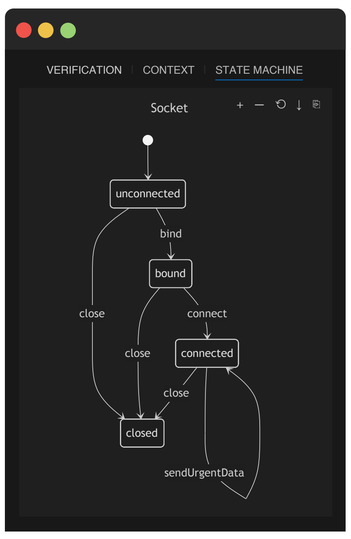
```java
@ExternalRefinementsFor("java.util.ArrayList")
@Ghost("int size")
public interface ArrayListRefinements<E> {
  @StateRefinement(to="s")
  public void ArrayList(
    size(ArrayList)  ghost

  @StateRefinement(to="size(this) == size(old(this)) + 1")
  public boolean add(E elem);

  public E get(@Refinement("0 <= _ && _ < size(this)") int index);
}
```

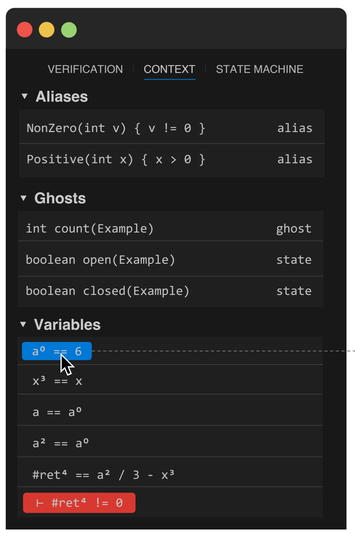
```
int index — ArrayListRefinements.get(int)
@Refinement("0 <= _ && _ < size(this)")
```

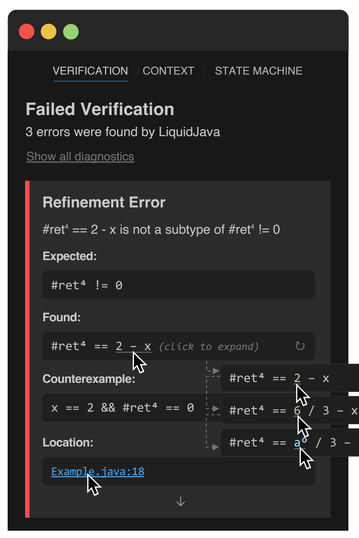
Syntax Highlighting **Autocomplete** **Hover Information**

Interactive View

Custom VS Code webview where developers can explore diagnostics in more detail, inspect the verification context at the cursor's position, and visualize typestate protocols through state machine diagrams.







Typestate Visualizer **Context Debugger** **Diagnostic Explorer**

Readability & Usability of Diagnostics

Improved **diagnostic messages** with clearer structure, better visual presentation, simplified expressions, and actionable feedback.

```
Refinement Error: #ret¹ == x / 2 && x > 0 is not a subtype of #ret¹ > 0
10 |     @Refinement(value="_ > 0", msg="result must be greater than 0")
11 |     public int half(@Refinement("_ > 0") int x) {
12 |         return x / 2;
   |         ^^^^^^^^^^^^ result must be greater than 0
13 |     }
14 |
 --> Counterexample: x == 1 && #ret¹ == 0

/Users/rcosta/liquidjava/examples/src/main/java/Example.java:12
```

x > 0 (refinement)

x = 1 (counterexample)

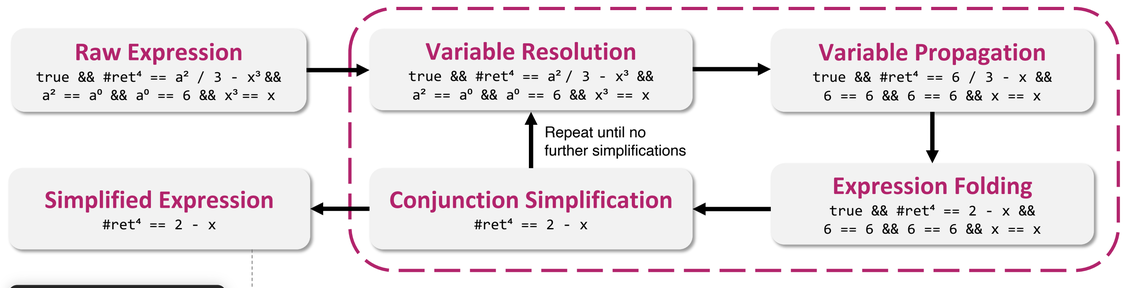
1 / 2 = 0 (integer division)

0 > 0 (refinement violation)

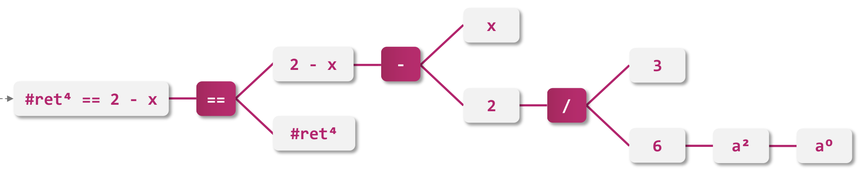
New Design **Counterexample** **Custom Message**

Expression Simplification

Simplification pipeline that saves each step as a node in a tree data structure, enabling the interactive exploration of simplifications in the diagnostic explorer of the VS Code extension.

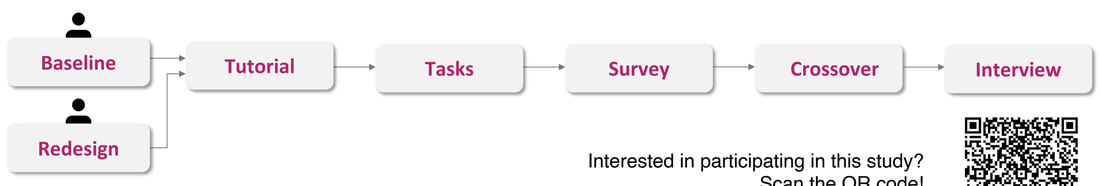



```java
@Refinement("_ != 0")
int example(int x) {
  int a = 6;
  return a / 3 - x;
}
```



Planned User Study

We aim to recruit 30 Java developers and split them into **baseline** and **redesign** groups to evaluate the impact of these improvements.



Interested in participating in this study? Scan the QR code!

Research Questions

RQ1. How does improved verification feedback affect developers' ability to **resolve** different types of verification errors?
RQ2. How does improved verification feedback affect developers' **understanding** of verification state and outcomes?
RQ3. What aspects of verification feedback do developers find **most valuable** for diagnosing and resolving errors?

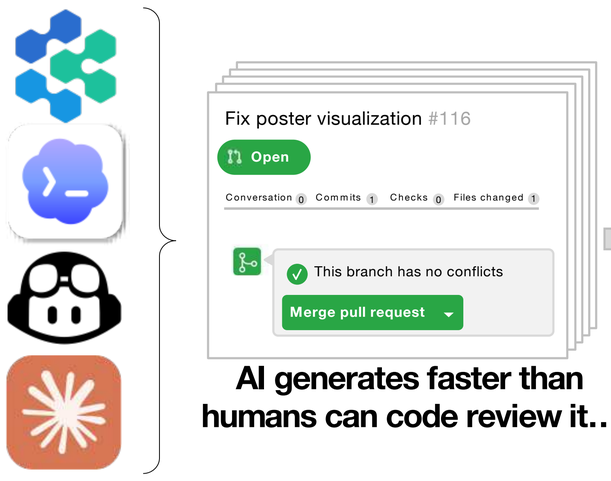
Acknowledgements:
This work was funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the project CMU-Portugal, ref. PRT/BD/154254/2021, and under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025, which includes the LASIGE Seed project ref. LASIGE-SP-2025.07.
This work was also funded by the European Union - NextGenerationEU through the PRR - Recovery and Resilience Plan under the LASIGE Research Unit, ref. UID/PRR/00408/2025, DOI https://doi.org/10.54499/UID/PRR/00408/2025 and ref. UID/PRR2/00408/2025.

LGTM! Characteristics of Auto-Merged LLM-based Agentic Pull-Requests

Ruben Branco*†, Paulo Canelas*†‡, Catarina Gamboa*†‡, Alcides Fonseca†

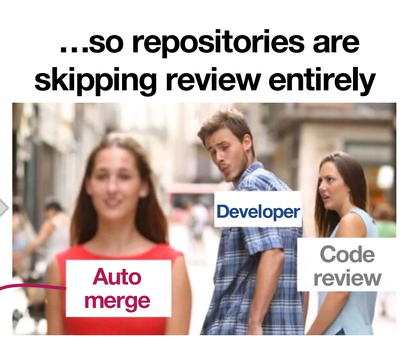
† LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal
‡ Software and Societal Systems Department, Carnegie Mellon University, US
* Authors contributed equally to this work

reliable software systems



AI generates faster than humans can code review it…

…so repositories are skipping review entirely



Merging a Pull Request (PR) without any reviews, review comments, or comments from anyone other than the author or bots.

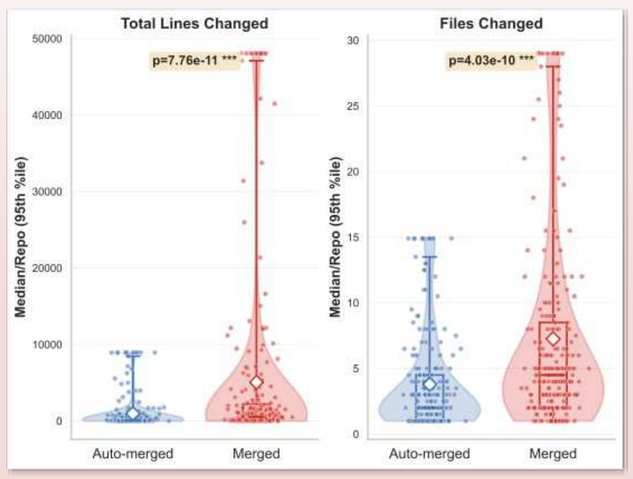
RQ1. What properties of agentic pull requests and repositories relate to auto-merges?

RQ2. What characteristics distinguish human and agent authored auto-merged PR?

AIDev dataset	5 agents	33,000 pull requests	2,800 repositories	≥100 stars

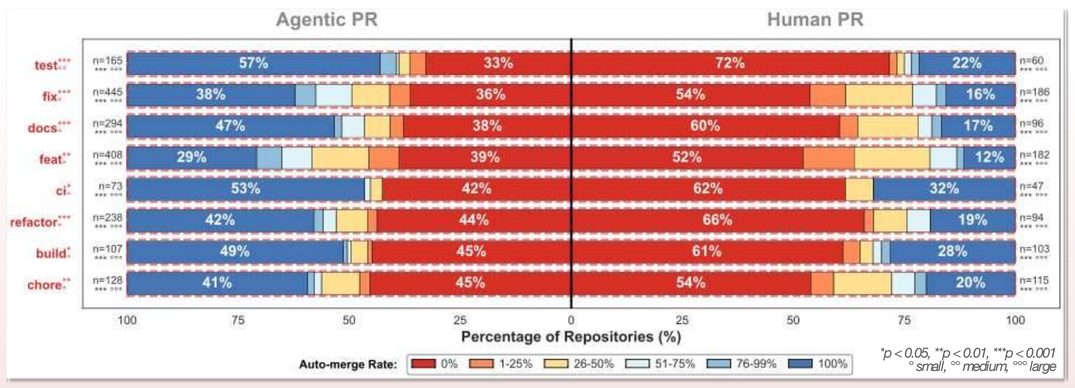
Answer to RQ1.

Auto-merge is bimodal: repos either do it for all agentic PRs or none. Auto-merged PRs are smaller, in less mature repos, and for lower-risk tasks.



Agentic auto-merge follows a **bimodal distribution**, repositories either merge or avoid it.

Auto-merge PRs are **significantly smaller** across all size metrics: lines, files, additions, and deletions.

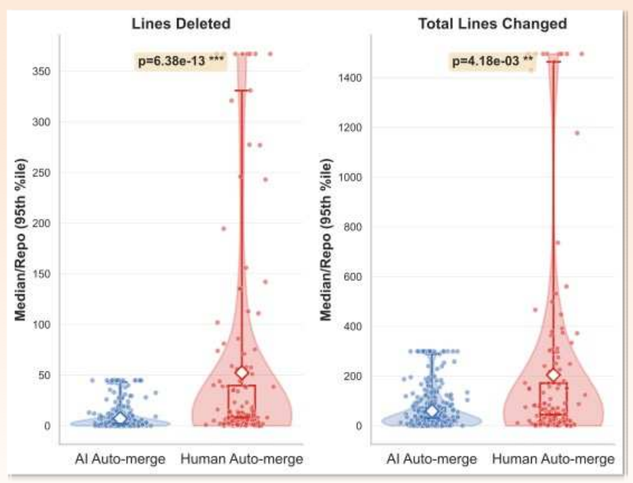


0% **auto-merge rates occur in more mature** repositories with more stars, forks, CI checks, and higher health scores.

Tests are highly trusted, features are less: tests and CI reach 100% auto-merge in over half of repos, while features reach only 29%, likely reflecting their higher risk.

For all task types, more repositories reach 100% auto-merge for agentic PRs than for human PRs.

Auto-merge is all-or-nothing for agentic PRs. When maintainers allow it, AI code skips review more often than human code, unless agents delete.



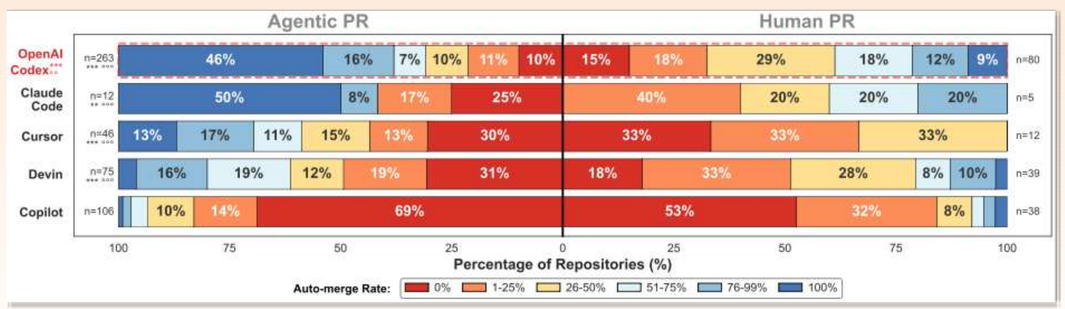
Agentic auto-merged PRs are **smaller and delete less** code than human ones.

Agentic PRs with deletions are auto-merged less often, suggesting **caution toward agent-removed code.**

In the same repositories, only OpenAI Codex shows significantly different auto-merge rates than human PRs.

Answer to RQ2.

Agent PRs are smaller and achieve higher auto-merge rates than human PRs. However, maintainers show caution toward agentic PRs that delete existing code.



Not all agents are the same: Codex and Claude have higher auto-merge rates. AI assistants and editors, Copilot, Cursor, and Devin have less.

Actionable Takeaways

1. **Keep agent PRs small and single-purpose**
   Auto-merged agentic PRs are consistently smaller and more focused, so developers should scope AI-generated changes to one clear task, with minimal files and lines changed.

2. **Start with low-risk, developer-facing tasks**
   Developer-facing tasks (tests, CI, builds) are the safest bet for agentic auto-merge. Customer-facing changes are riskier and should receive human review.

3. **Treat code deletion as a review trigger**
   Maintainers are notably more cautious when agents remove existing code, so PRs with deletions may benefit from extra explanation, or mandatory human review.

Mining Challenge #116

More details in the paper

https://doi.org/10.1145/3793302.3793621

Acknowledgements:
This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the project CMU-Portugal, ref. SFRH/BD/151469/2021 and PRT/BD/154254/2021, a PhD Scholarship (RB) https://doi.org/10.54499/2022.10727.BD, and under the LASIGE Research Unit, ref. UID/00408/2025, DOI https://doi.org/10.54499/UID/00408/2025. This work was also funded by the European Union, NextGenerationEU through the PRR - Recovery and Resilience Plan under the LASIGE Research Unit, ref. UID/PRR/00408/2025, DOI https://doi.org/10.54499/UID/PRR/00408/2025 and ref. UID/PRR2/00408/2025.